Algebraic quantum field theory of linear fields

In algebraic quantum field theory [2] the observables measurable in a region O of spacetime form a C*-algebra $\mathcal{A}(O)$. The observables are taken bounded, such as the exponentials $e^{\mathrm{i}\phi(f)}$ of smeared fields rather than the unbounded fields $\phi(f)$, which avoids questions of domains; closing in norm makes each $\mathcal{A}(O)$ a C*-algebra, which for the free field is determined by the canonical commutation relations up to isomorphism, independently of the Hilbert space (Remark 2.12).

These algebras sit inside one another: larger regions have larger algebras, algebras of spacelike separated regions commute, and the algebra of a region is already contained in the algebra of any region that determines it causally. Dimock [1] formulated these requirements for a Lorentzian manifold as five axioms and verified them for the free scalar field on a globally hyperbolic spacetime. These notes give a formalized version of that verification in an abstract form: the spacetime enters only through the Green data described next, primitivity is proved given an irreducible representation, and Dimock's form of the time-slice axiom follows on paper.

The analysis the verification rests on is the existence of retarded and advanced Green operators for the Klein–Gordon operator, a theorem of hyperbolic partial differential equations [3]. Here it enters as a hypothesis, the Green data of Chapter 7, modelled on the Green-hyperbolic operators of [11]. The hypothesis is proved on a lattice spacetime, the running example (Chapters 3 and 5), for the free massless field on $3+1$ Minkowski space (Chapter 8), and for the Klein–Gordon field there with any real m^2, negative values included (Chapter 9).

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About these notes

These notes are the mathematical account of the Lean library LinearAQFT and the blueprint from which it was built. The library depends on Mathlib, on LocalAlgebrasLean, which holds regions with their orthogonality and nets of local algebras, and on BosonicLean, which holds the CCR algebra, Weyl systems, their relabelling and pullback, the simplicity theorem, irreducible representations and primitivity; the declaration index at the end names the declarations of LinearAQFT and BosonicLean.

Terminology. Standard terms keep their usual meaning: net of local algebras; Weyl operator; Weyl system (here required to generate its algebra); CCR algebra; vector state; retarded and advanced Green operators; commutator function; causal propagator; Cauchy surface; globally hyperbolic; Cauchy temporal function; Green-hyperbolic operator [11]; the retarded potential; spherical means; orthochronous Lorentz and Poincaré maps; the Klein–Gordon operator; local operators; the tachyonic case; chronological order; Huygens' principle. The axioms are those of [1, p. 220]: Axioms 1–3 (isotony with generation, primitivity, causality) are essentially the Haag–Kastler axioms [2], Axiom 4 (time slice) goes back to Haag and Schroer [19], and Axiom 5 is covariance. Results of [1] are cited by their numbers there, for example Axiom 4 or Theorem 4 of [1].

The *time-slice property* is condition (N3) of Definition 1.11 for the causal index at hand; the *time-slice axiom* is Axiom 4 of [1], (N3) for the dependence of Example 1.4. For Green data the first is proved (Proposition 7.9 and Theorem 4.7). On paper it implies the second for the Green data of Remark 7.2, whose cutoffs are given by all Cauchy temporal functions (Remark 7.6), and for the instances of Chapters 8 and 9, whose cutoffs are inertial slabs, when the Cauchy surface

is flat. For a curved Cauchy surface that implication can fail, and the axiom then rests on all Cauchy temporal functions as cutoffs, which are not formalized (Remark 8.13).

Coined terms. These terms are coined for these notes and defined where they first appear: causal index (Definition 1.2); covariance pair (Definition 1.14); family of generators and generated net (Definition 1.20); relabelling unitary (Definition 2.13); right Weyl operator (Definition 2.16); phase-space data (Definition 4.1); the net of a Weyl system (Definition 4.6); the CCR net (Definition 4.10); the pullback of a Weyl system (Definition 5.7); the Cauchy map (Definition 5.9); the Cauchy-data net (Definition 5.12); the average M_n^u (Definition 6.3); Green data with their Green identities $PG^{\pm}f = f$, $G^{\pm}P_{\mathcal{D}}f = f$ (not Green's identities of vector calculus), cutoffs and slabs (Definition 7.1); the derived causal index (Definition 7.4); the derived phase-space data (Definition 7.7); a symmetry of Green data (Definition 7.12); the future and past cone measures (Definition 8.4); future unit vectors and inertial slabs (Definition 8.9); the extended Green operators and functions supported in the future or past of a compact set (Definition 9.3).

Reading conventions. C*-algebras are unital and complex; a $*$-homomorphism is unital. For a subset s of a C*-algebra A, $\mathrm{alg}(s)$ is the smallest $*$-subalgebra of A containing s, and $\overline{\mathrm{alg}}(s)$ is its norm closure, the C*-subalgebra generated by s. Subalgebras and $*$-subalgebras contain the unit. Two elements commute when $ab = ba$. Inner products are linear in the second argument. π and e also keep their numerical meaning. $[x = y]$ is 1 if $x = y$ and 0 otherwise.

A statement marked Lean ✓ has a checked proof in LinearAQFT or in its dependency BosonicLean whose correspondence with the statement here has been independently reviewed; the declaration index at the end names the checked declarations. Every Example carries exactly one status: Lean ✓ as above, script ✓ when a check script verifies it exhaustively, or paper when it is checked on paper only. Definitions, Remarks and paragraphs of motivation carry no marks; they are checked on paper, and "on paper" in them marks a claim that goes beyond the marked statements they cite. A sentence tagged formalization records a choice made for the formalization that a proof on paper would not need.

Notation

Each row lists symbols and their meanings, with the chapter where each meaning is defined in parentheses; a second meaning of a symbol is used only in the place its row names.

$\mathcal{R}$, O, $\leq$	a set of regions, a region, inclusion (1); the causal relation of points (1, 7, 8)
$O_1 \perp O_2$, $O \triangleleft O'$	orthogonality (spacelike separation), causal dependence (1)
$J(O)$, S_{t_1}	points comparable with a point of O (1, 7); the slices $t \in \{t_1, t_1 + 1\}$ of $\mathbb{Z}^2$ (1)
κ, Aut	an isomorphism of causal indices; the automorphism group (1)
A, $\mathcal{A}$	a C*-algebra, a net in it (1)
a, b	elements of a C*-algebra (1); translations in $\mathbb{Z}^2$ (5); integers in Example 1.7; $a = (a_t, a_x) \in \mathbb{Z}^2$ (7); $a \in \mathbb{R}^4$, the translation part of $\gamma = (\Lambda, a)$ (8)
$\mathrm{alg}(s)$, $\overline{\mathrm{alg}}(s)$	$*$-subalgebra generated by s, its norm closure (conventions)
$Z(s)$	elements commuting with every element of s and its adjoint (1)
Y	a set of points, finite on the lattice (1, 7)
α	a $*$-isomorphism of C*-algebras (1)
G, $\mathcal{A}_G$	a family of generators, the net it generates (1)
e, s, s'	a nilpotent matrix; sets of elements (1)
m^2, h	the mass parameter; a function on $\mathbb{Z}^2$, often a solution (1, 3); h an element of $\mathcal{F}$ (7); m^2 any real number, $P_{m^2} = P + m^2$ (9)
(M, g), Σ	a globally hyperbolic spacetime, a Cauchy surface (1)
π, H, T	a representation, its Hilbert space, a bounded operator (1)
Γ, κ_γ, α_γ	a group and its actions on regions and on A (1)

$\sigma_x, \sigma_y, \sigma_z$	the Pauli matrices in $M_2(\mathbb{C})$ (Example 1.17)
ξ, A, a, b, c	a nonzero vector of H (1); a self-adjoint operator and coordinates $(a, b) \in \mathbb{R}^2$ in Example 2.7 (2); a translation vector of $\mathbb{Z}^2$ (3); see also the rows above
g, ζ, z	a Lorentzian metric (1), a test function (3–9), a smooth function on $\mathbb{R}$ in Remark 8.13; a bijection of $\mathbb{Z}^2$ (5); the Fourier variable (5), a coordinate (8)
p, q, δ_p	points of $\mathbb{Z}^2$ (1), of M (7); the indicator of $\{p\}$ (3)
V, σ, u, w	a real vector space, an alternating form, vectors of V (2)
$\ell^2(V)$, ψ, v, δ_v	square-summable functions on V, one of them, a point of V, a basis vector (2)
$\langle \cdot, \cdot \rangle$	the inner product of $\ell^2(V)$ (2); the real pairing $\sum_p f(p)h(p)$ on $\mathbb{Z}^2$ (3); the continuum pairing $\int fh$ (Remark 2.1); the pairing of Green data (7)
$W(u)$, $\mathrm{CCR}(V, \sigma)$	a Weyl operator, the C*-algebra they generate (2)
L, U_L, α_L	a symplectic bijection, its unitary, its $*$-isomorphism (2)
$R(w)$, τ	a right Weyl operator, the vector state (2)
$\delta(\cdot)$	the Dirac delta (2, 8, 9)
λ, ν, $\mathcal{J}$	a real number (2, 6, 7, 9), the dilation factor (8, 9); a tolerance (6), the half-width of a slab (7); a complex structure (6)
$\mathcal{D}$, f, g	test functions on $\mathbb{Z}^2$ (3), in general (4)
K, $G^{\pm}$	the retarded kernel (3); the Green operators (3); the Green operators of Green data (7), those of the wave operator (8); any Green operators of a local operator (Section 8.6), and the extended Green operators (9)
Δ	the commutator function, $Ef = \sum_q f(q)\Delta(\cdot - q)$ (3)
k, $\mathcal{T}$	the test function $(1 - \chi)G^+ f + \chi G^- f$ (3, 7); a Cauchy temporal function (7)
$\mathcal{D}(O)$, $[f]$	test functions localized in O; the phase-space element of f (4)
$\theta = (\kappa, \kappa_*, L)$, α_θ	an isomorphism of phase-space data; its $*$-isomorphism of CCR algebras (4)
$\mathcal{A}_W$	the net of a Weyl system (A, W) (4)
r, ε	the time reversal $(t, x) \mapsto (-t, x)$ of the lattice; the sign $(-1)^{t+x}$ (4); r also a radius (8); the time reversal of M (9)
κ_a, κ_{a*}, L_a, θ_a	translation by $a \in \mathbb{Z}^2$ of regions, of test functions, of V; the isomorphism (5)
ρ, κ_ρ, θ_ρ	the spatial reflection $(t, x) \mapsto (t, -x)$, its action on regions, its isomorphism (5)
α_a	the automorphism of $\mathrm{CCR}(V, \sigma)$ of the translation by a (5)
V_S, $d = (d_0, d_1)$	the space of Cauchy data, an element of it (5)
σ_S, W_S	the Wronskian; a Weyl system over (V_S, σ_S) (5)
Λ, Λ_{t_1}, F	a linear map preserving the forms; the Cauchy map at t_1; the change of slice (5); Λ an orthochronous Lorentz map (8)
ω, β	a state; an inverse temperature (Remark 2.20)
M_n^u, $\mu_n(u, v)$	the average over n conjugations by $W(u)$; the mean of its phases (6)
I, τ_A	a two-sided ideal; a functional on A with $\tau_A(W(v)) = [v = 0]$ (6)
n, N	the length of an average; a number of vectors (6)
$\hat{b}(v)$	the coefficient of $W(v)$ in a finite combination of Weyl elements (6)
ϕ, Π, E, P	field, momentum, causal propagator, Klein–Gordon operator (Remark 2.1); E and P of Green data (7); P any local operator (Section 8.6), the wave operator (8, 9); the lattice P and E (3)
M, $J^{\pm}$	the points of Green data (of a spacetime, its points); causal future and past (7)
$\mathcal{F}$, ι, $P_{\mathcal{D}}$	functions, the map of test functions into functions, P on test functions (7)
χ, S_χ	a cutoff and its slab (7); on the lattice, the time cutoff (3)
γ, $\gamma_{\mathcal{D}}$, $\gamma_{\mathcal{F}}$, L_γ	a symmetry of Green data; its action on test functions, functions, V (7)

η, e_0	the Minkowski form, the time direction (8); in Remark 2.15 a $*$-homomorphism (2)
∇^2, $\mu^{\pm}$	the Laplacian on $\mathbb{R}^3$; the future and past cone measures (8), not the phases μ_n (6)
n, τ_n, ϑ	a future unit vector, its time function, the smooth transition function (8); n also a length (6); τ alone is the vector state (2)
$f_{t,x}$, Φ, ϕ	a translate and reflection of f; a function on $[0,\infty)$; a function on $\mathbb{R}^3$ (8); ϕ also the field (Remark 2.1)
u, v, w; s, t, x, y, z; e_1; D; ℓ	vectors of $\mathbb{R}^4$; coordinates; a unit vector of $\mathbb{R}^3$; the double cone $\{\lvert t\rvert + \lvert\vec{x}\rvert < 1\}$; a thickness (8)
$\mathsf{A}f$, $\bar{f}$, ρ, Q	the spherical mean, its radial profile, the radius $\lvert y\rvert$, an orthogonal operator (8); ρ also the reflection (5)
$I^{\pm}$	the open future and past cones (8); I also an ideal (6)
$\mathcal{P}^{\uparrow}$, $\gamma = (\Lambda, a)$	the orthochronous Poincaré group, one of its maps $p \mapsto \Lambda p + a$ (8)
e_i, e^i, d, dist	the standard basis of $\mathbb{R}^4$ and its dual basis, with e_1 not the unit vector of $\mathbb{R}^3$ of Chapter 8; a vector of $\mathbb{R}^3$; the distance of the norm $\max(\lvert q^0\rvert, \lvert\vec{q}\rvert)$ (9); d also Cauchy data (5)
$k(p-q)$, $\mu^{\pm}$	the convolution kernel of Remark 9.28; the cone measures (8), not the phases μ_n (6)
$G^{\pm}_{m^2}$, $(G^{\pm})^j$, j	the Green operators of P_{m^2}; the iterates of the extended Green operators; a series index (9)
$\mathcal{F}^{\pm}$, ψ	smooth functions supported in the future or past of a compact set; a cutoff test function (9); ψ also a vector of $\ell^2(V)$ (2)
t_0, C, T, s	a time before which a function vanishes; a bound; a time bound; an elapsed time (9); T also a bounded operator (1), s also a coordinate (8), t_0 also a time coordinate (7)
C^+, ∂C^+, $\ll$	the closed future cone, its boundary (the light-cone surface); chronological order (9); C alone is a bound
K, K', A, B, U	subsets of M, compact where stated; U a neighbourhood or an open ball (9); K also the retarded kernel (3), A also a C*-algebra (1), U_L a unitary (2)
h_1, h_2, H_j, $\hat{d}$, y_1, z_1, q_0, p', ξ, k	functions on M; the terms of a series; vectors of $\mathbb{R}^3$; points of M; a real number; a summation index in Lemma 9.24 (9)
n, J_1, I_1	an integer exponent (9); the Bessel function and the modified Bessel function (Remark 9.28)
f_λ, f_0, p_0, c, m_0^2	the dilation $f(\cdot/\lambda)$; the bump, point, value $(G^+)^2 f_0(p_0)$ and small parameter of Theorem 9.26 (9); c also a constant level of τ_n (8)

Roadmap

These notes are also the blueprint of a Lean library, and their chapters are ordered by logical dependence. The lattice appears in Chapter 1 as an example; its propagators (Chapter 3) come before the general net (Chapter 4), and the lattice net is completed in Chapter 5, apart from primitivity (Chapter 6). Chapter 1 states the axioms for any set of regions carrying inclusion, orthogonality and dependence, and reduces them to properties of generators (Proposition 1.21); Chapter 2 constructs the Weyl operators and the CCR algebra. Figure 1 shows which chapters use which.

The main result is an abstract form of Theorem 4 of [1], which is proved there for globally hyperbolic spacetimes: whenever Green data exist (retarded and advanced Green operators with cutoffs, Definition 7.1), the net generated by the Weyl operators of localized test functions is generating and causal, has the time-slice property for the dependence derived from the cutoffs, and every symmetry of the Green data gives a covariance pair (Axioms 1, 3, 4 and 5; Theorem 7.11 and Proposition 7.13). The globally hyperbolic setting of [1] and its form of the time-slice axiom are reached from it on paper only (Remarks 7.2 and 7.6). Chapter 3 constructs the propagators on the lattice, and Chapter 4 proves isotony, generation, causality and time slice for any phase-space data, and covariance for their CCR net; with Example 4.2 these hold on the lattice without

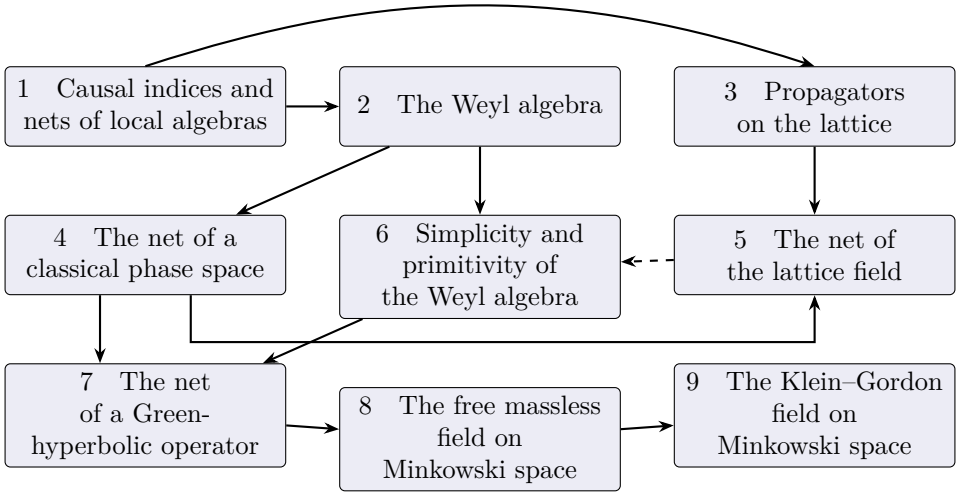


Figure 1: Dependence of the chapters. An arrow means that the later chapter uses the earlier one; a chapter also uses every chapter from which a chain of arrows leads to it, and arrows implied by such chains are omitted. Chapter 6, for instance, rests mainly on Chapter 2. Chapter 7 constructs the classical phase space from Green data, which gives the theorem for every instance of Green data, over the derived causal index. Chapter 4 uses Chapter 3 only in its running example, Example 4.2, and in Remark 4.4. Chapter 6 uses Chapters 4 and 5 only in Remark 6.15 and its last section. Chapter 7 uses Chapters 3 and 5 only in remarks, motivating paragraphs and its last section. Chapter 8 uses Chapter 7 and, through it, Chapters 4 and 6; it does not use the lattice chapters. Chapter 9 uses Chapters 7 and 8 and, through them, Chapters 4 and 6. Uses only in examples, remarks, motivating paragraphs or in a last section are listed above; the one from Chapter 5 to Chapter 6 is drawn dashed, and chains of arrows are followed along solid arrows only.

assumptions. Chapter 5 states the lattice data as numbered results, adds covariance under the group of lattice translations (Theorem 5.6), and shows that the net defined through Cauchy data is the net of a pulled-back Weyl system, isomorphic for the CCR algebra to the lattice CCR net (Theorem 5.13).

Primitivity, Axiom 2 of [1], is a property of the Weyl algebra alone: for nondegenerate σ, as for the phase space of a field, Chapter 6 proves that the algebra is simple (Theorem 6.9) and deduces primitivity from the existence of an irreducible representation (Theorems 6.14 and 6.16). Chapter 7 derives the phase-space data from Green operators and cutoffs (Proposition 7.9), which gives the axioms (Theorem 7.11), with the lattice as an instance (Theorem 7.15); a continuum instance needs such Green data, sketched in general (Remark 7.2), and, with all Cauchy temporal functions as cutoffs, has the time-slice axiom in the form of [1] (Remark 7.6).

Chapter 8 constructs Green data for the free massless field on $3 + 1$ Minkowski space, with slabs between parallel hyperplanes as cutoffs, which gives the axioms for that field (primitivity given an irreducible representation) with the time-slice property for those slabs (Theorem 8.12, Remark 8.13) and its covariance under the orthochronous Poincaré group (Theorem 8.18).

Chapter 8 ends by stating its construction for every local operator with Green operators (Theorem 8.21; in the formalization the results of Chapter 8 on cutoffs, uniqueness and covariance are derived from it). Chapter 9 constructs the Green operators of the Klein–Gordon operator $P + m^2$ for every real m^2 as a series in the Green operators of the wave operator, which gives the same results for that field (Theorem 9.18), with the massless field as the case $m^2 = 0$ (Corollary 9.19); Huygens' principle holds exactly when $m^2 = 0$ (Theorem 9.27). The results that correspond to [1] depend on few chapters: the axioms for phase-space data (Theorem 6.16) and their covariance (Theorem 4.11) use Chapters 1, 2, 4 and 6 only; the lattice instance (Corollary 6.18) adds

Chapters 3 and 5; the massless field (Theorems 8.12 and 8.18) adds Chapters 7 and 8, and in the formalization the construction of Theorem 8.21, but neither the lattice chapters nor the series of Chapter 9.

1 Causal indices and nets of local algebras

A net assigns to each region of spacetime the observables that can be measured inside it. The axioms of [1, §1] refer to the regions only through inclusion, spacelike separation and causal dependence, so they can be stated for any set of regions carrying these three relations; covariance under isometries becomes covariance under the isomorphisms of these relations, a larger class (Remark 1.16). For a net generated by a family of elements, every axiom except primitivity then reduces to a property of the generators (Proposition 1.21).

1.1 Regions and their causal relations

The running example is the lattice $\mathbb{Z}^2$ with points (t, x), time t and space x, and the partial order

$$(t, x) \leq (t', x') \quad \Longleftrightarrow \quad |x' - x| \leq t' - t,$$

under which (t', x') lies in the discrete future light cone of (t, x). For a set O of points let $J(O)$ be the set of points comparable with some point of O, that is, lying in its future or past cone, and for $t_1 \in \mathbb{Z}$ let $S_{t_1} = \{(t, x) : t = t_1 \text{ or } t = t_1 + 1\}$ be a pair of consecutive time slices. For the discrete wave equation with mass parameter $m^2 \in \mathbb{R}$, $h(t+1, x) = h(t, x+1) + h(t, x-1) - h(t-1, x) - m^2 h(t, x)$, the values of a solution h on S_{t_1} determine it everywhere, by this recursion forwards and its solution for $h(t-1, x)$ backwards.

Remark 1.1 (Why a lattice). The free field on a globally hyperbolic spacetime is built from the retarded and advanced Green operators of the Klein–Gordon operator, whose existence is a theorem on hyperbolic partial differential equations [3]; these notes take it as a hypothesis (Chapter 7); for the massless field on $3+1$ Minkowski space it is proved in Chapter 8, and for the Klein–Gordon field there with any real m^2 in Chapter 9. On $\mathbb{Z}^2$ the recursion above determines the Green operators slice by slice, so the hypothesis becomes a theorem, and the lattice is a spacetime on which the propagator hypothesis holds. The recursion moves by at most one site per time step, which gives the light cone, and a pair of slices carries the Cauchy data: these are the two features of spacetime that the axioms use. The lattice is a spacetime in its own right, not a discretization whose continuum limit is studied.

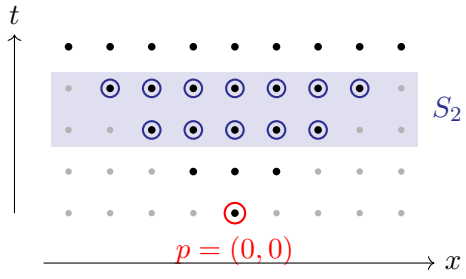


Figure 2: The future of $p = (0, 0)$ in the lattice $\mathbb{Z}^2$ (black points: $|x| \leq t$) and the slices S_2 (shaded). The circled points form $J(\{p\}) \cap S_2$.

The three relations among sets of lattice points are these. Inclusion $O \subseteq O'$. Orthogonality: $O_1 \perp O_2$ when no point of O_1 is comparable with a point of O_2. Dependence: $O \triangleleft O'$ when for every finite set $Y \subseteq O$ some pair of slices meets the causal shadow of Y only inside O', that is, $J(Y) \cap S_{t_1} \subseteq O'$ for some t_1, which may depend on Y. Observables will be localized in finite sets, which is why finite parts of O suffice; for finite O one takes $Y = O$. For the point $p = (0, 0)$:

- $\{p\} \perp \{(0, 1)\}$ and $\{p\} \perp \{(1, 2)\}$, since $|1| > 0$ and $|2| > 1$; but $\{p\}$ and $\{(1, 1)\}$ are not orthogonal, since $(1, 1)$ lies on the future light cone of p.
- $J(\{p\}) \cap S_2$ consists of the points $(2, x)$ with $|x| \leq 2$ and $(3, x)$ with $|x| \leq 3$ (Figure 2), so $\{p\} \triangleleft O'$ for every O' containing these twelve points.

- $\{p\} \triangleleft \{p\}$ fails: for every t_1 the set $J(\{p\}) \cap S_{t_1}$ contains at least four points, so dependence is not reflexive.
- For the slice $O = \{(0, x) : x \in \mathbb{Z}\}$ and $O' = \{(t, x) : t \geq 1,\ |x| \leq t^2\}$, $O \triangleleft O'$: a finite $Y \subseteq O$ lies in $|x| \leq R$, and $J(Y) \cap S_{t_1}$ lies in $|x| \leq R + t_1 + 1 \leq t_1^2$ for large t_1. No single pair of slices serves all of O, since $J(O)$ is all of $\mathbb{Z}^2$.

Definition 1.2 (Causal index)**.** Let $(\mathcal{R}, \leq)$ be a preordered set, whose elements are called regions. A *causal index* on $\mathcal{R}$ consists of two relations, *orthogonality* $O_1 \perp O_2$ and *dependence* $O \triangleleft O'$ (read: O is causally dependent on O'), such that for all regions:

(i) $O_1 \perp O_2$ implies $O_2 \perp O_1$;

(ii) $O_0 \leq O_1$ and $O_1 \perp O_2$ imply $O_0 \perp O_2$;

(iii) $O_0 \leq O$ and $O \triangleleft O'$ imply $O_0 \triangleleft O'$;

(iv) $O \triangleleft O'$ and $O' \leq O_1$ imply $O \triangleleft O_1$.

From (i) and (ii), orthogonality is also antitone in its second argument: $O_3 \leq O_2$ and $O_1 \perp O_2$ imply $O_1 \perp O_3$.

Example 1.3 (The lattice)**.** Lean ✓ The subsets of $\mathbb{Z}^2$ ordered by inclusion, with $\perp$ and $\triangleleft$ as above, form a causal index. Condition (i) holds because comparability is symmetric, and (ii) because, for $O_0 \subseteq O_1$, a comparable pair in $O_0 \times O_2$ would lie in $O_1 \times O_2$. For (iii), a finite subset of $O_0 \subseteq O$ is a finite subset of O. For (iv), enlarging O' preserves the inclusion $J(Y) \cap S_{t_1} \subseteq O'$.

Example 1.4 (A globally hyperbolic spacetime)**.** paper On a globally hyperbolic Lorentzian manifold (M, g) take as regions the open sets with compact closure. Let $O_1 \perp O_2$ when no causal curve joins a point of O_1 to a point of O_2; this is the spacelike separation of [1]. Let $O \triangleleft O'$ when there is a smooth spacelike Cauchy surface Σ such that every endless causal curve through a point of O meets Σ in O'; this is the form of causal dependence used in the proof of Theorem 4 of [1, p. 226]. Conditions (i)–(iv) hold because a causal curve traversed backwards is causal, a curve through $O_0 \subseteq O$ passes through O, and a curve meeting Σ in O' meets it in every region containing O'.

Remark 1.5 (Dependence through curves). The other formulation of [1], every endless causal curve through O meets O' [1, p. 220], satisfies (i)–(iv) for the same reasons.

In the lattice, comparability replaces causal curves and a pair of slices replaces a Cauchy surface, chosen for each finite part of O (Remark 7.6 compares the two forms).

Definition 1.6 (Isomorphism of causal indices)**.** An *isomorphism* from a causal index on $\mathcal{R}$ to a causal index on $\mathcal{R}'$ is a bijection $\kappa : \mathcal{R} \to \mathcal{R}'$ such that $\kappa O \leq \kappa O' \iff O \leq O'$, $\kappa O_1 \perp \kappa O_2 \iff O_1 \perp O_2$ and $\kappa O \triangleleft \kappa O' \iff O \triangleleft O'$ for all regions. The isomorphisms of a causal index to itself form a group Aut under composition, $(\kappa_1 \kappa_2)(O) = \kappa_1(\kappa_2 O)$, with the identity as unit.

Identities, composites and inverses of isomorphisms are isomorphisms, by inspection of the three equivalences.

Example 1.7 (Symmetries of the lattice)**.** paper Each point map below sends regions to regions, and the induced map on subsets of $\mathbb{Z}^2$ is an automorphism of the causal index of Example 1.3.

- A translation $(t, x) \mapsto (t + a, x + b)$ preserves the order of points, hence J and $\perp$, and maps S_{t_1} to $S_{t_1 + a}$.
- The reflection $(t, x) \mapsto (t, -x)$ preserves the order of points and each slice.

- The time reversal $(t, x) \mapsto (-t, x)$ reverses the order of points, so it preserves comparability, J and $\perp$, and maps S_{t_1} to S_{-t_1-1}.

Regions are ordered by inclusion, which all three maps preserve. Each map is a bijection of $\mathbb{Z}^2$ that sends finite sets to finite sets and pairs of slices to pairs of slices, with $J(\kappa Y) = \kappa J(Y)$, and its inverse is of the same kind, so $\lhd$ is preserved in both directions.

Remark 1.8 (Time orientation). A time reversal, where the spacetime has one (the lattice of Example 1.7, Minkowski space), is a symmetry of the causal index. It exchanges the retarded and advanced propagators of the wave equation [3], hence $E \mapsto -E$ and $\sigma \mapsto -\sigma$, and by Remark 2.15 no $*$-isomorphism sends each Weyl operator $W(u)$ to its time reverse. Covariance of the free field is therefore required only for maps that preserve the time orientation. The argument excludes the map of test functions induced by a time reversal, not every linear map over it: Remark 4.4 gives a symmetry of the massless lattice field over the time reversal.

1.2 Nets and their axioms

Definition 1.9 (Net)**.** Let A be a C*-algebra and $\mathcal{R}$ a preordered set. A *net* in A over $\mathcal{R}$ assigns to each region O a norm-closed $*$-subalgebra $\mathcal{A}(O)$ of A, the *local algebra* of O, such that $O \leq O'$ implies $\mathcal{A}(O) \subseteq \mathcal{A}(O')$ (*isotony*).

Each local algebra, being norm closed, is itself a C*-algebra, and it contains the unit of A.

Remark 1.10 (The global algebra). In [1] the global algebra is the closure of the union of the local algebras; here the net lives in a given algebra A, and the generation axiom below states that A is the C*-algebra generated by the local algebras, which is that closure when the regions are directed (any two lie in a third).

A net does not carry a causal index; each axiom takes one as an argument, so that one net can be tested against several causal structures on the same regions. formalization

Definition 1.11 (The axioms)**.** Let $\mathcal{A}$ be a net in A over a causal index on $\mathcal{R}$.

(N1) $\mathcal{A}$ is *generating* if the C*-subalgebra generated by $\bigcup_O \mathcal{A}(O)$ is A.

(N2) $\mathcal{A}$ is *causal* if $O_1 \perp O_2$ implies $ab = ba$ for all $a \in \mathcal{A}(O_1)$ and $b \in \mathcal{A}(O_2)$.

(N3) $\mathcal{A}$ has the *time-slice property* if $O \lhd O'$ implies $\mathcal{A}(O) \subseteq \mathcal{A}(O')$.

A representation $\pi : A \to B(H)$ on a complex Hilbert space H is *irreducible* if every bounded operator T with $T\pi(a) = \pi(a)T$ for all $a \in A$ is a multiple of the identity. The algebra A is *primitive* if it has an injective irreducible representation on a nonzero Hilbert space.

The Hilbert spaces allowed in primitivity range over a fixed universe of Hilbert spaces, determined by A. This is no restriction: if π is irreducible and $\xi \neq 0$, the closure of $\pi(A)\xi$ is a nonzero closed subspace invariant under $\pi(A)$, and the projection onto it commutes with $\pi(A)$, so it is the identity and $H = \overline{\pi(A)\xi}$. Hence H has a dense subset indexed by A, and H is unitarily equivalent to a Hilbert space in the universe of A. formalization

Remark 1.12 (Irreducibility). Irreducibility is defined here by a trivial commutant. For representations of C*-algebras this is equivalent to the absence of closed invariant subspaces other than 0 and H (see, e.g., [5, Ch. 2]): the commutant of $\pi(A)$ is a von Neumann algebra, so it equals $\mathbb{C}1$ exactly when its only projections are 0 and 1, and its projections are the projections onto closed invariant subspaces.

Remark 1.13 (The time-slice axiom). Axiom 4 of [1, p. 220] reads: if O is causally dependent on O', then $\mathcal{A}(O) \subseteq \mathcal{A}(O')$. This is (N3) when dependence is taken in the sense of the spacetime; (N3) for a given causal index is called the time-slice property. There the meaning of causal

dependence is left open, naming as a strong form that every endless causal curve through O meets O' (Remark 1.5); the proof of Theorem 4 of [1, p. 226] uses the form of Example 1.4; the lattice and Green data use a pointwise form, which the form of Example 1.4 implies (Remark 7.6) for the Green data of Remark 7.2, and for those of Chapters 8 and 9 when the Cauchy surface is flat (Remark 8.13). The time-slice axiom of locally covariant field theory, by which the algebra of a neighbourhood of a Cauchy surface is the whole algebra, is a different statement; it is not assumed here. For Green data, (N3) gives it for the slabs of the cutoffs, and, for the Green data of Remark 7.2, for neighbourhoods of every smooth spacelike Cauchy surface (after Theorem 7.11).

Axiom of [1, §1]	content	here (defined; proved)
Axiom 1	nets of local observables, $\mathcal{A} = \overline{\bigcup_O \mathcal{A}(O)}$	Definition 1.9 and (N1); Theorems 4.7 (for Green data with Proposition 7.9) and 9.18
Axiom 2	primitivity	Definition 1.11; Theorem 6.14, 6.16
Axiom 3	first causality (spacelike commutativity)	(N2); as for Axiom 1
Axiom 4	second causality (time slice)	(N3); as for Axiom 1, over the causal index at hand
Axiom 5	covariance	Definition 1.14; Theorems 4.11, 5.6, 8.18, 9.18(c)

Table 2: The axioms of [1] and where they are defined or proved.

Definition 1.14 (Covariance)**.** Let $\mathcal{A}$ be a net in A over a causal index on $\mathcal{R}$, and $\mathcal{A}'$ a net in A' over a causal index on $\mathcal{R}'$. A pair (κ, α) of an isomorphism κ of the causal indices and a $*$-isomorphism $\alpha : A \to A'$ is a *covariance pair* from $\mathcal{A}$ to $\mathcal{A}'$ if

$$\alpha(\mathcal{A}(O)) = \mathcal{A}'(\kappa O) \qquad \text{for every region } O. \tag{1.1}$$

Two nets over the same causal index are *isomorphic* if some (id, α) is a covariance pair between them. A net $\mathcal{A}$ is *covariant under* a group Γ, given group homomorphisms $\gamma \mapsto \kappa_\gamma$ into Aut and $\gamma \mapsto \alpha_\gamma$ into the $*$-automorphisms of A, if each $(\kappa_\gamma, \alpha_\gamma)$ is a covariance pair from $\mathcal{A}$ to itself.

Lemma 1.15 (Covariance pairs compose)**.** Lean ✓ *The pair* $(\mathrm{id}, \mathrm{id})$ *is a covariance pair from a net to itself. If* (κ_1, α_1) *is a covariance pair from* $\mathcal{A}$ *to* $\mathcal{A}'$ *and* (κ_2, α_2) *one from* $\mathcal{A}'$ *to* $\mathcal{A}''$*, then* $(\kappa_2 \circ \kappa_1, \alpha_2 \circ \alpha_1)$ *is one from* $\mathcal{A}$ *to* $\mathcal{A}''$*, and* $(\kappa_1^{-1}, \alpha_1^{-1})$ *is one from* $\mathcal{A}'$ *to* $\mathcal{A}$*.*

Proof. The first claim is (1.1) for identities. For the second, $\alpha_2\alpha_1(\mathcal{A}(O)) = \alpha_2(\mathcal{A}'(\kappa_1 O)) = \mathcal{A}''(\kappa_2\kappa_1 O)$. For the third, put $O = \kappa_1^{-1}O'$ in $\alpha_1(\mathcal{A}(O)) = \mathcal{A}'(\kappa_1 O)$ and apply α_1^{-1}: $\mathcal{A}(\kappa_1^{-1}O') = \alpha_1^{-1}(\mathcal{A}'(O'))$. □

Remark 1.16 (Covariance as an assignment)*.* Axiom 5 of [1] asks for more than a single pair: an assignment $\kappa \mapsto \alpha_\kappa$ on the isometries between spacetimes of one isometry class, with $\alpha_{\mathrm{id}} = \mathrm{id}$ and $\alpha_{\kappa_1\kappa_2} = \alpha_{\kappa_1}\alpha_{\kappa_2}$, each (κ, α_κ) a covariance pair. Such an assignment is a construction on a family of nets. By Lemma 1.15, $(\kappa_1\kappa_2, \alpha_{\kappa_1}\alpha_{\kappa_2})$ is a covariance pair whenever the factors are, so the composition law compares two covariance pairs over the same isometry $\kappa_1\kappa_2$.

1.3 Nets generated by families of elements

The local algebras of the free field are generated by explicit unitaries, the Weyl operators of test functions localized in the region. Whether two such algebras commute is therefore a question about their generators. The answer needs one more hypothesis than the commutation of the generators themselves (Examples 1.17 and 1.18).

Example 1.17 (Two qubits). Lean ✓ In $A = M_2(\mathbb{C}) \otimes M_2(\mathbb{C})$ let $s = \{\sigma_x \otimes 1, \sigma_z \otimes 1\}$ and $s' = \{1 \otimes \sigma_x, 1 \otimes \sigma_z\}$. Since $\sigma_x\sigma_z = -\mathrm{i}\sigma_y$, the $*$-algebra generated by σ_x, σ_z contains $1, \sigma_x, \sigma_y, \sigma_z$, which span $M_2(\mathbb{C})$. Hence $\overline{\mathrm{alg}}(s) = M_2(\mathbb{C}) \otimes 1$ and $\overline{\mathrm{alg}}(s') = 1 \otimes M_2(\mathbb{C})$, and every element of the first commutes with every element of the second.

Example 1.18 (An element commuting with itself). Lean ✓ In $A = M_2(\mathbb{C})$ let $e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $s = s' = \{e\}$. Every element of s commutes with every element of s'. But $ee^* = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $e^*e = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, so e does not commute with e^*, and $\overline{\mathrm{alg}}(s)$, which contains e and e^*, is all of $M_2(\mathbb{C})$ and is not commutative. The generated algebras do not commute.

Commutation with adjoints is the hypothesis that Example 1.18 lacks. For a subset s of A let

$$Z(s) = \{x \in A : \ xb = bx \text{ and } xb^* = b^*x \text{ for all } b \in s\},$$

the elements commuting with every element of s and with its adjoint.

Lemma 1.19 (Commutation of generated algebras). Lean ✓ *Let A be a C*-algebra and $s, s' \subseteq A$, and suppose that every $a \in s$ commutes with every $b \in s'$ and with every b^*, $b \in s'$. Then every element of $\overline{\mathrm{alg}}(s)$ commutes with every element of $\overline{\mathrm{alg}}(s')$.*

Proof. The set $Z(s)$ is a C*-subalgebra of A for every s. It is a unital subalgebra. It is closed under the adjoint: taking adjoints in $xb = bx$ and $xb^* = b^*x$ gives $b^*x^* = x^*b^*$ and $bx^* = x^*b$. It is norm closed, since each equation $xb = bx$ survives norm limits.

The hypothesis says $s \subseteq Z(s')$, so $\overline{\mathrm{alg}}(s) \subseteq Z(s')$. Fix $x \in \overline{\mathrm{alg}}(s)$. Both x and x^* lie in $\overline{\mathrm{alg}}(s) \subseteq Z(s')$, so every $b \in s'$ commutes with x and with x^*; that is, $s' \subseteq Z(\{x\})$. Hence $\overline{\mathrm{alg}}(s') \subseteq Z(\{x\})$, and every element of $\overline{\mathrm{alg}}(s')$ commutes with x. □

In Example 1.17 every generator is self-adjoint, so the hypothesis reduces to commutation of the generators. In Example 1.18 it fails, because e does not commute with e^*.

Definition 1.20 (Generated net). A *family of generators* in A over $\mathcal{R}$ assigns to each region O a subset $G(O)$ of A, with $G(O) \subseteq G(O')$ whenever $O \leq O'$. The net *generated* by G is $\mathcal{A}_G(O) = \overline{\mathrm{alg}}(G(O))$.

It is a net: each $\overline{\mathrm{alg}}(G(O))$ is a norm-closed $*$-subalgebra, and $s_0 \subseteq s_1$ implies $\overline{\mathrm{alg}}(s_0) \subseteq \overline{\mathrm{alg}}(s_1)$.

Proposition 1.21 (Axioms from generators). Lean ✓ *Let G be a family of generators in a C*-algebra A over a causal index on a preordered set $\mathcal{R}$ of regions.*

(a) If, whenever $O_1 \perp O_2$, every $a \in G(O_1)$ commutes with every $b \in G(O_2)$ and with every b^*, $b \in G(O_2)$, then $\mathcal{A}_G$ is causal.

(b) If, whenever $O \triangleleft O'$, $G(O) \subseteq \mathcal{A}_G(O')$, then $\mathcal{A}_G$ has the time-slice property.

(c) If $\overline{\mathrm{alg}}(\bigcup_O G(O)) = A$, then $\mathcal{A}_G$ is generating.

(d) Let G' be a family of generators in A' over a causal index on $\mathcal{R}'$, κ an isomorphism of the causal indices and $\alpha : A \to A'$ a $*$-isomorphism with $\alpha(G(O)) = G'(\kappa O)$ for every O. Then (κ, α) is a covariance pair from $\mathcal{A}_G$ to $\mathcal{A}_{G'}$.

Proof. (a) Lemma 1.19 with $s = G(O_1)$ and $s' = G(O_2)$.

(b) $\mathcal{A}_G(O')$ is a C*-subalgebra containing $G(O)$, hence it contains $\overline{\mathrm{alg}}(G(O))$.

(c) The C*-subalgebra generated by $\bigcup_O \mathcal{A}_G(O)$ contains $\bigcup_O G(O)$, hence contains $\overline{\mathrm{alg}}(\bigcup_O G(O)) = A$.

(d) A $*$-homomorphism maps the $*$-algebra generated by a set onto the $*$-algebra generated by its image, so $\alpha(\mathrm{alg}(G(O))) = \mathrm{alg}(G'(\kappa O))$. It remains to pass to closures. A $*$-homomorphism φ between C*-algebras does not increase norms: the spectrum of $\varphi(b)$ is contained in that of b, the norm of a self-adjoint element is its spectral radius, and so $\|\varphi(a)\|^2 = \|\varphi(a^*a)\| \leq \|a^*a\| = \|a\|^2$. Applied to α and α^{-1}, this makes α a homeomorphism, which maps the closure of a set onto the closure of its image. Hence $\alpha(\overline{\mathrm{alg}}(G(O))) = \overline{\mathrm{alg}}(G'(\kappa O))$.

□

Remark 1.22 (Moving generators). In (b) the generators of O are required to lie in the closed algebra of O', not in $G(O')$. For the free field the Weyl operators are labelled by the classes $[f]$ of test functions modulo the kernel of E (Remark 2.1), and the stronger inclusion holds: if $O \lhd O'$, every f localized in O has an f' localized in O' with $Ef' = Ef$ (on the lattice this is Corollary 3.11), so $[f] = [f']$ and the generator $W([f])$ of O is a generator of O'. This is the step of the proof of Theorem 4 of [1, p. 226] that gives time slice. The classes and the net are constructed in Chapter 4 (Example 4.2 and Definition 4.6).

2 The Weyl algebra

The observables of the free field are generated by the Weyl operators $W(u)$, formally $e^{\mathrm{i}\phi(u)}$, labelled by the vectors u of a real vector space V with an alternating form σ, the classical phase space of the field. On the Hilbert space $\ell^2(V)$ these operators are unitary, and the C*-algebra they generate serves as the algebra A of the axioms for the free field. The Hilbert space $\ell^2(V)$ carries the regular representation of the additive group V, twisted by the phases of σ. It serves to show that Weyl operators exist; it is not the vacuum (Fock) representation of the field, which these notes do not construct (Remarks 2.21 and 2.22).

Remark 2.1 (Signs). Spacetime signature is mostly plus, and the Klein–Gordon operator is $P = -\Box + m^2$ ($= \partial_t^2 - \nabla^2 + m^2$ in flat space). The causal propagator is $E = E^{\mathrm{adv}} - E^{\mathrm{ret}}$ for this P, so that $[\phi(f), \phi(g)] = \mathrm{i}\langle f, Eg\rangle$ and, at equal times, $[\phi(t,x), \Pi(t,y)] = \mathrm{i}\delta(x-y)$ for the momentum $\Pi = \partial_t\phi$. On the lattice this sign is Proposition 3.7(e). With $[f]$ the class of a test function f modulo the kernel of E, $\sigma([f],[g]) = \langle f, Eg\rangle$ and $W(u) = e^{\mathrm{i}\phi(u)}$, the Baker–Campbell–Hausdorff formula $e^X e^Y = e^{X+Y}e^{[X,Y]/2}$ for a central commutator gives $W(u)W(w) = e^{-\mathrm{i}\sigma(u,w)/2}W(u+w)$, which is Proposition 2.4 below.

2.1 Weyl operators on $\ell^2(V)$

Throughout, V is a real vector space of any dimension, regarded as a discrete set, and σ is a real bilinear form on V that is *alternating*: $\sigma(u,u) = 0$, hence $\sigma(w,u) = -\sigma(u,w)$. σ may be degenerate. $\ell^2(V)$ is the Hilbert space of square-summable $\psi : V \to \mathbb{C}$, and δ_v is the indicator function of $\{v\}$.

Definition 2.2 (Weyl operator)**.** For $u \in V$, $(W(u)\psi)(v) = e^{\mathrm{i}\sigma(u,v)/2}\,\psi(v+u)$.

Lemma 2.3 (Isometry)**.** Lean ✓ *The Weyl operators on $\ell^2(V)$ (Definition 2.2) satisfy $\|W(u)\psi\| = \|\psi\|$ for all $u \in V$, $\psi \in \ell^2(V)$, and $W(0) = 1$.*

Proof. $|(W(u)\psi)(v)|^2 = |\psi(v+u)|^2$, and $v \mapsto v+u$ is a bijection of V, so the two sums of squares have the same terms. □

Proposition 2.4 (Weyl relation)**.** Lean ✓ *For a real vector space V with alternating form σ, the Weyl operators on $\ell^2(V)$ satisfy $W(u)W(w) = e^{-\mathrm{i}\sigma(u,w)/2}\,W(u+w)$ for all $u, w \in V$.*

Proof. $(W(u)W(w)\psi)(v) = e^{\mathrm{i}\sigma(u,v)/2}e^{\mathrm{i}\sigma(w,v+u)/2}\psi(v+u+w)$ and $(W(u+w)\psi)(v) = e^{\mathrm{i}\sigma(u+w,v)/2}\psi(v+u+w)$; by bilinearity the exponents differ by $\mathrm{i}\sigma(w,u)/2 = -\mathrm{i}\sigma(u,w)/2$. □

Proposition 2.5 (Unitarity)**.** Lean ✓ *For $u \in V$ the Weyl operator $W(u)$ on $\ell^2(V)$ satisfies $W(u)W(-u) = 1$, $W(u)^* = W(-u)$, and $W(u)$ is unitary.*

Proof. Proposition 2.4 and $\sigma(u,u) = 0$ give $W(u)W(-u) = W(0) = 1$. An isometry a satisfies $a^*a = 1$, by polarization of $\langle a\psi, a\psi\rangle = \langle\psi,\psi\rangle$. Hence $W(u)^* = W(u)^*W(u)W(-u) = W(-u)$, and $W(u)W(u)^* = W(u)W(-u) = 1$. □

Definition 2.6 (CCR algebra)**.** $\mathrm{CCR}(V,\sigma) = \overline{\mathrm{alg}}\{W(u) : u \in V\}$, the C*-subalgebra of the bounded operators on $\ell^2(V)$ generated by the Weyl operators.

Example 2.7 (One degree of freedom)**.** paper For $V = \mathbb{R}^2$ with $\sigma((a,b),(a',b')) = ab' - ba'$, the operators $W(a,0)$ and $W(0,b)$ satisfy $W(a,0)W(0,b) = e^{-\mathrm{i}ab}W(0,b)W(a,0)$ by Proposition 2.4 applied twice. This is the Weyl form $e^{\mathrm{i}a\hat{q}}e^{\mathrm{i}b\hat{p}} = e^{-\mathrm{i}ab}e^{\mathrm{i}b\hat{p}}e^{\mathrm{i}a\hat{q}}$ of $[\hat{q},\hat{p}] = \mathrm{i}$. On $\ell^2(\mathbb{R}^2)$, however, $\lambda \mapsto W(\lambda u)$ is not continuous: for $\lambda \neq 0$ and $u \neq 0$, $W(\lambda u)\delta_0 = \delta_{-\lambda u}$ is orthogonal to δ_0, so $\|W(\lambda u)\delta_0 - \delta_0\| = \sqrt{2}$. The unitary group $\lambda \mapsto W(\lambda u)$ is therefore not strongly continuous, so it has no self-adjoint generator, since $\lambda \mapsto e^{\mathrm{i}\lambda A}$ is strongly continuous for every self-adjoint A [8, Theorem VIII.7]: no field operator $\phi(u)$ exists in this representation.

Example 2.8 (Zero form)**.** paper For $\sigma = 0$ the Weyl operators commute, $W(u)W(w) = W(u+w)$, and $\mathrm{CCR}(V,0)$ is commutative: the closure of the commutative algebra spanned by the $W(u)$ is commutative.

2.2 Weyl systems

Definition 2.9 (Weyl system)**.** A *Weyl system* over (V,σ) is a unital C*-algebra A with a map $W : V \to A$ such that each $W(u)$ is unitary, $W(u)W(w) = e^{-\mathrm{i}\sigma(u,w)/2}W(u+w)$, and $\overline{\mathrm{alg}}\{W(u) : u \in V\} = A$.

Lemma 2.10 (Consequences of the relation)**.** Lean ✓ *In a Weyl system $W(0) = 1$, $W(u)^* = W(-u)$ and $W(u)W(w) = e^{-\mathrm{i}\sigma(u,w)}W(w)W(u)$; in particular $W(u)$ and $W(w)$ commute when $\sigma(u,w) = 0$.*

Proof. The relation with $u = w = 0$ gives $W(0)^2 = W(0)$, and multiplying by $W(0)^*$ gives $W(0) = 1$. Then $W(u)W(-u) = e^{\mathrm{i}\sigma(u,u)/2}W(0) = 1$, so $W(-u) = W(u)^{-1} = W(u)^*$. Writing both $W(u)W(w)$ and $W(w)W(u)$ as multiples of $W(u+w)$ gives the phase $e^{-\mathrm{i}\sigma(u,w)/2}/e^{-\mathrm{i}\sigma(w,u)/2} = e^{-\mathrm{i}\sigma(u,w)}$. □

Proposition 2.11 (The concrete system)**.** Lean ✓ $\mathrm{CCR}(V,\sigma)$ *with the Weyl operators is a Weyl system.*

Proof. Unitarity and the relation are Propositions 2.5 and 2.4. For generation, the inclusion of $\mathrm{CCR}(V,\sigma)$ into the bounded operators is an isometric $*$-homomorphism with closed image; it maps the $*$-subalgebra of $\mathrm{CCR}(V,\sigma)$ generated by the $W(u)$ onto the one of the bounded operators generated by them, so the closure of the former is the preimage of $\mathrm{CCR}(V,\sigma)$, which is all of it. □

Remark 2.12 (Uniqueness and covariance). The zero algebra satisfies the three conditions of Definition 2.9; $\mathrm{CCR}(V,\sigma)$ is not the zero algebra, since $\ell^2(V) \neq 0$. Covariance is not part of the definition: for $\mathrm{CCR}(V,\sigma)$ it is Proposition 2.14, and for another Weyl system a family of $*$-isomorphisms with $\alpha_L(W(u)) = W'(Lu)$ has to be assumed. For nondegenerate σ (Definition 6.1) and both algebras nonzero such a family exists by the uniqueness theorem for the CCR algebra [6] (see also [7] and [4, Thm. 5.2.8]), which [1, p. 225] uses by citation; these notes do not prove it, and it is not formalized. For degenerate σ it can fail: for $\sigma = 0$ and $V \neq 0$, $A = \mathbb{C}$ with $W \equiv 1$ is a Weyl system, and $\mathrm{CCR}(V,0)$ is not isomorphic to it, since the vectors $W(u)\delta_0 = \delta_{-u}$ (Proposition 2.17) show that $\mathrm{CCR}(V,0)$ is not one-dimensional.

2.3 Relabelling

Let (V',σ') be a second space and $L : V \to V'$ a linear bijection that is *symplectic*: $\sigma'(Lu,Lv) = \sigma(u,v)$.

Definition 2.13 (Relabelling unitary)**.** $(U_L\psi)(v') = \psi(L^{-1}v')$ for $\psi \in \ell^2(V)$ and $v' \in V'$.

Proposition 2.14 (Relabelling)**.** Lean ✓ *Let $L : V \to V'$ be a linear bijection with $\sigma'(Lu,Lv) = \sigma(u,v)$ and U_L its relabelling unitary (Definition 2.13). Then:*

(a) $U_L : \ell^2(V) \to \ell^2(V')$ is unitary, $U_L^* = U_{L^{-1}}$, $U_{\mathrm{id}} = 1$ and $U_{L_2L_1} = U_{L_2}U_{L_1}$.

(b) $U_L W(u) U_L^* = W'(Lu)$, where W' are the Weyl operators of (V',σ').

(c) $\alpha_L(a) = U_L a U_L^*$ restricts to a $*$-isomorphism $\alpha_L : \mathrm{CCR}(V,\sigma) \to \mathrm{CCR}(V',\sigma')$ with $\alpha_L(W(u)) = W'(Lu)$, $\alpha_{\mathrm{id}} = \mathrm{id}$ and $\alpha_{L_2L_1} = \alpha_{L_2}\alpha_{L_1}$.

Proof. (a) U_L reindexes along the bijection L^{-1}, so it preserves norms, and $U_{L^{-1}}$ is its inverse; an invertible isometry is unitary. The composition law is $\psi(L_1^{-1}L_2^{-1}v'') = \psi((L_2L_1)^{-1}v'')$.

(b) With $v = L^{-1}v'$, $(U_L W(u) U_L^* \psi)(v') = e^{\mathrm{i}\sigma(u,v)/2}\psi(L(v+u)) = e^{\mathrm{i}\sigma'(Lu,v')/2}\psi(v'+Lu)$.

(c) Conjugation by U_L is a continuous $*$-isomorphism of the bounded operators. By (b) it maps the Weyl operators of V onto those of V', hence the generated $*$-algebra into $\mathrm{CCR}(V',\sigma')$, and by continuity its closure as well; the same holds for L^{-1}, which gives the inverse. The identities follow from (a).

□

Remark 2.15 (Time reversal). If $\sigma'(Lu, Lv) = -\sigma(u,v)$, as for time reversal, no $*$-homomorphism η with $\eta(W(u)) = W'(Lu)$ into a nonzero algebra exists unless $\sigma = 0$: applying η to Proposition 2.4 gives $e^{-\mathrm{i}\sigma(u,w)/2} = e^{\mathrm{i}\sigma(u,w)/2}$, while $\sigma(\lambda u, w) = \lambda\sigma(u,w)$ takes every real value when $\sigma(u,w) \neq 0$. This is the algebraic form of Remark 1.8. The relation does not exclude an antilinear map, which conjugates the phase, as in Wigner's treatment of time reversal [9, Ch. 26]; such a map is not a $*$-homomorphism of complex algebras and lies outside Definition 1.14.

2.4 Right Weyl operators and the vector state

The operators and the state of this section are tools for Chapter 6. The right Weyl operators commute with the whole algebra and move δ_0 to every basis vector, which makes the vector state at δ_0 faithful (Remark 2.21); and $\tau(W(v)) = [v = 0]$ picks out the coefficient of 1 in a finite linear combination of Weyl operators.

Definition 2.16 (Right Weyl operator)**.** $(R(w)\psi)(v) = e^{-\mathrm{i}\sigma(w,v)/2}\,\psi(v+w)$, the Weyl operator of the form $-\sigma$.

Proposition 2.17 (Right operators)**.** Lean ✓ *For $w \in V$ the right Weyl operator $R(w)$ on $\ell^2(V)$ (Definition 2.16) is unitary, $R(w)^* = R(-w)$, $R(w)R(w') = e^{\mathrm{i}\sigma(w,w')/2}R(w+w')$, and $R(w)$ commutes with every element of $\mathrm{CCR}(V,\sigma)$. Moreover $W(u)\delta_0 = \delta_{-u}$ and $R(w)\delta_0 = \delta_{-w}$.*

Proof. The first three claims are Propositions 2.4 and 2.5 for $-\sigma$. The exponents of $(W(u)R(w)\psi)(v)$ and $(R(w)W(u)\psi)(v)$ differ by $-\mathrm{i}(\sigma(w,u)+\sigma(u,w))/2 = 0$, so $R(w)$ commutes with every $W(u)$, and so does $R(w)^* = R(-w)$. By Lemma 1.19, with s the set of Weyl operators and $s' = \{R(w)\}$, $R(w)$ commutes with $\mathrm{CCR}(V,\sigma)$. Finally $(W(u)\delta_0)(v) = e^{\mathrm{i}\sigma(u,v)/2}[v+u=0]$, and at $v = -u$ the phase is $e^{-\mathrm{i}\sigma(u,u)/2} = 1$; the same for $R(w)$. □

Definition 2.18 (Vector state)**.** $\tau(a) = \langle \delta_0, a\,\delta_0\rangle$ for $a \in \mathrm{CCR}(V,\sigma)$.

Proposition 2.19 (Values of the vector state)**.** Lean ✓ *The vector state τ of $\mathrm{CCR}(V,\sigma)$ (Definition 2.18) is linear, $\tau(1) = 1$, $\tau(a^*a) = \|a\delta_0\|^2$, $|\tau(a)| \le \|a\|$, and $\tau(W(v)) = 1$ if $v = 0$ and 0 otherwise.*

Proof. $\tau(a^*a) = \langle a\delta_0, a\delta_0\rangle$, and $|\tau(a)| \le \|a\delta_0\| \le \|a\|$ by the Cauchy–Schwarz inequality. By Proposition 2.17, $\tau(W(v)) = \langle \delta_0, \delta_{-v}\rangle$. □

Remark 2.20 (The tracial state). τ is a *tracial state* of the CCR algebra [16]: $\tau(W(u)W(w)) = e^{-\mathrm{i}\sigma(u,w)/2}[u+w=0]$ is symmetric in u and w, since $\sigma(u,-u) = 0$, so $\tau(ab) = \tau(ba)$ by linearity and continuity. The vector δ_0 is cyclic, since $W(u)\delta_0 = \delta_{-u}$, so $\ell^2(V)$ is the GNS representation of τ. For any dynamics the KMS condition at $\beta = 0$ reduces to $\omega(ab) = \omega(ba)$, so tracial states are the equilibrium states at infinite temperature [17, Remark 2]; τ is not a vacuum. For nondegenerate σ it is the only tracial state: a tracial state ω has $\omega(W(v)) = \omega(W(u)W(v)W(u)^*) = e^{-\mathrm{i}\sigma(u,v)}\omega(W(v))$ by Lemma 2.10; for $v \neq 0$ nondegeneracy gives w with $\sigma(w,v) \neq 0$, and a real multiple u of w has $\sigma(u,v) \notin 2\pi\mathbb{Z}$, so $\omega(W(v)) = 0 = \tau(W(v))$. By linearity and continuity $\omega = \tau$.

Remark 2.21 (Faithfulness and reducibility). If $a \in \mathrm{CCR}(V,\sigma)$ and $a\delta_0 = 0$, then $a = 0$: for every w, $a\delta_{-w} = aR(w)\delta_0 = R(w)a\delta_0 = 0$ by Proposition 2.17, and a bounded operator vanishing on the orthonormal basis $\{\delta_v\}$ is zero. Hence $\tau(a^*a) = \|a\delta_0\|^2$ vanishes only for $a = 0$. For $V \neq 0$ the representation on $\ell^2(V)$ is reducible: for $w \neq 0$, $R(w)$ commutes with $\mathrm{CCR}(V,\sigma)$ but is not a multiple of 1, since $R(w)\delta_0 = \delta_{-w}$ is orthogonal to δ_0. Primitivity therefore needs another representation (Remark 6.15); the first claim is restated as Lemma 6.8.

Remark 2.22 (Fock representations). A Fock representation is the usual alternative to $\ell^2(V)$. It needs a complex structure on V compatible with σ, that is, a choice of vacuum, which a curved spacetime does not single out [1, p. 219]. For nondegenerate σ its Weyl operators generate a C*-algebra isomorphic to $\mathrm{CCR}(V,\sigma)$ by the uniqueness theorem of Remark 2.12, so the net and its axioms are the same. It differs in two ways. It is irreducible, and $\lambda \mapsto W(\lambda u)$ is strongly continuous, so field operators exist by Stone's theorem. And a bounded symplectic map is implemented by a unitary only when its antilinear part is Hilbert–Schmidt, by Shale's theorem [15], whereas on $\ell^2(V)$ every symplectic bijection is (Proposition 2.14). Theorem 6.14 needs an irreducible representation, and a Fock representation is one (Remark 6.15).

3 Propagators on the lattice

The Weyl operators of the free field are labelled by test functions, two test functions giving the same operator when E maps them to the same solution, and their commutation relations are fixed by E, the difference of the advanced and retarded Green operators (Remark 2.1). On the lattice $\mathbb{Z}^2$ both Green operators are given by an explicit kernel, and the two properties of E that the axioms need, vanishing at spacelike separation and a time-slice identity, become statements about finitely many terms.

3.1 The discrete Klein–Gordon equation

Points of $\mathbb{Z}^2$ are $p = (t, x)$, ordered and compared as in Chapter 1. A *test function* is a finitely supported $f : \mathbb{Z}^2 \to \mathbb{R}$; they form a real vector space $\mathcal{D}$. The operator P below acts on all functions $h : \mathbb{Z}^2 \to \mathbb{R}$, and a solution is an h with $Ph = 0$. For $m^2 \in \mathbb{R}$ (of either sign) let

$$(Ph)(t,x) = h(t+1,x)+h(t-1,x)-h(t,x+1)-h(t,x-1)+m^2h(t,x), \qquad \langle f, h\rangle = \sum_p f(p)\,h(p).$$

The operator P is the discrete form of $-\Box + m^2 = \partial_t^2 - \partial_x^2 + m^2$, and it maps $\mathcal{D}$ into $\mathcal{D}$.

Lemma 3.1 (Symmetry of P). Lean ✓ *The lattice Klein–Gordon operator P satisfies $\langle Pf, h\rangle = \langle f, Ph\rangle$ for every test function $f \in \mathcal{D}$ and every $h : \mathbb{Z}^2 \to \mathbb{R}$.*

Proof. Each term of P is a translation or a multiple of the identity, and the substitution $p \mapsto p-c$ in the finite sum $\sum_p f(p+c)h(p)$ moves a translation by c from f to a translation by $-c$ on h. The translations occur in pairs $\pm c$ with equal coefficients. □

Definition 3.2 (Retarded kernel). $K : \mathbb{Z}^2 \to \mathbb{R}$ is given by $K(t, x) = 0$ for $t \le 0$, $K(1, x) = [x = 0]$, and, for $t \ge 1$,

$$K(t+1,x) = -K(t-1,x) + K(t,x+1) + K(t,x-1) - m^2K(t,x).$$

Lemma 3.3 (The kernel). Lean ✓ *The retarded kernel K of Definition 3.2 satisfies:*

(a) $PK = \delta_{(0,0)}$.

(b) $K(t,x) \neq 0$ *implies* $|x| \le t-1$.

(c) $K(t,-x) = K(t,x)$.

(d) If $m^2 = 0$, *then* $K(t,x) \neq 0$ *implies that* $t + x$ *is odd.*

Proof. (a) For $t \ge 1$, $(PK)(t,x) = 0$ is the recursion. At $t = 0$ only the term $K(1,x) = [x = 0]$ survives, and for $t \le -1$ every term vanishes.

Parts (b)–(d) follow by induction on $t \ge 1$ over the pair of slices $t-1$, t, starting from $K(0,\cdot) = 0$ and $K(1,\cdot)$. The inductive steps:

(b) each of the values $K(t-1,x)$, $K(t,x\pm1)$, $K(t,x)$ that determine $K(t+1,x)$ vanishes unless $|x| \le t$.

(c) $K(1,\cdot)$ and the recursion are invariant under $x \mapsto -x$.

(d) for $m^2 = 0$, $K(t+1,x)$ is a sum of values at the points $(t-1,x)$ and $(t,x\pm1)$, each with the parity of $t+1+x$, and $K(1,x) \neq 0$ only at $x = 0$.

□

Part (b) says that K lives strictly inside the forward light cone: a signal moves at most one site per time step.

3.2 Green operators and the commutator function

Definition 3.4 (Green operators)**.** For $f \in \mathcal{D}$ and $p = (t, x)$,

$$G^+ f(p) = \sum_q f(q)\, K(p - q), \qquad G^- f(p) = \sum_q f(q)\, K(q_t - t,\, x - q_x), \qquad Ef = G^- f - G^+ f,$$

where $q = (q_t, q_x)$. Thus $Ef(p) = \sum_q f(q)\Delta(p - q)$ with the commutator function $\Delta(t, x) = K(-t, x) - K(t, x)$.

The operator G^+ has support in the future of the source and is called *retarded*, as in physics; G^- is *advanced*, and $E = G^- - G^+$ is advanced minus retarded, as in Remark 2.1. Bär, Ginoux and Pfäffle name them the other way round: their advanced Green operator G_+ is the one with support in the causal future [3, Def. 3.4.1], and their $G = G_+ - G_-$ [3, §3.4] equals $-E$. In [1, p. 222], $E = E^+ - E^-$ with the support of $E^+ f$ in the causal future of $\operatorname{supp} f$, which is $-E$ here; the relation $[\phi(f), \phi(f')] = \mathrm{i}^{-1}\langle f, (E^+ - E^-)f'\rangle$ of [1, Thm. 2, p. 224], written with Dimock's $E^+ - E^- = -E$, agrees with Remark 2.1.

Lemma 3.5 (Uniqueness)**.** Lean ✓ *If $h : \mathbb{Z}^2 \to \mathbb{R}$ satisfies $Ph = 0$ for the lattice Klein–Gordon operator P and h vanishes at all times before some t_0, or at all times after some t_0, then $h = 0$.*

Proof. $Ph = 0$ expresses h on the slice $t+1$ through the slices t and $t-1$, and h on the slice $t-1$ through the slices t and $t+1$. Induction in the respective direction gives $h = 0$. □

Proposition 3.6 (Green operators)**.** Lean ✓ *For every test function $f \in \mathcal{D}$, the Green operators $G^\pm$ of Definition 3.4 satisfy:*

(a) $PG^+ f = PG^- f = f$*;*

(b) $G^+ f(p) \neq 0$ *implies* $|x - q_x| \leq t - q_t - 1$ *for some* $q \in \operatorname{supp} f$*, and* $G^- f(p) \neq 0$ *implies* $|x - q_x| \leq q_t - t - 1$ *for some* $q \in \operatorname{supp} f$*;*

(c) $G^+ Pf = G^- Pf = f$*.*

Proof. (a) P commutes with translations and with the finite sum, so $PG^+ f = \sum_q f(q)\delta_q = f$ by Lemma 3.3(a); the reflected kernel $(t, x) \mapsto K(-t, x)$ also satisfies $P(\cdot) = \delta_{(0,0)}$, since P is invariant under $t \mapsto -t$.

(b) This is Lemma 3.3(b).

(c) $G^+ Pf - f$ is annihilated by P by (a) and vanishes at all times before the support of f by (b) (if t_f is the earliest time of $\operatorname{supp} f$, then Pf vanishes before $t_f - 1$, and the strict inequality in (b) makes $G^+ Pf$ vanish before t_f), so it is zero by Lemma 3.5; G^- is the same with the future.

□

Proposition 3.7 (The commutator function)**.** Lean ✓ *For $f, g \in \mathcal{D}$, the causal propagator E and the commutator function Δ of Definition 3.4 satisfy:*

(a) $PEf = 0$ *and* $EPf = 0$*;*

(b) $\Delta(-t, -x) = -\Delta(t, x)$*, hence* $\langle f, Eg\rangle = -\langle g, Ef\rangle$*;*

(c) if $\operatorname{supp} f \perp \operatorname{supp} g$*, then* $\langle f, Eg\rangle = 0$*;*

(d) if $m^2 = 0$*, then* $\Delta(t, x) = 0$ *whenever* $t + x$ *is even;*

(e) $\Delta(-1, x) = [x = 0]$ *and* $\Delta(0, x) = 0$*.*

Proof. (a) This is Proposition 3.6(a),(c).

(b) By Lemma 3.3(c), $\Delta(-t,-x) = K(t,x) - K(-t,x) = -\Delta(t,x)$, and $\langle f, Eg\rangle = \sum_{p,q} f(p)\Delta(p-q)g(q)$.

(c) A nonzero $\Delta(p-q)$ forces $|x_p - x_q| \le |t_p - t_q| - 1$ by Lemma 3.3(b), so p and q are comparable.

(d) This is Lemma 3.3(d) for both terms of Δ.

(e) $\Delta(-1,x) = K(1,x)$ and $\Delta(0,x) = K(0,x) - K(0,x)$.

□

Part (e) is the canonical commutation relation: with $\sigma([f],[g]) = \langle f, Eg\rangle$ as in Remark 2.1, it gives $[\phi(0,x), \phi(1,y) - \phi(0,y)] = \mathrm{i}[x = y]$, the lattice form of $[\phi, \Pi] = \mathrm{i}\delta$. The sign of E is therefore the one of canonical quantization.

Example 3.8 (Opposite parity). script ✓ For $m^2 = 0$, the recursion gives $K(2,\pm 1) = 1$, $K(2,0) = 0$ and $K(3,0) = 1$, so $\Delta(3,0) = -1$ and $\langle \delta_{(0,0)}, E\delta_{(3,0)}\rangle = \Delta(-3,0) = 1$. By Proposition 3.7(d), a pair of points with a nonzero value of Δ at $m^2 = 0$ must have $t + x$ of opposite parity, as here.

3.3 Time slice

Theorem 3.9 (Time slice). Lean ✓ *Let $f \in \mathcal{D}$, $t_1 \in \mathbb{Z}$, $h = Ef$, and let χ be the indicator of the times $t \ge t_1 + 1$. Then $f' = P((1-\chi)h)$ is a test function with* $\operatorname{supp} f' \subseteq S_{t_1} \cap J(\operatorname{supp} f)$ *and $Ef' = Ef$.*

Proof. Since $Ph = 0$, a term-by-term evaluation gives $f'(t_1, x) = -h(t_1 + 1, x)$, $f'(t_1 + 1, x) = h(t_1, x)$, and $f' = 0$ at all other times. If $f'(p) \neq 0$, then $h(p') \neq 0$ for the other point p' of S_{t_1} with the same x, and Proposition 3.6(b) gives $q \in \operatorname{supp} f$ with $|x - q_x| \le |t_{p'} - q_t| - 1 \le |t_p - q_t|$; so $p \in J(\operatorname{supp} f)$, a finite set within S_{t_1}. Next, $G^+ f' + \chi h$ is annihilated by P and vanishes for $t \le t_1$, since f' is supported at times $\ge t_1$; and $G^- f' - (1-\chi)h$ is annihilated by P and vanishes for $t \ge t_1 + 1$, since f' is supported at times $\le t_1 + 1$. By Lemma 3.5 both vanish, and $Ef' = (1-\chi)h + \chi h = h$. □

The value $f'(t_1, x) = -h(t_1 + 1, x)$ can be nonzero where $h(t_1, x) = 0$; that p still lies in $J(\operatorname{supp} f)$ comes from the strict inequality in Lemma 3.3(b).

Remark 3.10 (Time slice from the Green operators alone). Given the support statement, the identity $Ef' = Ef$ also follows from the identities $PG^{\pm} f = f$ and $G^{\pm} Pf = f$ on test functions and the support bounds of Proposition 3.6, without applying Lemma 3.5 to χh and $(1-\chi)h$; these notes prove the identities $G^{\pm} Pf = f$ with that lemma, but they also follow from the formal adjointness $\langle g, G^+ f\rangle = \langle G^- g, f\rangle$ for $f, g \in \mathcal{D}$, with the pairing $\sum_p f(p)h(p)$ written in either order when one argument is finitely supported, a consequence of Lemma 3.3(c): $\langle g, G^+ Pf\rangle = \langle G^- g, Pf\rangle = \langle PG^- g, f\rangle = \langle g, f\rangle$ by Lemma 3.1 and Proposition 3.6(a), and likewise for G^-. The function $k = (1-\chi)G^+ f + \chi G^- f$ is a test function: $G^+ f$ vanishes before the earliest time of $\operatorname{supp} f$ and $G^- f$ after the latest, and each time slice meets $J(\operatorname{supp} f)$ in finitely many points. Since $PG^{\pm} f = f$,

$$f' = P((1-\chi)G^- f) - P((1-\chi)G^+ f) = f - P(\chi G^- f) - P((1-\chi)G^+ f) = f - Pk,$$

and $E(f - f') = EPk = 0$ by Proposition 3.7(a).

Corollary 3.11 (Dependence moves test functions). Lean ✓ *If $O \triangleleft O'$ in the causal index of Example 1.3 and* $\operatorname{supp} f \subseteq O$, *there is $f' \in \mathcal{D}$ with* $\operatorname{supp} f' \subseteq O'$ *and $Ef' = Ef$.*

Proof. $\operatorname{supp} f$ is a finite subset of O, so there is t_1 with $J(\operatorname{supp} f) \cap S_{t_1} \subseteq O'$; apply Theorem 3.9. □

4 The net of a classical phase space

The observables of the free field are Weyl operators labelled by the elements of its classical phase space, and an observable belongs to a region when the source that generates the phase-space element can be chosen inside that region. This chapter states that information for an arbitrary causal index and proves that the net generated by the Weyl operators of localized sources satisfies generation, causality and time slice in every Weyl system (isotony holds for every generated net), and covariance in the CCR algebra, with the composition law of Axiom 5 of [1].

4.1 Phase-space data

Definition 4.1 (Phase-space data)**.** Fix a causal index on a preordered set $\mathcal{R}$ of regions. *Phase-space data* consist of real vector spaces $\mathcal{D}$ (test functions) and V (the phase space), a monotone map $O \mapsto \mathcal{D}(O)$ from regions to linear subspaces of $\mathcal{D}$ such that every $f \in \mathcal{D}$ lies in some $\mathcal{D}(O)$, an alternating bilinear form σ on V, and a surjective linear map $f \mapsto [f]$ from $\mathcal{D}$ onto V, which assigns to each test function the element of the phase space it generates, such that:

(P1) if $O_1 \perp O_2$, $f \in \mathcal{D}(O_1)$ and $g \in \mathcal{D}(O_2)$, then $\sigma([f],[g]) = 0$;

(P2) if $O \lhd O'$ and $f \in \mathcal{D}(O)$, there is $f' \in \mathcal{D}(O')$ with $[f'] = [f]$.

"Phase-space data" is a term of these notes; the object of the literature is a symplectic space of solutions together with the test functions localized in each region [3]. The map $f \mapsto [f]$ is only required to be surjective, so V may be a quotient of $\mathcal{D}$ or a space of solutions, and σ may be degenerate: nothing in this chapter uses more, and Chapter 6 adds nondegeneracy (Definition 6.1) as a hypothesis. For the free field, V with σ is the covariant phase space of physics, and $\sigma([f],[g]) = \langle f, Eg\rangle$ is the Peierls bracket of the smeared fields $\phi(f)$ and $\phi(g)$ [10], whose quantization is $[\phi(f),\phi(g)] = \mathrm{i}\langle f, Eg\rangle$ (Remark 2.1).

Example 4.2 (The lattice field)**.** Lean ✓ On the lattice causal index of Example 1.3 take $\mathcal{D}$ the finitely supported $f : \mathbb{Z}^2 \to \mathbb{R}$, $\mathcal{D}(O) = \{f : \operatorname{supp} f \subseteq O\}$, $V = \mathcal{D}/\ker E$ with E the causal propagator of Definition 3.4, $[f]$ the class of f, and $\sigma([f],[g]) = \langle f, Eg\rangle$. Every f lies in $\mathcal{D}(\operatorname{supp} f)$, since every subset of $\mathbb{Z}^2$ is a region. The form is well defined and alternating by Proposition 3.7(b): $\langle f, Eg\rangle = -\langle g, Ef\rangle$ vanishes when $Ef = 0$ or $Eg = 0$, and $\langle f, Ef\rangle = 0$. Condition (P1) is Proposition 3.7(c), since orthogonality is antitone in both arguments, and (P2) is Corollary 3.11, since $Ef' = Ef$ means $[f'] = [f]$. The form is nondegenerate: $\sigma([f],[\delta_p]) = -\langle \delta_p, Ef\rangle = -(Ef)(p)$, so $\sigma([f],[\delta_p]) = 0$ for all p gives $Ef = 0$.

Definition 4.3 (Isomorphism of phase-space data)**.** An *isomorphism* $\theta = (\kappa, \kappa_*, L)$ from phase-space data $(\mathcal{D}, V, \sigma, [\cdot])$ to $(\mathcal{D}', V', \sigma', [\cdot]')$ consists of an isomorphism κ of the causal indices (Definition 1.6), a linear bijection $\kappa_* : \mathcal{D} \to \mathcal{D}'$ and a linear bijection $L : V \to V'$ with

$$\kappa_*(\mathcal{D}(O)) = \mathcal{D}'(\kappa O), \qquad [\kappa_* f]' = L[f], \qquad \sigma'(Lu, Lv) = \sigma(u, v).$$

The last condition makes L symplectic, as in Chapter 2. It is where the time orientation enters: a time reversal exchanges the retarded and advanced Green operators (Remark 1.8), so the triple formed by it, the pushforward of test functions and the induced map L has $\sigma'(Lu, Lv) = -\sigma(u, v)$ and is not an isomorphism when $\sigma \neq 0$. Other linear maps over a time reversal are not excluded: Remark 4.4 gives one that is an isomorphism.

Remark 4.4 (Time reversal of the massless lattice field). Let $m^2 = 0$ in Example 4.2, let $r(t,x) = (-t,x)$ be the time reversal of Example 1.7, let $\varepsilon(t,x) = (-1)^{t+x}$, and put $\kappa_* f = \varepsilon \cdot (f \circ r)$. Then $\kappa_*(\mathcal{D}(O)) = \mathcal{D}(rO)$, and $\kappa_*\kappa_* f = f$. Since G^- is defined by the time-reflected kernel, $G^{\pm}(f \circ r) = (G^{\mp} f) \circ r$, hence $E(f \circ r) = -(Ef) \circ r$. By Proposition 3.7(d), $\Delta(p - q) \neq 0$ implies

$\varepsilon(p)\varepsilon(q) = -1$, hence $E(\varepsilon g) = -\varepsilon\, Eg$. Together, $E\kappa_* f = \varepsilon \cdot ((Ef) \circ r)$. So κ_* maps $\ker E$ onto itself, $L[f] = [\kappa_* f]$ is a well-defined linear bijection of V, and

$$\sigma(L[f], L[g]) = \langle \varepsilon \cdot (f \circ r), \varepsilon \cdot ((Eg) \circ r) \rangle = \langle f, Eg \rangle = \sigma([f], [g]).$$

Thus $\theta = (r, \kappa_*, L)$ is an isomorphism of phase-space data with $\theta\theta = \mathrm{id}$, and by Theorem 4.11 (r, α_θ) is a covariance pair of the CCR net of the massless lattice field with $\alpha_\theta^2 = \mathrm{id}$, so the net is covariant under the group $\{\mathrm{id}, r\}$. For $m^2 \neq 0$ the sign ε no longer anticommutes with E (the recursion gives $K(2,0) = -m^2$, so $\Delta(2,0) = m^2$ at a point with $t + x$ even, and $E(\varepsilon\delta_{(0,0)})(2,0) = m^2 \neq -m^2 = -\varepsilon(2,0)\,(E\delta_{(0,0)})(2,0)$), and these notes do not decide whether another isomorphism over the time reversal exists.

Lemma 4.5 (Identity and composite). Lean ✓ *The identity maps form an isomorphism of phase-space data, and if $\theta_1 = (\kappa_1, \kappa_{1*}, L_1)$ and $\theta_2 = (\kappa_2, \kappa_{2*}, L_2)$ are isomorphisms, so is $\theta_2\theta_1 = (\kappa_2\kappa_1, \kappa_{2*}\kappa_{1*}, L_2L_1)$.*

Proof. $\kappa_{2*}\kappa_{1*}(\mathcal{D}(O)) = \kappa_{2*}(\mathcal{D}'(\kappa_1 O)) = \mathcal{D}''(\kappa_2\kappa_1 O)$; $[\kappa_{2*}\kappa_{1*} f]'' = L_2[\kappa_{1*} f]' = L_2L_1[f]$; and $\sigma''(L_2L_1u, L_2L_1v) = \sigma'(L_1u, L_1v) = \sigma(u, v)$. □

4.2 The net of a Weyl system

From here on, phase-space data $(\mathcal{D}, V, \sigma, [\cdot])$ on a causal index are fixed; primed data $(\mathcal{D}', V', \sigma', [\cdot]')$ are a second such choice, possibly on another causal index.

Definition 4.6 (The net of a Weyl system). Let (A, W) be a Weyl system over (V, σ) (Definition 2.9). Its generators are $G(O) = \{W([f]) : f \in \mathcal{D}(O)\}$, a family of generators in the sense of Definition 1.20, and its *net* is the net they generate, $\mathcal{A}_W(O) = \overline{\mathrm{alg}}\, G(O)$.

Theorem 4.7 (Generation, causality, time slice). Lean ✓ *For phase-space data $(\mathcal{D}, V, \sigma, [\cdot])$ on a causal index and every Weyl system (A, W) over (V, σ), the net $\mathcal{A}_W$ is generating, causal and has the time-slice property.*

Proof. The three properties follow from Proposition 1.21 once the generators are checked. Generation: $\bigcup_O G(O) = W(V)$, since every $u \in V$ is some $[f]$ and every f lies in some $\mathcal{D}(O)$; and $W(V)$ generates A. Causality: for $O_1 \perp O_2$, $f \in \mathcal{D}(O_1)$, $g \in \mathcal{D}(O_2)$, (P1) gives $\sigma([f], [g]) = 0$, so $W[f]$ and $W[g]$ commute by Lemma 2.10; the adjoint $W[g]^* = W[-g]$ is again a generator of O_2, so the same argument applies to it. Time slice: for $O \triangleleft O'$ and $f \in \mathcal{D}(O)$, (P2) gives $f' \in \mathcal{D}(O')$ with $W[f] = W[f'] \in G(O')$. □

For the lattice field of Example 4.2 and the CCR algebra this gives generation, causality and time slice without assumptions on the spacetime; Chapter 5 completes that instance with translation covariance.

4.3 Covariance

Theorem 4.8 (Covariance pairs from Weyl systems). Lean ✓ *Let $\theta = (\kappa, \kappa_*, L)$ be an isomorphism of phase-space data, (A, W) and (A', W') Weyl systems over (V, σ) and (V', σ'), and $\alpha : A \to A'$ a $*$-isomorphism with $\alpha(W(u)) = W'(Lu)$ for all u. Then (κ, α) is a covariance pair from $\mathcal{A}_W$ to $\mathcal{A}'_{W'}$.*

Proof. By Proposition 1.21(d) it suffices that $\alpha(G(O)) = G'(\kappa O)$, and $\alpha(G(O)) = \{W'(L[f]) : f \in \mathcal{D}(O)\} = \{W'([\kappa_* f]') : f \in \mathcal{D}(O)\} = \{W'([f']') : f' \in \mathcal{D}'(\kappa O)\}$. □

Only the first two conditions of Definition 4.3 enter here, since α is given; the symplectic condition is preserved under composition (Lemma 4.5) and is used in Theorem 4.11, where it makes α_θ exist. When $A' \neq 0$ it also follows from the hypotheses: applying α to the Weyl relation gives $e^{-\mathrm{i}\sigma(u,w)/2} = e^{-\mathrm{i}\sigma'(Lu,Lw)/2}$, and the scaling argument of Remark 2.15 gives $\sigma'(Lu, Lw) = \sigma(u, w)$.

Corollary 4.9 (Isomorphic Weyl systems)**.** Lean ✓ *If (A, W) and (A', W') are Weyl systems over (V, σ), both nets formed from the same phase-space data, and $\alpha : A \to A'$ is a $*$-isomorphism with $\alpha \circ W = W'$, the nets $\mathcal{A}_W$ and $\mathcal{A}'_{W'}$ are isomorphic.*

Proof. Theorem 4.8 with the identity isomorphism. □

For nondegenerate σ and nonzero algebras such an α always exists, by the uniqueness theorem for the CCR algebra [6] (see also [7]); the paper [1, p. 225] uses it, by citation, for the independence of the net from the representation in its Theorem 3. These notes do not prove it, and it is not formalized here; the corollary carries α as a hypothesis.

Definition 4.10 (The CCR net)**.** The *CCR net* of phase-space data is $\mathcal{A}_W$ for the Weyl system $(\mathrm{CCR}(V, \sigma), W)$ of Proposition 2.11. For an isomorphism $\theta = (\kappa, \kappa_*, L)$ let $\alpha_\theta : \mathrm{CCR}(V, \sigma) \to \mathrm{CCR}(V', \sigma')$ be conjugation by the relabelling unitary U_L (Proposition 2.14).

Theorem 4.11 (Covariance of the CCR net)**.** Lean ✓ *For every isomorphism $\theta = (\kappa, \kappa_*, L)$ from phase-space data $(\mathcal{D}, V, \sigma, [\cdot])$ to $(\mathcal{D}', V', \sigma', [\cdot]')$, (κ, α_θ) is a covariance pair from the CCR net of the first to that of the second, $\alpha_{\mathrm{id}} = \mathrm{id}$, and $\alpha_{\theta_2\theta_1} = \alpha_{\theta_2}\alpha_{\theta_1}$.*

Proof. Proposition 2.14 gives $\alpha_\theta(W(u)) = W'(Lu)$, $\alpha_{\mathrm{id}} = \mathrm{id}$ and $\alpha_{L_2L_1} = \alpha_{L_2}\alpha_{L_1}$, and the symplectic map of $\theta_2\theta_1$ is L_2L_1 (Lemma 4.5). Theorem 4.8 gives the covariance pair. □

This is Axiom 5 of [1, p. 220] for the CCR net: an assignment $\theta \mapsto \alpha_\theta$ with $\alpha_{\mathrm{id}} = \mathrm{id}$ and $\alpha_{\theta_2\theta_1} = \alpha_{\theta_2}\alpha_{\theta_1}$, each (κ, α_θ) a covariance pair (Remark 1.16), indexed by the isomorphisms of phase-space data. An assignment indexed by the isometries themselves, as in [1], is obtained by taking for each isometry the triple of the induced map of regions, the pushforward of test functions and the induced map on V; pushforwards compose, and L is determined by κ_* because $[\cdot]$ is surjective, so this choice respects composition.

For the free field these include the maps induced by the isometries that preserve the time orientation, which commute with the Green operators by their uniqueness on a globally hyperbolic spacetime [3, Cor. 3.4.3], a result formalized here only for the massless field on Minkowski space (Theorem 8.14, Proposition 8.17) and for the Klein–Gordon field there (Theorem 9.18(b), (c)); the converse fails, since $(\mathrm{id}, -\mathrm{id}, -\mathrm{id})$ is an isomorphism that no isometry induces when $\mathcal{D} \neq 0$. Covariance under a group of such isomorphisms is Theorem 5.6 for the lattice translations and Theorem 8.18 and Theorem 9.18(c) for the orthochronous Poincaré group.

5 The net of the lattice field

This chapter restates Example 4.2 as numbered results, including nondegeneracy, shows that no condition is vacuous, and constructs isomorphisms of phase-space data for the lattice translations and the spatial reflection. With Chapter 4 this gives the lattice CCR net with generation, causality, time slice and translation covariance, for every m^2 and with no hypotheses. The last subsection shows that the net defined through Cauchy data, as in [1, §2], is the net of a pulled-back Weyl system, and for the CCR algebra over Cauchy data is isomorphic to the lattice CCR net.

Throughout this chapter, $m^2 \in \mathbb{R}$ is fixed, $\mathcal{D}$ is the space of test functions on $\mathbb{Z}^2$, and three consequences of Chapter 3 are used repeatedly. By Proposition 3.7(e), and for the third value by (b) with the evenness below,

$$\Delta(-1,x) = [x=0], \qquad \Delta(0,x) = 0, \qquad \Delta(1,x) = -[x=0]. \tag{5.1}$$

By Lemma 3.3(b),(c), applied to each term of $\Delta(t,x) = K(-t,x) - K(t,x)$, $\Delta(t,x) \neq 0$ implies $|x| \leq |t| - 1$, and $\Delta(t,-x) = \Delta(t,x)$. By Proposition 3.7(a), $PEf = 0$.

5.1 Phase-space data

Proposition 5.1 (The lattice phase-space data)**.** Lean ✓ *For $m^2 \in \mathbb{R}$, the lattice test functions $\mathcal{D}$ and the causal propagator E of Definition 3.4, let $\mathcal{D}(O) = \{f \in \mathcal{D} : \operatorname{supp} f \subseteq O\}$, $V = \mathcal{D}/\ker E$ with $[f]$ the class of f, and $\sigma([f],[g]) = \langle f, Eg\rangle$. Then E is linear, σ is a well-defined alternating bilinear form, and $(\mathcal{D}, V, \sigma, [\cdot])$ are phase-space data over the lattice causal index (Example 1.3).*

Proof. $(Ef)(p) = \sum_q f(q)\Delta(p-q)$ and $\langle f,h\rangle$ are finite sums, linear in f and in h. The form $(f,g) \mapsto \langle f, Eg\rangle$ vanishes when $Eg = 0$, and when $Ef = 0$ because $\langle f, Eg\rangle = -\langle g, Ef\rangle$ (Proposition 3.7(b)); so it descends to $V \times V$, and $\langle f, Ef\rangle = -\langle f, Ef\rangle$ makes it alternating. $\mathcal{D}(O)$ is a subspace, monotone in O, and $f \in \mathcal{D}(\operatorname{supp} f)$. (P1) is Proposition 3.7(c), since no point of a subset of O_1 is comparable with a point of a subset of O_2 when $O_1 \perp O_2$. (P2) is Corollary 3.11: $Ef' = Ef$ means $f' - f \in \ker E$, that is, $[f'] = [f]$. □

V is taken as the quotient $\mathcal{D}/\ker E$ rather than the space $E\mathcal{D}$ of solutions; the two are isomorphic through $[f] \mapsto Ef$. The quotient makes $f \mapsto [f]$ a quotient map, and the form descends by the universal property. formalization

Proposition 5.2 (Nondegeneracy)**.** Lean ✓ *For the lattice phase-space data of Proposition 5.1, $\sigma([f],[\delta_p]) = -(Ef)(p)$ for all $f \in \mathcal{D}$ and $p \in \mathbb{Z}^2$. By definition of V, $[f] = 0$ if and only if $Ef = 0$, and $\sigma(u,w) = 0$ for all w implies $u = 0$.*

Proof. $\sigma([f],[\delta_p]) = -\langle \delta_p, Ef\rangle = -(Ef)(p)$. $[f] = 0$ means $f \in \ker E$. If $\sigma([f],w) = 0$ for all w, then $(Ef)(p) = 0$ for all p. □

Example 5.3 (The conditions are not vacuous)**.** Lean ✓ (a) $\sigma \neq 0$: the recursion gives $K(2,\pm 1) = 1$, $K(2,0) = -m^2$ and $K(3,0) = -K(1,0) + K(2,1) + K(2,-1) - m^2K(2,0) = 1 + m^4$, so $\sigma([\delta_{(0,0)}],[\delta_{(3,0)}]) = \Delta(-3,0) = -\Delta(3,0) = 1 + m^4$, which is 1 at $m^2 = 0$ (Example 3.8). (b) The regions $\{(0,0)\}$ and $\{(0,1)\}$ are orthogonal, and $[\delta_{(0,0)}]$, $[\delta_{(0,1)}]$ are nonzero: by (5.1), $\sigma([\delta_{(0,x)}],[\delta_{(1,x)}]) = \Delta(-1,0) = 1$. (c) $O = \{(5,0)\}$ and $O' = J(O) \cap S_0$, of $11 + 9 = 20$ points, satisfy $O \lhd O'$ and $O \not\subseteq O'$, so the time-slice property is not isotony: the algebra of a point is contained in that of a region, far from it in time, that does not contain the point.

5.2 Translations and the spatial reflection

For $a \in \mathbb{Z}^2$ let $\kappa_a O = O + a$ and $(\kappa_{a*} f)(p) = f(p - a)$. Let $\rho(t, x) = (t, -x)$, $\kappa_\rho O = \rho O$ and $(\kappa_{\rho*} f)(p) = f(\rho p)$.

Lemma 5.4 (Covariance of E)**.** Lean ✓ *For $a \in \mathbb{Z}^2$ and the reflection $\rho(t, x) = (t, -x)$, the lattice causal propagator E satisfies $(E\kappa_{a*} f)(p) = (Ef)(p - a)$ and $(E\kappa_{\rho*} f)(p) = (Ef)(\rho p)$ for $f \in \mathcal{D}$.*

Proof. Substitute $q \mapsto q+a$, respectively $q \mapsto \rho q$, in $\sum_q f(q)\Delta(p-q)$; the second uses $\Delta(\rho(p-q)) = \Delta(p - q)$, the evenness of Δ in x. □

Proposition 5.5 (Isomorphisms of the lattice data)**.** Lean ✓ *For the lattice phase-space data of Proposition 5.1:*

(a) For each $a \in \mathbb{Z}^2$, $\theta_a = (\kappa_a, \kappa_{a*}, L_a)$ with $\kappa_a O = O + a$, $\kappa_{a*} f = f(\cdot - a)$ and $L_a[f] = [\kappa_{a*} f]$ is an isomorphism of these data to themselves (Definition 4.3), and $L_0 = \mathrm{id}$, $L_{a+b} = L_a L_b$.

(b) $\theta_\rho = (\kappa_\rho, \kappa_{\rho*}, L_\rho)$ with $\kappa_\rho O = \rho O$, $\kappa_{\rho*} f = f \circ \rho$ for the reflection $\rho(t, x) = (t, -x)$, and $L_\rho[f] = [\kappa_{\rho*} f]$ is an isomorphism of these data to themselves.

Proof. Both are cases of one construction: a bijection ζ of $\mathbb{Z}^2$ with $\Delta(\zeta p - \zeta q) = \Delta(p - q)$ for all p, q, together with an automorphism κ of the causal index with $\kappa O = \zeta O$. Applying the hypothesis to $\zeta^{-1} p$ and q gives $E(f \circ \zeta^{-1}) = (Ef) \circ \zeta^{-1}$, as in Lemma 5.4. So the pushforward $f \mapsto f \circ \zeta^{-1}$, a linear bijection with $\mathcal{D}(O) \mapsto \mathcal{D}(\zeta O)$, maps $\ker E$ onto $\ker E$; $L[f] = [f \circ \zeta^{-1}]$ is a well-defined linear bijection, and $\langle f \circ \zeta^{-1}, E(g \circ \zeta^{-1})\rangle = \langle f, Eg\rangle$. For (a), $\zeta(p) = p + a$; $L_0 = \mathrm{id}$ and $L_{a+b} = L_a L_b$ because the pushforwards compose. For (b), $\zeta = \rho$ ($\rho^{-1} = \rho$). In both cases κ is an automorphism by Example 1.7, which lists the translations and the reflection. □

Example 1.7 lists the spatial reflection among the automorphisms of the lattice causal index; what is new here is the isomorphism of phase-space data over it. The time reversal r carries one for $m^2 = 0$, with a sign twist (Remark 4.4), and for $m^2 \neq 0$ these notes leave open whether it carries one; the reflection carries one for every m^2, without a sign twist.

5.3 The net

Theorem 5.6 (The net of the lattice field)**.** Lean ✓ *For every $m^2 \in \mathbb{R}$ the CCR net $\mathcal{A}$ of the lattice phase-space data is generating, causal and has the time-slice property, and it is covariant under the translation group $\mathbb{Z}^2$: with $\alpha_a = \alpha_{\theta_a}$ (Definition 4.10), $a \mapsto \kappa_a$ and $a \mapsto \alpha_a$ are group homomorphisms and every (κ_a, α_a) is a covariance pair.*

Proof. Theorem 4.7 for Proposition 5.1 gives the first three properties, and Theorem 4.11 with Proposition 5.5(a) gives the covariance pairs. $a \mapsto \kappa_a$ is a homomorphism since $O + 0 = O$ and $O + (a + b) = (O + b) + a$. The $*$-isomorphism α_θ depends on θ only through L, and $L_0 = \mathrm{id}$, $L_{a+b} = L_a L_b$, so Theorem 4.11 gives $\alpha_0 = \mathrm{id}$ and $\alpha_{a+b} = \alpha_a \alpha_b$. □

The theorem has no hypotheses: the propagators are the explicit ones of Chapter 3, and Example 5.3 shows that $\sigma \neq 0$, that orthogonal regions carry nonzero elements of V, and that dependence does not reduce to inclusion. No hypothesis on m^2 is needed either: the axioms say nothing about energy, so $m^2 < 0$, where the classical field grows exponentially, is included. The theorem is the lattice form of Axioms 1, 3 and 4 of [1] and of the translation part of Axiom 5; primitivity (Axiom 2) is Corollary 6.18, given an irreducible representation. By Theorem 4.11 and Proposition 5.5(b), $(\kappa_\rho, \alpha_{\theta_\rho})$ is a covariance pair as well.

5.4 The Cauchy-data net

In [1, §2, pp. 222–225] the net is built from a representation of the Weyl relations over Cauchy data: $e^{\mathrm{i}\phi(f)}$ is represented by the Weyl operator of the Cauchy data of Ef on a Cauchy surface. On the lattice two consecutive time slices play the part of a Cauchy surface (Chapter 3), and this subsection proves that the net so defined is the net of a pulled-back Weyl system, isomorphic for the CCR algebra to the net of Theorem 5.6.

Definition 5.7 (Pullback of a Weyl system)**.** Let (V,σ) and (V_S,σ_S) be real vector spaces with alternating forms, and $\Lambda : V \to V_S$ a linear map with $\sigma_S(\Lambda u,\Lambda w)=\sigma(u,w)$. The *pullback* of a Weyl system (A,W_S) over (V_S,σ_S) along Λ is the pair $(A, W_S\circ\Lambda)$; it is a Weyl system when Λ is onto (Proposition 5.8).

Proposition 5.8 (Pullback)**.** Lean ✓ *Let $\Lambda : V \to V_S$ and (A,W_S) be as in Definition 5.7. Then $W_S\circ\Lambda$ consists of unitaries and satisfies the Weyl relation over (V,σ). If Λ is onto, $(A, W_S\circ\Lambda)$ is a Weyl system over (V,σ), and for phase-space data with phase space (V,σ) its net is $O \mapsto \overline{\mathrm{alg}}\{W_S(\Lambda[f]) : f \in \mathcal{D}(O)\}$.*

Proof. $W_S(\Lambda u)W_S(\Lambda w) = e^{-\mathrm{i}\sigma_S(\Lambda u,\Lambda w)/2}W_S(\Lambda u+\Lambda w) = e^{-\mathrm{i}\sigma(u,w)/2}W_S(\Lambda(u+w))$. If Λ is onto, the ranges of $W_S\circ\Lambda$ and W_S coincide, so they generate A. The net is Definition 4.6. □

The relabelling of Proposition 2.14 acts along symplectic bijections and changes the algebra; the pullback keeps the algebra A of W_S.

Definition 5.9 (Cauchy data and the Cauchy map)**.** *Cauchy data* are pairs $d=(d_0,d_1)$ of finitely supported functions $\mathbb{Z}\to\mathbb{R}$; they form a real vector space V_S with the *Wronskian* $\sigma_S(d,d') = \sum_x\big(d_0(x)d_1'(x) - d_1(x)d_0'(x)\big)$, an alternating form. For $t_1\in\mathbb{Z}$ the *Cauchy map* is $\Lambda_{t_1}[f] = \big((Ef)(t_1,\cdot),\,(Ef)(t_1+1,\cdot)\big)$.

Λ_{t_1} is well defined and linear: $(Ef)(t,x)\neq 0$ requires some $q\in\operatorname{supp} f$ with $|x-x_q|\le|t-t_q|-1$, so each slice of Ef is finitely supported, and Ef depends only on $[f]$. The data are the field on two consecutive slices. With the discrete momentum $\Pi = d_1 - d_0$ the Wronskian is $\sum_x(d_0\Pi' - \Pi d_0')$, the lattice form of $\int(\phi\,\partial_t\phi' - \partial_t\phi\,\phi')$ on a Cauchy surface, and its sign is that of Remark 2.1.

Lemma 5.10 (Two slices determine a solution)**.** Lean ✓ *If $h:\mathbb{Z}^2\to\mathbb{R}$ satisfies $Ph=0$ for the lattice Klein–Gordon operator P and $h(t_1,\cdot)=h(t_1+1,\cdot)=0$ for some $t_1\in\mathbb{Z}$, then $h=0$.*

Proof. $(Ph)(t,x)=0$ gives $h(t+1,x)$ from $h(t,\cdot)$ and $h(t-1,x)$, and equally $h(t-1,x)$ from $h(t,\cdot)$ and $h(t+1,x)$. By induction on $j\ge 0$ over pairs of consecutive slices, h vanishes on the pairs (t_1+j,t_1+j+1) and (t_1-j,t_1-j+1). □

This sharpens Lemma 3.5: two consecutive vanishing slices suffice, with no condition in the past or future.

Theorem 5.11 (The Cauchy map is a symplectic bijection)**.** Lean ✓ *For $t_1\in\mathbb{Z}$, the lattice phase-space data (V,σ) of Proposition 5.1 and Cauchy data d (Definition 5.9), let $f'=f'_{t_1}(d)$ be the test function with $f'(t_1,x)=-d_1(x)$, $f'(t_1+1,x)=d_0(x)$, and zero elsewhere. Then $\Lambda_{t_1}[f']=d$, Λ_{t_1} is injective, and $\sigma_S(\Lambda_{t_1}u,\Lambda_{t_1}w)=\sigma(u,w)$. So Λ_{t_1} is a symplectic bijection $V\to V_S$.*

Proof. By (5.1), $(Ef')(t_1,x) = \sum_y\big(-d_1(y)\Delta(0,x-y)+d_0(y)\Delta(-1,x-y)\big) = d_0(x)$ and $(Ef')(t_1+1,x) = \sum_y\big(-d_1(y)\Delta(1,x-y)+d_0(y)\Delta(0,x-y)\big) = d_1(x)$. If $\Lambda_{t_1}[f]=0$, then $h=Ef$ has $Ph=0$ and vanishes on two consecutive slices, so $Ef=0$ by Lemma 5.10 and $[f]=0$ by Proposition 5.2. For the form, let $u=[f]$, $w=[g]$, $d=\Lambda_{t_1}u$ and $d'=\Lambda_{t_1}w$. Injectivity and the first statement give $u=[f'_{t_1}(d)]$, so

$$\begin{aligned}\sigma(u,w) = \langle f'_{t_1}(d), Eg\rangle &= \sum_x\big(-d_1(x)(Eg)(t_1,x) + d_0(x)(Eg)(t_1+1,x)\big)\\ &= \sum_x\big(d_0(x)d_1'(x) - d_1(x)d_0'(x)\big).\end{aligned}$$

□

The preimage is explicit, and the symplectic property needs no summation by parts; the sign of σ_S is the one that makes Λ_{t_1} symplectic for $\sigma([f],[g]) = \langle f, Eg\rangle$. For $d = \Lambda_{t_1}[f]$ the preimage $f'_{t_1}(d)$, with $f'(t_1,\cdot) = -(Ef)(t_1+1,\cdot)$ and $f'(t_1+1,\cdot) = (Ef)(t_1,\cdot)$, is the test function $P((1-\chi)Ef)$ of Theorem 3.9: the time-slice construction is the inverse of the Cauchy map.

Definition 5.12 (The Cauchy-data net)**.** For a Weyl system (A, W_S) over (V_S, σ_S) and $t_1 \in \mathbb{Z}$, the *Cauchy-data net* is $O \mapsto \overline{\mathrm{alg}}\{W_S(\Lambda_{t_1}[f]) : \operatorname{supp} f \subseteq O\}$.

Theorem 5.13 (The Cauchy-data net is the lattice net)**.** Lean ✓

(a) For every Weyl system (A, W_S) over (V_S, σ_S) and every t_1, the Cauchy-data net is the net of the pullback $W_S \circ \Lambda_{t_1}$ over the lattice phase-space data; hence it is generating, causal and has the time-slice property.

(b) For $A = \mathrm{CCR}(V_S, \sigma_S)$, the Cauchy-data net is isomorphic to the CCR net $\mathcal{A}$ of Theorem 5.6 through $\alpha_{\Lambda_{t_1}}$, and it is covariant under the translations with the automorphisms $\alpha_{\Lambda_{t_1}} \alpha_a \alpha_{\Lambda_{t_1}}^{-1}$.

Proof. (a) By Theorem 5.11, Λ_{t_1} is onto and symplectic, so Proposition 5.8 applies and the two nets have the same generators; Theorem 4.7 gives the properties. (b) The relabelling isomorphism $\alpha_{\Lambda_{t_1}} : \mathrm{CCR}(V,\sigma) \to \mathrm{CCR}(V_S,\sigma_S)$ of Proposition 2.14 satisfies $\alpha_{\Lambda_{t_1}}(W(u)) = W_S(\Lambda_{t_1} u)$, so it carries the Weyl system of $\mathrm{CCR}(V,\sigma)$ to the pullback, a Weyl system over the same (V,σ). Corollary 4.9 and (a) give the isomorphism of nets. Covariance pairs are transported by conjugation (Lemma 1.15), and conjugation preserves the homomorphism property. □

Proposition 5.14 (Change of slice)**.** Lean ✓ *For the lattice mass parameter m^2 and every $t_1 \in \mathbb{Z}$, let $F(d_0, d_1) = (d_1,\ -d_0 + d_1(\cdot+1) + d_1(\cdot-1) - m^2 d_1)$. Then $\Lambda_{t_1+1} = F \circ \Lambda_{t_1}$, and $\sigma_S(Fd, Fd') = \sigma_S(d, d')$.*

Proof. With $h = Ef$, the first component of $\Lambda_{t_1+1}[f]$ is $h(t_1+1,\cdot)$, the second of $\Lambda_{t_1}[f]$; the second, by $(Ph)(t_1+1,x) = 0$, is $h(t_1+2,x) = -h(t_1,x) + h(t_1+1,x+1) + h(t_1+1,x-1) - m^2 h(t_1+1,x)$. Since Λ_{t_1} and Λ_{t_1+1} are symplectic bijections, $F = \Lambda_{t_1+1}\Lambda_{t_1}^{-1}$ preserves σ_S. □

F is one step of the leapfrog scheme for P, and the proof shows why it is symplectic without a matrix computation: it is the composite of two symplectic bijections, the Cauchy maps at consecutive times. In Fourier variables, with z on the unit circle dual to x, F is the matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & z+z^{-1}-m^2 \end{smallmatrix}\right)$.

Remark 5.15 (Dynamics on Cauchy data). From Lemma 5.4, $\Lambda_{t_1} L_{(1,0)} = \Lambda_{t_1-1}$ and $(\Lambda_{t_1} L_{(0,b)}[f])_i = (\Lambda_{t_1}[f])_i(\cdot - b)$. With Proposition 5.14, the unit time translation acts on Cauchy data as $\Lambda_{t_1} L_{(1,0)} \Lambda_{t_1}^{-1} = F^{-1}$, and a spatial translation by b as the shift $d \mapsto d(\cdot - b)$. For the CCR algebra over Cauchy data the Cauchy-data nets at consecutive times are related by the relabelling of F: their generators are $W_S(F\Lambda_{t_1}[f]) = \alpha_F(W_S(\Lambda_{t_1}[f]))$, so the net at t_1+1 is α_F applied to the net at t_1.

6 Simplicity and primitivity of the Weyl algebra

Axiom 2 of [1] asks for a faithful irreducible representation of the algebra of the net. In a simple C*-algebra the kernel of a representation on a nonzero Hilbert space is a proper two-sided ideal, hence zero, so primitivity reduces to simplicity and the existence of one irreducible representation. For nondegenerate σ the CCR algebra is simple: averaging a finite combination of Weyl operators over conjugations by Weyl operators removes, in the limit, every term except the coefficient of 1, which the vector state reads off (Proposition 6.7).

Throughout this chapter, V is a real vector space, σ an alternating bilinear form on V, and (A, W) a Weyl system over (V, σ) (Definition 2.9). In Section 6.2, $A = \mathrm{CCR}(V, \sigma)$ with its Weyl operators (Proposition 2.11) and τ is its vector state (Definition 2.18); Section 6.3 concerns an arbitrary C*-algebra A.

Definition 6.1 (Nondegenerate form; simple algebra)**.** σ is *nondegenerate* if $\sigma(u, w) = 0$ for all $w \in V$ implies $u = 0$. A *two-sided ideal* of a C*-algebra A is a linear subspace I with $ab, ba \in I$ for all $a \in A$ and $b \in I$; it is not assumed closed. A is *simple* if $A \neq 0$ and its only two-sided ideals are 0 and A.

Nondegeneracy enters only through Lemma 6.6.

6.1 Averages over conjugations

Lemma 6.2 (Conjugation)**.** Lean ✓ *In a Weyl system (A, W) over (V, σ), $W(u)W(v)W(u)^* = e^{-\mathrm{i}\sigma(u,v)}\, W(v)$ for all $u, v \in V$.*

Proof. By Lemma 2.10, $W(u)W(v) = e^{-\mathrm{i}\sigma(u,v)}W(v)W(u)$, and $W(u)W(u)^* = 1$. □

Definition 6.3 (Average)**.** For $u \in V$ and $n \geq 1$, the *average* $\mathsf{M}_n^u : A \to A$ is

$$\mathsf{M}_n^u(a) = \frac{1}{n} \sum_{j=0}^{n-1} W(ju)\, a\, W(ju)^*,$$

and for $v \in V$ the mean phase is $\mu_n(u, v) = \frac{1}{n} \sum_{j=0}^{n-1} e^{-\mathrm{i}j\sigma(u,v)}$.

Since $\sigma(u, u) = 0$, the Weyl relation gives $W(u)W(ju) = W((j + 1)u)$, so $W(ju) = W(u)^j$ and $\mathsf{M}_n^u(a)$ is the mean of a over the first n powers of conjugation by $W(u)$. For $a = W(v)$ each conjugation multiplies by the phase of Lemma 6.2, and the average multiplies by the mean $\mu_n(u, v)$ of these phases. In Lean the average is defined for $n = 0$ as well, as 0. formalization

Lemma 6.4 (Properties of the average)**.** Lean ✓ *Let (A, W) be a Weyl system over (V, σ), $u \in V$, $n \geq 1$, and M_n^u the average of Definition 6.3.*

(a) M_n^u is linear, $\mathsf{M}_n^u(1) = 1$, and $\|\mathsf{M}_n^u(a)\| \leq \|a\|$.

(b) If I is a two-sided ideal of A and $a \in I$, then $\mathsf{M}_n^u(a) \in I$.

(c) $\mathsf{M}_n^u(W(v)) = \mu_n(u, v)\, W(v)$ for every $v \in V$.

Proof. (a) Each term $W(ju)aW(ju)^*$ is linear in a, equals 1 for $a = 1$, and has norm $\|a\|$, since multiplication by a unitary preserves the norm in a C*-algebra; the triangle inequality bounds the mean of n such terms by $\|a\|$.

(b) Each term is a product with a in the middle, and I is closed under sums and multiplication by scalars.

(c) By Lemma 6.2 with ju in place of u, the j-th term is $e^{-\mathrm{i}j\sigma(u,v)}W(v)$.

□

Lemma 6.5 (Averaged phases). Lean ✓ *For an alternating form σ on V and $u, v \in V$ with $\sigma(u,v) \notin 2\pi\mathbb{Z}$, the mean phase $\mu_n(u,v)$ of Definition 6.3 satisfies $|\mu_n(u,v)| \le \dfrac{2}{n\,|1 - e^{-\mathrm{i}\sigma(u,v)}|}$, so $\mu_n(u,v) \to 0$ as $n \to \infty$.*

Proof. The sum is geometric with ratio $e^{-\mathrm{i}\sigma(u,v)}$, which differs from 1 exactly when $\sigma(u,v) \notin 2\pi\mathbb{Z}$. So $n\mu_n(u,v) = (1 - e^{-\mathrm{i}n\sigma(u,v)})/(1 - e^{-\mathrm{i}\sigma(u,v)})$, and the numerator has modulus at most 2. □

To make $\mu_n(u, v_l) \to 0$ for several v_l at once, one u must avoid $\sigma(u, v_l) \in 2\pi\mathbb{Z}$ for all of them.

Lemma 6.6 (Avoiding $2\pi\mathbb{Z}$). Lean ✓ *Let σ be a nondegenerate alternating form on a real vector space V and $v_1, \dots, v_N \in V$ nonzero. There is $u \in V$ with $\sigma(u, v_l) \notin 2\pi\mathbb{Z}$ for every l.*

Proof. For each l nondegeneracy gives w with $\sigma(v_l, w) \neq 0$; put $w_l = w$, so that $\sigma(w_l, v_l) = -\sigma(v_l, w) \neq 0$. For real λ let $u(\lambda) = \lambda w_1 + \lambda^2 w_2 + \cdots + \lambda^N w_N$. Then $\sigma(u(\lambda), v_l)$ is a polynomial in λ whose coefficient of λ^l is $\sigma(w_l, v_l) \neq 0$, so the product over l of these polynomials is nonzero, and some λ_0 gives $u_0 = u(\lambda_0)$ with $\sigma(u_0, v_l) \neq 0$ for all l. For $0 < \lambda < 2\pi / \max_l |\sigma(u_0, v_l)|$ each $\lambda\sigma(u_0, v_l)$ lies in $(-2\pi, 2\pi)$ and is nonzero, so $u = \lambda u_0$ works. □

The Lean proof takes u_0 off a finite union of proper subspaces and avoids a countable set of λ. formalization

Proposition 6.7 uses two properties of the vector state only: it is bounded by the norm, and $\tau(W(v)) = [v = 0]$ (Proposition 2.19). It is stated for a functional τ_A on any Weyl system with these two properties; τ is the case $A = \mathrm{CCR}(V, \sigma)$.

Proposition 6.7 (Approximation). Lean ✓ *Let (A, W) be a Weyl system over (V, σ) with σ nondegenerate, and $\tau_A : A \to \mathbb{C}$ a linear functional with $|\tau_A(a)| \le \|a\|$ and $\tau_A(W(v)) = [v = 0]$ for all v. For every $a \in A$ and $\nu > 0$ there are $u \in V$ and $n \ge 1$ with $\|\mathsf{M}_n^u(a) - \tau_A(a)1\| < \nu$.*

Proof. The linear span of the Weyl elements contains $1 = W(0)$ and is closed under products and adjoints, by the Weyl relation and $W(u)^* = W(-u)$; so it is $\mathrm{alg}\{W(u) : u \in V\}$, which is dense in A by Definition 2.9. Choose b in the span with $\|a - b\| < \nu/3$, and write $b = \hat{b}(0)\,1 + \sum_{l=1}^{N} \hat{b}(v_l) W(v_l)$ with distinct nonzero v_l and complex coefficients $\hat{b}(\cdot)$; then $\tau_A(b) = \hat{b}(0)$, and the coefficients are unique, since $\tau_A(W(v)^* b) = \hat{b}(v)$. Choose u for $v_1, \dots, v_N$ by Lemma 6.6. By Lemma 6.4(a),(c),

$$\mathsf{M}_n^u(b) - \tau_A(b)1 = \sum_{l=1}^{N} \hat{b}(v_l)\,\mu_n(u, v_l)\,W(v_l),$$

whose norm is at most $\sum_l |\hat{b}(v_l)|\,|\mu_n(u, v_l)|$, since $\|W(v_l)\| = 1$, and so less than $\nu/3$ for large n by Lemma 6.5. Then

$$\|\mathsf{M}_n^u(a) - \tau_A(a)1\| \le \|\mathsf{M}_n^u(a - b)\| + \|\mathsf{M}_n^u(b) - \tau_A(b)1\| + |\tau_A(b - a)|\,\|1\| < \nu,$$

by Lemma 6.4(a), the bound on τ_A, and $\|1\| \le 1$. □

The vector u depends on a and ν through the approximant b: the proposition says that $\tau_A(a)1$ lies in the norm closure of the set of averages $\{\mathsf{M}_n^u(a) : u \in V,\ n \ge 1\}$, each a convex combination of unitary conjugates of a.

6.2 Simplicity of the CCR algebra

Lemma 6.8 (Faithfulness). Lean ✓ *If $a \in \mathrm{CCR}(V, \sigma)$ and $a\delta_0 = 0$, then $a = 0$. Hence $\tau(a^*a) = \|a\delta_0\|^2 > 0$ for every nonzero $a \in \mathrm{CCR}(V, \sigma)$, and $\mathrm{CCR}(V, \sigma) \neq 0$.*

Proof. The argument of Remark 2.21: $a\delta_{-w} = aR(w)\delta_0 = R(w)a\delta_0 = 0$ for every w by Proposition 2.17, and a bounded operator vanishing on the orthonormal basis is zero. The second claim is Proposition 2.19, and $1\delta_0 \neq 0$. □

Theorem 6.9 (Simplicity). Lean ✓ *If σ is nondegenerate, $\mathrm{CCR}(V,\sigma)$ is simple (Definition 6.1); the ideals are not assumed closed.*

Proof. $\mathrm{CCR}(V,\sigma) \neq 0$ by Lemma 6.8. Let $I \neq 0$ be a two-sided ideal and $a \in I$ nonzero. Then $a^*a \in I$ and $\tau(a^*a) > 0$ by Lemma 6.8. Proposition 6.7 with $\tau_A = \tau$ gives u, n with $\|\mathsf{M}_n^u(a^*a) - \tau(a^*a)1\| < \tau(a^*a)$, so the element $\tau(a^*a)^{-1}\mathsf{M}_n^u(a^*a)$ of I (Lemma 6.4(b)) lies within distance 1 of 1. It is invertible by the Neumann series, which converges since $\mathrm{CCR}(V,\sigma)$ is norm-closed in $B(\ell^2(V))$, so I contains 1 and equals $\mathrm{CCR}(V,\sigma)$. □

Simplicity of the Weyl algebra of a nondegenerate form is classical [7, 6][4, Thm. 5.2.8].

Remark 6.10 (Degenerate forms). For degenerate σ the theorem fails. If $\sigma(u_0,\cdot) = 0$ with $u_0 \neq 0$, then $W(u_0)$ commutes with every $W(v)$ and $W(v)^* = W(-v)$ (Lemma 2.10), hence is central (Lemma 1.19), and $W(u_0)-1$ generates the two-sided ideal $(W(u_0)-1)\,\mathrm{CCR}(V,\sigma)$. It is nonzero, since $(W(u_0)-1)\delta_0 = \delta_{-u_0} - \delta_0$. $W(u_0)$ acts as the translation $\psi \mapsto \psi(\cdot + u_0)$, and the unit vectors $\psi_N = N^{-1/2}\sum_{j<N}\delta_{ju_0}$, whose points ju_0 are distinct, satisfy $(W(u_0)-1)\psi_N = N^{-1/2}(\delta_{-u_0} - \delta_{(N-1)u_0})$, of norm $(2/N)^{1/2}$. If the ideal contained 1, then $(W(u_0)-1)b = 1$ for some b, and by centrality $1 = \|b(W(u_0)-1)\psi_N\| \leq \|b\|\,(2/N)^{1/2}$ for all N, which is impossible; so the ideal is proper.

Remark 6.11 (The vector state is not regular). A state ω is *regular* if $\lambda \mapsto \omega(W(\lambda u))$ is continuous for every u; a Fock vacuum is regular. $\tau(W(\lambda u)) = [\lambda u = 0]$ is discontinuous at $\lambda = 0$ for $u \neq 0$, so τ is not regular, and on $\ell^2(V)$ no field operator $\phi(u)$ exists (Example 2.7). The proof of Theorem 6.9 uses τ only as a faithful functional that reads off the coefficient of 1. Simplicity is a property of the C*-algebra; through the isomorphism of Remark 2.22, which rests on the uniqueness theorem cited in Remark 2.12, it holds for the C*-algebra generated by the Weyl operators of a Fock representation.

6.3 Primitivity

Proposition 6.12 (Representations of a simple algebra). Lean ✓ *Let A be a simple C*-algebra and $\pi : A \to B(H)$ a $*$-homomorphism, with H a nonzero Hilbert space. Then π is injective.*

Proof. $\ker\pi$ is a two-sided ideal, and $1 \notin \ker\pi$ since $\pi(1) = 1 \neq 0$. So $\ker\pi = 0$. □

Definition 6.13 (Irreducible-representation hypothesis). A C*-algebra A *has an irreducible representation* if there is an irreducible representation of A (Definition 1.11) on a nonzero Hilbert space, with the convention on Hilbert spaces stated after Definition 1.11.

Theorem 6.14 (Primitivity). Lean ✓ *A simple C*-algebra that has an irreducible representation is primitive. In particular, if σ is nondegenerate and $\mathrm{CCR}(V,\sigma)$ has an irreducible representation, then $\mathrm{CCR}(V,\sigma)$ is primitive.*

Proof. By Proposition 6.12 the irreducible representation is injective. The second statement follows with Theorem 6.9. □

Remark 6.15 (Existence of irreducible representations). Every nonzero C*-algebra has an irreducible representation: its states form a nonempty weak*-compact convex set, which has extreme points, the pure states, by the Krein–Milman theorem, and the GNS representation of a pure state is irreducible [5, §2.3, Thm. 2.3.19]. For $V \neq 0$ the representation on $\ell^2(V)$ is reducible (Remark 2.21). For nondegenerate σ and a complex structure on V compatible with σ, a Fock representation combined with the isomorphism of Remark 2.22 is an explicit one. On the lattice such a structure exists on the Cauchy data of Chapter 5: $\mathcal{J}(d_0,d_1) = (-d_1,d_0)$ satisfies $\mathcal{J}^2 = -1$, $\sigma_S(\mathcal{J}d,\mathcal{J}d') = \sigma_S(d,d')$ and $\sigma_S(d,\mathcal{J}d) = \sum_x (d_0(x)^2 + d_1(x)^2) > 0$ for $d \neq 0$, and the Cauchy map transports it to V. $\mathcal{J}$ does not commute with the change of slice F, so its Fock state is not invariant under time translation; it serves only to exhibit an irreducible representation. The hypothesis of Theorem 6.14 is kept because the existence theorem is not formalized in LinearAQFT. formalization

6.4 The axioms for the CCR net

Theorem 6.16 (Axioms 1–4 for the CCR net)**.** Lean ✓ *Let* $(\mathcal{D}, V, \sigma, [\cdot])$ *be phase-space data (Definition 4.1) with nondegenerate* σ*, and suppose* $\mathrm{CCR}(V, \sigma)$ *has an irreducible representation. Then the CCR net (Definition 4.10) is generating, causal and has the time-slice property, and its algebra* $\mathrm{CCR}(V, \sigma)$ *is primitive.*

Proof. Theorem 4.7 for the Weyl system of Proposition 2.11, and Theorem 6.14. □

Remark 6.17 (Theorem 4 of [1]). Theorem 4 of [1] asserts the axioms for the free scalar field on a globally hyperbolic spacetime [1, pp. 225–226]. With Theorem 4.11 for Axiom 5, Theorem 6.16 is its counterpart in these notes, with five differences:

- the spacetime and its propagators enter through phase-space data, derived from Green data in Chapter 7; continuum Green data are formalized for the massless field on $3+1$ Minkowski space with inertial slabs as cutoffs (Chapter 8) and for the Klein–Gordon field there with any real m^2 (Chapter 9), and not for the general Klein–Gordon field of Remark 7.2;
- covariance is indexed by isomorphisms of phase-space data (Theorem 4.11);
- the existence of an irreducible representation is a hypothesis; it holds for every nonzero C*-algebra by the theorem cited in Remark 6.15, and is kept because that theorem is not formalized;
- the time-slice property is proved for the dependence of the causal index; the form of [1] follows on paper for the Green data of Remark 7.2 (Remark 7.6), and for the Minkowski instances of Chapters 8 and 9 directly from the inertial slabs when the Cauchy surface is flat, and otherwise through all Cauchy temporal functions as cutoffs, which are not formalized (Remark 8.13);
- covariance is required only for maps preserving the time orientation (Remark 1.8).

For a concrete spacetime the counterpart is Theorem 9.18: the Klein–Gordon field on $3 + 1$ Minkowski space, with Axioms 1, 3 and 5, Axiom 4 over the derived causal index, and Axiom 2 given an irreducible representation.

Corollary 6.18 (The lattice)**.** Lean ✓ *For every* $m^2 \in \mathbb{R}$*, the CCR algebra of the lattice field is simple, and the CCR net of the lattice field is generating, causal, has the time-slice property, and is covariant under the translations of* $\mathbb{Z}^2$*. If the algebra has an irreducible representation, it is primitive.*

Proof. The lattice form is nondegenerate (Proposition 5.2). Apply Theorem 6.9, Theorem 6.16 and Theorem 5.6. □

7 The net of a Green-hyperbolic operator

Chapter 4 takes phase-space data as given. For a free field they come from the retarded and advanced Green operators of the field equation: V is the space of test functions modulo the kernel of $E = G^- - G^+$, and $\sigma([f],[g]) = \langle f, Eg\rangle$. This chapter isolates what that construction uses, as *Green data*, derives the causal index and the phase-space data from them, and proves nondegeneracy; the conclusions of Theorem 6.16 then hold for every instance, over the derived causal index, with primitivity given an irreducible representation. The lattice of Chapter 3 is an instance whose derived data are those of Chapter 5.

Formal adjointness in (G6) forces P to be formally self-adjoint on test functions, $\langle P_{\mathcal{D}} f, g\rangle = \langle P_{\mathcal{D}} f, G^+ P_{\mathcal{D}} g\rangle = \langle P_{\mathcal{D}} g, G^- P_{\mathcal{D}} f\rangle = \langle P_{\mathcal{D}} g, f\rangle$, so on a spacetime Green data are modelled on the formally self-adjoint Green-hyperbolic operators of [11, Def. 3.2], the class used for the CCR quantization in [14, §3.1, Def. 3.1, Thm. 3.10]; Remark 7.2 sketches the Klein–Gordon instance.

7.1 Green data

Definition 7.1 (Green data)**.** *Green data* consist of:

(G1) a set M of points with a relation $p \le q$ (q lies in the causal future of p); for $O \subseteq M$, $J^+(O) = \{q : \exists p \in O,\ p \le q\}$, $J^-(O) = \{q : \exists p \in O,\ q \le p\}$ and $J(O) = J^+(O) \cup J^-(O)$;

(G2) a real vector space $\mathcal{F}$ (functions) with a map $\operatorname{supp} : \mathcal{F} \to 2^M$ such that $\operatorname{supp} 0 = \emptyset$, $\operatorname{supp}(h + h') \subseteq \operatorname{supp} h \cup \operatorname{supp} h'$ and $\operatorname{supp}(\lambda h) \subseteq \operatorname{supp} h$ for $\lambda \in \mathbb{R}$;

(G3) a real vector space $\mathcal{D}$ (test functions) with a linear map $\iota : \mathcal{D} \to \mathcal{F}$, written $f \mapsto f$ when no confusion arises, and $\operatorname{supp} f = \operatorname{supp} \iota f$ for $f \in \mathcal{D}$;

(G4) a linear map $P : \mathcal{F} \to \mathcal{F}$ and a linear map $P_{\mathcal{D}} : \mathcal{D} \to \mathcal{D}$ with $\iota P_{\mathcal{D}} = P \iota$;

(G5) a bilinear pairing $\langle f, h\rangle$ for $f \in \mathcal{D}$ and $h \in \mathcal{F}$ that vanishes when $\operatorname{supp} f \cap \operatorname{supp} h = \emptyset$, and is nondegenerate in h: if $\langle f, h\rangle = 0$ for all $f \in \mathcal{D}$, then $h = 0$;

(G6) linear maps $G^\pm : \mathcal{D} \to \mathcal{F}$ satisfying the *Green identities* $PG^\pm f = f$, $G^\pm P_{\mathcal{D}} f = f$, $\operatorname{supp} G^\pm f \subseteq J^\pm(\operatorname{supp} f)$, and *formal adjointness* $\langle f, G^+ g\rangle = \langle g, G^- f\rangle$ for $f, g \in \mathcal{D}$;

(G7) a set of *cutoffs* χ, each carrying a linear map $\chi : \mathcal{F} \to \mathcal{F}$ and a *slab* $S_\chi \subseteq M$ (two cutoffs may share the map and differ in the slab), such that for every $f \in \mathcal{D}$, with $E = G^- - G^+$:

- (c1) $\operatorname{supp} P(\chi E f) \subseteq S_\chi \cap J(\operatorname{supp} f)$;
- (c2) $(1 - \chi) G^+ f + \chi G^- f = \iota k$ for some $k \in \mathcal{D}$.

The relation $\le$ need not be transitive, and ι need not be injective, which costs nothing: if $\iota f = 0$ then $\langle g, Ef\rangle = -\langle f, Eg\rangle = 0$ for all g by (G5) and Lemma 7.8(c), so $Ef = 0$. P is not required to be local; (c1) is where finite propagation speed enters: in the continuum through the locality of P (Remark 7.2), on the lattice through the strict light cone of the kernel (Lemma 3.3(b)). On the lattice, χ is the step at t_1 and k is the function of Remark 3.10.

Remark 7.2 (The continuum). For the Klein–Gordon operator on a globally hyperbolic spacetime M of dimension at least 2, $3+1$ Minkowski space included, with M also the set of points:

- $p \le q$ iff $q \in J^+(p)$; $\mathcal{F} = C^\infty(M)$ and $\mathcal{D} = C_c^\infty(M)$ with the usual support; $\langle f, h\rangle = \int_M f h \, d\mathrm{vol}$, nondegenerate by the fundamental lemma of the calculus of variations;
- $G^\pm$ are the retarded and advanced Green operators, which exist and satisfy (G6) by [3, Cor. 3.4.3, Lemma 3.4.4]; the lemma pairs the Green operators of P with those of its formal adjoint, which is P itself;

- a Cauchy temporal function $\mathcal{T}$ exists, and every smooth spacelike Cauchy surface is a level set of one [12, Thm. 1.1], [13, Thm. 1.2]. The cutoffs are indexed by all Cauchy temporal functions $\mathcal{T}$, all $t_1 < t_2$ and all smooth functions of $\mathcal{T}$ equal to 0 for $\mathcal{T} < t_1$ and to 1 for $\mathcal{T} > t_2$; χ is multiplication by that function, with slab $\mathcal{T}^{-1}[t_1, t_2]$. The family must be carried to itself by the isometries for them to act on it (Definition 7.12); on Minkowski space the smaller family of inertial slabs is (Proposition 8.17). Then (c1) holds because $P(\chi Ef) = [P, \chi]Ef$ and $[P, \chi]$ is a differential operator supported where the cutoff function is not locally constant; (c2) holds because $J^+(\operatorname{supp} f) \cap \mathcal{T}^{-1}(-\infty, t_2]$ and $J^-(\operatorname{supp} f) \cap \mathcal{T}^{-1}[t_1, \infty)$ are compact [3, Cor. A.5.4].

This instance is not formalized in LinearAQFT in this generality; the massless field on $3+1$ Minkowski space, with inertial slabs as cutoffs, is (Chapter 8), and so is the Klein–Gordon field on $3+1$ Minkowski space with any real m^2 and the same cutoffs (Chapter 9). Its dependence is compared with that of Example 1.4 in Remark 7.6.

Remark 7.3 (Sign conventions). The literature names and signs the Green operators differently. The table relates these notes (Remark 2.1) to [1] and to [3], as compared after Definition 3.4.

these notes	[1]	[3]
$G^+ = E^{\mathrm{ret}}$, supported in J^+	E^+ [1, p. 222]	G_+, called advanced [3, Def. 3.4.1]
$E = G^- - G^+$	$E^+ - E^- = -E$ [1, p. 222]	$G_+ - G_- = -E$ [3, §3.4]
$\sigma([f],[g]) = \langle f, Eg\rangle$	$[\phi(f), \phi(g)] = \mathrm{i}^{-1}\langle f, (E^+ - E^-)g\rangle$ [1, Thm. 2, p. 224]	not compared here

7.2 The derived causal index

Definition 7.4 (Derived causal index)**.** For Green data (Definition 7.1), on the subsets of M ordered by inclusion:

- $O_1 \perp O_2$ if no point of O_1 is comparable, in either order, with a point of O_2;
- $O \triangleleft O'$ if every $f \in \mathcal{D}$ with $\operatorname{supp} f \subseteq O$ has a cutoff χ with $S_\chi \cap J(\operatorname{supp} f) \subseteq O'$.

Lemma 7.5 (Causal index)**.** Lean ✓ *For Green data, the relations $\perp$ and $\triangleleft$ of Definition 7.4 on the subsets of M, ordered by inclusion, form a causal index (Definition 1.2).*

Proof. $\perp$ is symmetric because comparability is. If $O_0 \subseteq O_1$ and $O_1 \perp O_2$, then $O_0 \perp O_2$. If $O_0 \subseteq O$, a test function supported in O_0 is supported in O; if $O' \subseteq O''$, $S_\chi \cap J(\operatorname{supp} f) \subseteq O'$ gives $S_\chi \cap J(\operatorname{supp} f) \subseteq O''$. □

The slab may depend on the test function; one slab for the whole region, $S_\chi \cap J(O) \subseteq O'$, suffices, since J is monotone, but on a spacetime it is not necessary (Remark 7.6). The time-slice argument moves one test function at a time, so this is the dependence it uses. On the lattice, where the supports of test functions are the finite sets, it is Example 1.3. With no cutoffs no pair is dependent and the time-slice property is vacuous; every instance below has cutoffs.

Remark 7.6 (Dependence on a spacetime). For the Green data of Remark 7.2, on open regions with compact closure:

- *The dependence of Example 1.4 implies that of Definition 7.4.* Let every endless causal curve through O meet the Cauchy surface Σ in O', and let $\operatorname{supp} f \subseteq O$. Then $J(O) \cap \Sigma \subseteq O'$: a point q of $J(O) \cap \Sigma$ lies on a causal curve from a point of O, which extends to an endless causal curve meeting the acausal Σ only at q. Then $Y = \operatorname{supp} f$ is compact, $J(Y)$ is closed, and $J(Y) \cap \Sigma$ is compact [3, Lemma A.5.1, Cor. A.5.4] and contained in $J(O) \cap \Sigma \subseteq O'$. Take $\mathcal{T}$ with $\Sigma = \mathcal{T}^{-1}(0)$ [13, Thm. 1.2]. The compact sets $\big(J(Y) \cap \mathcal{T}^{-1}[-\nu, \nu]\big) \setminus O'$ [3, Lemma A.5.1, Cor. A.5.4] decrease, as $\nu \downarrow 0$, to $(J(Y) \cap \Sigma) \setminus O' = \emptyset$, so one of them is empty, and the slab $\mathcal{T}^{-1}[-\nu, \nu]$ serves for f.

- *One slab for the whole region is not necessary.* In 1+1 Minkowski space let $O = \{|t|+|x| < 1\}$, $\Sigma = \{t = 0\}$ and $O' = \{|x| < 1,\ |t| < (1-|x|)^2\}$. Every endless causal curve through O meets Σ in O', so $O \triangleleft O'$. But $J(O) = \{t > |x| - 1\} \cup \{t < 1 - |x|\}$ contains every $(t_0, 1)$ with $t_0 \neq 0$, and every slab $\mathcal{T}^{-1}[t_1, t_2]$ meets the timelike line $x = 1$ in a segment of positive length, hence in such a point, which is not in O'.

The CCR net of the Green data of Remark 7.2, with all Cauchy temporal functions as cutoffs, therefore has the time-slice axiom in the form of Example 1.4; with fewer cutoffs, such as the inertial slabs of Chapter 8, the implication can fail (Remark 8.13). None of this remark is formalized.

7.3 The derived phase-space data

Definition 7.7 (Derived phase-space data)**.** For Green data, the *derived phase-space data* are: $E = G^- - G^+ : \mathcal{D} \to \mathcal{F}$; $\mathcal{D}(O) = \{f \in \mathcal{D} : \operatorname{supp} f \subseteq O\}$; $V = \mathcal{D}/\ker E$ with classes $[f]$; $\sigma([f],[g]) = \langle f, Eg\rangle$.

Lemma 7.8 (The causal propagator)**.** Lean ✓ *For Green data, $E = G^- - G^+$ and $f, g \in \mathcal{D}$:*

(a) $\operatorname{supp} Ef \subseteq J(\operatorname{supp} f)$*;*

(b) $EP_{\mathcal{D}} f = 0$ *and* $PEf = 0$*;*

(c) $\langle f, Eg\rangle = -\langle g, Ef\rangle$*.*

Proof. (a) $\operatorname{supp}(G^- f - G^+ f) \subseteq \operatorname{supp} G^- f \cup \operatorname{supp} G^+ f$ by (G2), and (G6).

(b) $EP_{\mathcal{D}} f = G^- P_{\mathcal{D}} f - G^+ P_{\mathcal{D}} f = f - f = 0$ and $PEf = PG^- f - PG^+ f = f - f = 0$.

(c) By formal adjointness applied to (f, g) and to (g, f), $\langle f, G^- g\rangle - \langle f, G^+ g\rangle = \langle g, G^+ f\rangle - \langle g, G^- f\rangle$.

□

Proposition 7.9 (Phase-space data)**.** Lean ✓ *For Green data, the derived data $(\mathcal{D}, V, \sigma, [\cdot])$ of Definition 7.7 are phase-space data (Definition 4.1) over the derived causal index.*

Proof. Localization. $\mathcal{D}(O)$ is a subspace by (G2), monotone in O, and $f \in \mathcal{D}(\operatorname{supp} f)$.

The form. $(f, g) \mapsto \langle f, Eg\rangle$ vanishes for $g \in \ker E$, and for $f \in \ker E$ by Lemma 7.8(c), so it descends to $V \times V$; it is alternating by the same lemma. The map $f \mapsto [f]$ is linear and surjective.

(P1). If $O_1 \perp O_2$, $f \in \mathcal{D}(O_1)$ and $g \in \mathcal{D}(O_2)$, then $\operatorname{supp} Eg \subseteq J(O_2)$ contains no point of O_1, so $\langle f, Eg\rangle = 0$ by (G5).

(P2). Let $O \triangleleft O'$ and $f \in \mathcal{D}(O)$, and take χ with $S_\chi \cap J(\operatorname{supp} f) \subseteq O'$. With k from (c2) and $f' = f - P_{\mathcal{D}} k$,

$$\iota f' = f - P((1-\chi)G^+ f + \chi G^- f) = -P(\chi Ef),$$

by $PG^+ f = f$ and linearity. By (c1), $\operatorname{supp} f' \subseteq S_\chi \cap J(\operatorname{supp} f) \subseteq O'$, and by Lemma 7.8(b), $Ef' = Ef$, so $[f'] = [f]$. □

Proposition 7.10 (Nondegeneracy)**.** Lean ✓ *For Green data, the derived form $\sigma([f],[g]) = \langle f, Eg\rangle$ on $V = \mathcal{D}/\ker E$ is nondegenerate (Definition 6.1).*

Proof. If $\sigma([f], w) = 0$ for all w, then $\langle g, Ef\rangle = -\sigma([f],[g]) = 0$ for every g, so $Ef = 0$ by (G5) and $[f] = 0$. □

Theorem 7.11 (The axioms for Green data)**.** Lean ✓ *For Green data such that $\mathrm{CCR}(V, \sigma)$ has an irreducible representation, the CCR net of the derived phase-space data is generating, causal and has the time-slice property over the derived causal index, and its algebra is primitive.*

Proof. Propositions 7.9 and 7.10, and Theorem 6.16. □

Without the hypothesis the proof still gives generation, causality and time slice (Theorem 4.7), and simplicity of $\mathrm{CCR}(V,\sigma)$ (Theorem 6.9); only primitivity uses the irreducible representation. Every subset of M is a region, so $M \triangleleft S_\chi$ for every cutoff, and the time-slice property gives $\mathcal{A}(S_\chi) = \mathcal{A}(M)$: the algebra of a slab of the cutoff family is the whole algebra. This is a form of the time-slice axiom of locally covariant field theory mentioned in Remark 1.13, for these slabs. For the Green data of Remark 7.2, (N3) and isotony also give $\mathcal{A}(U) = \mathcal{A}(M)$ for every open neighbourhood U of a smooth spacelike Cauchy surface: the first argument of Remark 7.6 uses only that the support of a test function is compact, so it gives $M \triangleleft U$.

7.4 Symmetries

Definition 7.12 (Symmetry of Green data)**.** A *symmetry* γ of Green data consists of a bijection γ of M with $\gamma p \le \gamma q \iff p \le q$; linear bijections $\gamma_\mathcal{D}$ of $\mathcal{D}$ and $\gamma_\mathcal{F}$ of $\mathcal{F}$ with $\iota\gamma_\mathcal{D} = \gamma_\mathcal{F}\iota$, $\operatorname{supp}\gamma_\mathcal{F} h = \gamma(\operatorname{supp} h)$, $G^\pm\gamma_\mathcal{D} = \gamma_\mathcal{F} G^\pm$ and $\langle \gamma_\mathcal{D} f, \gamma_\mathcal{F} h\rangle = \langle f, h\rangle$; and a bijection $\chi \mapsto \gamma\chi$ of the cutoffs with $S_{\gamma\chi} = \gamma(S_\chi)$.

Proposition 7.13 (Symmetries act on the phase space)**.** Lean ✓ *A symmetry γ of Green data gives an isomorphism of the derived phase-space data (Definition 4.3) with regions $\kappa_\gamma O = \gamma(O)$, test functions $\gamma_\mathcal{D}$, and the linear bijection $L_\gamma[f] = [\gamma_\mathcal{D} f]$. The identity maps form a symmetry with $L = \mathrm{id}$, and the composite of two symmetries is a symmetry with $L_{\gamma_2\gamma_1} = L_{\gamma_2} L_{\gamma_1}$.*

Proof. γ preserves comparability, so it preserves $\perp$; it maps shadows to shadows and slabs to slabs, so $S_\chi \cap J(Y) \subseteq O'$ iff $S_{\gamma\chi} \cap J(\gamma Y) \subseteq \gamma O'$ for every $Y \subseteq M$. Since $\operatorname{supp}\gamma_\mathcal{D} f = \gamma(\operatorname{supp} f)$, $\gamma_\mathcal{D}\mathcal{D}(O) = \mathcal{D}(\gamma O)$; since every cutoff is some $\gamma\chi$ and every test function some $\gamma_\mathcal{D} f$, $O \triangleleft O'$ iff $\gamma O \triangleleft \gamma O'$. From $E\gamma_\mathcal{D} = \gamma_\mathcal{F} E$ and injectivity of $\gamma_\mathcal{F}$, $\gamma_\mathcal{D} \ker E = \ker E$, so L_γ is well defined and bijective, with inverse $[f] \mapsto [\gamma_\mathcal{D}^{-1} f]$, and $\langle \gamma_\mathcal{D} f, E\gamma_\mathcal{D} g\rangle = \langle \gamma_\mathcal{D} f, \gamma_\mathcal{F} E g\rangle = \langle f, Eg\rangle$. The identity and composition statements hold on classes. □

The inverse maps of a symmetry form a symmetry, so the symmetries form a group. With Theorem 4.11, $\gamma \mapsto (\kappa_\gamma, \alpha_{L_\gamma})$ gives covariance pairs with the composition law of Axiom 5. Time reversal exchanges J^+ and J^-, hence G^+ and G^-, so it is not a symmetry in this sense (compare Remark 4.4). Maps between two sets of Green data are defined in the same way; they are not needed here.

7.5 The lattice

Proposition 7.14 (Lattice Green data)**.** Lean ✓ *For every $m^2 \in \mathbb{R}$ the following are Green data: $M = \mathbb{Z}^2$ with the order of Chapter 1; $\mathcal{F}$ the functions $\mathbb{Z}^2 \to \mathbb{R}$ with $\operatorname{supp} h = \{p : h(p) \neq 0\}$; $\mathcal{D}$ the finitely supported functions and ι the inclusion; P the lattice Klein–Gordon operator and $P_\mathcal{D}$ its restriction to $\mathcal{D}$; the pairing of Chapter 3; $G^\pm$ of Definition 3.4; and the cutoffs χ_{t_1}, $t_1 \in \mathbb{Z}$, with $\chi_{t_1} h(t,x) = h(t,x)$ for $t \ge t_1 + 1$ and 0 otherwise, and $S_{\chi_{t_1}} = S_{t_1}$.*

Proof. (G1)–(G4) are immediate, and (G5) holds with $\langle \delta_p, h\rangle = h(p)$. (G6): Proposition 3.6 gives $PG^\pm f = f$, $G^\pm P_\mathcal{D} f = f$ and the supports. Formal adjointness: $\langle f, G^+ g\rangle = \sum_{p,q} f(p)g(q)K(p-q)$ and $\langle g, G^- f\rangle = \sum_{p,q} g(q)f(p)K(t_p - t_q, x_q - x_p)$ agree because K is even in x (Lemma 3.3(c)). (c1): $P(\chi_{t_1} Ef) = PEf - P((1-\chi_{t_1})Ef) = -f'$ with f' the test function of Theorem 3.9. (c2): $(1-\chi_{t_1})G^+ f$ vanishes before the earliest time of $\operatorname{supp} f$ and after t_1, and at each time for $|x|$ beyond the spatial extent of $\operatorname{supp} f$ plus the elapsed time (Lemma 3.3(b)); likewise $\chi_{t_1} G^- f$. So k is finitely supported. □

Theorem 7.15 (Comparison with Chapter 5)**.** Lean ✓ *For the lattice Green data, the derived causal index is that of Example 1.3, E is the causal propagator of Chapter 3, and the identity maps of regions, test functions and $\mathcal{D}/\ker E$ form an isomorphism from the derived phase-space data to those of Proposition 5.1.*

Proof. Comparability is the lattice's, $J(O)$ is the lattice shadow and $S_{\chi_{t_1}} = S_{t_1}$, so $\perp$ agrees, and $\triangleleft$ agrees because the supports of test functions are the finite sets (the indicator of a finite Y has support Y). $E = G^- - G^+$ is the definition of Chapter 3, and $\mathcal{D}(O)$, V and σ are given by the same formulas as in Proposition 5.1. □

Proposition 7.16 (Translations)**.** Lean ✓ *For $a \in \mathbb{Z}^2$, the translation $\gamma p = p + a$, with $\gamma_{\mathcal{D}} f = f(\cdot - a)$, $\gamma_{\mathcal{F}} h = h(\cdot - a)$ and $\gamma\chi_{t_1} = \chi_{t_1+a_t}$, is a symmetry of the lattice Green data of Proposition 7.14, and under the comparison $L_\gamma = L_a$ (Proposition 5.5).*

Proof. The order, supports and pairing are translation invariant, $G^{\pm}$ commute with translations because their kernels depend on $p - q$, and $S_{t_1+a_t} = S_{t_1} + a$. On classes, $L_\gamma[f] = [f(\cdot - a)] = L_a[f]$. □

By Theorem 7.15 and Proposition 7.16, with Propositions 7.9 and 7.13, the generation, causality, time slice and translation covariance of Theorem 5.6 follow from the Green data alone; primitivity, Corollary 6.18, carries the irreducible-representation hypothesis of Theorem 7.11. A continuum instance needs Green data on a spacetime, sketched in Remark 7.2; with all Cauchy temporal functions as cutoffs it has the time-slice axiom in the form of Example 1.4 (Remark 7.6). Chapter 8 formalizes the massless field on Minkowski space with slabs between parallel hyperplanes as cutoffs, which gives that form for flat Cauchy surfaces (Remark 8.13), and Chapter 9 the Klein–Gordon field with any real m^2 and the same cutoffs.

Chapters 3–5 prove these lattice facts directly; this chapter recovers them from the Green data, and Corollary 6.18 collects them. Uniqueness of the Green operators is not a consequence of Green data: it needs P to be formally self-adjoint on pairs of functions whose supports meet in a compact set, and it is proved for each instance (Lemma 3.5, Theorems 8.14 and 8.21(b)).

8 The free massless field on Minkowski space

Chapter 7 reduces the axioms and covariance to Green data. This chapter constructs Green data for the massless wave operator $P = \partial_t^2 - \nabla^2$ on $3+1$ Minkowski space, the operator $P = -\Box + m^2$ of Remark 2.1 with $m = 0$. The retarded and advanced Green operators are given by the retarded potential and its advanced counterpart, and the cutoffs are slabs between parallel spacelike hyperplanes.

For the free massless field the hypothesis of the introduction, the existence of Green operators, thereby becomes a theorem. The CCR net is generating and causal, and it has the time-slice property for these slabs, which implies the time-slice axiom in the form of [1] (Example 1.4) for flat Cauchy surfaces (Remark 8.13, on paper). Its algebra is primitive, given an irreducible representation, and it is covariant under the orthochronous Poincaré group.

Remark 7.2 sketches the general case with all Cauchy temporal functions as cutoffs; that larger family is not formalized. Chapter 9 adds the term m^2, for every real m^2.

8.1 Minkowski space and the wave operator

Definition 8.1 (Minkowski space)**.** Points are $p = (p^0, \vec{p}) \in M = \mathbb{R} \times \mathbb{R}^3$, with the Euclidean norm $|\cdot|$ on $\mathbb{R}^3$ and the Minkowski form $\eta(u, w) = -u^0 w^0 + \vec{u} \cdot \vec{w}$ (mostly plus); $e_0 = (1, \vec{0})$. A vector v is *future causal* if $v^0 \geq |\vec{v}|$, and $p \leq q$ iff $q - p$ is future causal; $I^+ = \{v : v^0 > |\vec{v}|\}$ and $I^- = -I^+$ are the open future and past cones.

$\mathcal{D} = C_c^\infty(M)$ and $\mathcal{F} = C^\infty(M)$ are the real test functions and smooth functions, ι is the inclusion, and $\operatorname{supp} h = \overline{\{h \neq 0\}}$ is the closed support. The wave operator $P = \partial_0^2 - \sum_{i=1}^3 \partial_i^2 = -\eta^{\mu\nu}\partial_\mu\partial_\nu$ acts on $\mathcal{F}$ and on $\mathcal{D}$, and $\langle f, h\rangle = \int_M f(p)h(p)\,dp$ with Lebesgue measure on M. In coordinates (t, x) the spatial variable $x \in \mathbb{R}^3$ carries no arrow.

Remark 8.2 (Closed supports). With the closed support, $\mathcal{D}(O)$ is $C_c^\infty(O)$ for open O, as in [1]: a test function localized in O vanishes near the boundary of O.

With the open support $\{h \neq 0\}$ instead, a test function could be nonzero up to the boundary, and two things would change, as the double cone $D = \{|t| + |\vec{x}| < 1\}$ shows. A test function positive exactly on $D \cap \{t > 0\}$ would give a Weyl element outside the usual algebra of that region. And D would no longer depend on a thin layer $D \cap \{|t| < \ell\}$ around its base, since one test function could fill all of D.

The formalization uses the closed support. formalization

Proposition 8.3 (The Minkowski data)**.** Lean ✓ *The objects of Definition 8.1 satisfy (G1)–(G5) of Definition 7.1. In detail:*

(a) the causal relation $\leq$ on the set M of points is a partial order (reflexive, transitive and antisymmetric);

(b) the support map $h \mapsto \operatorname{supp} h$ from $\mathcal{F}$ to subsets of M satisfies $\operatorname{supp} 0 = \emptyset$, $\operatorname{supp}(h + h') \subseteq \operatorname{supp} h \cup \operatorname{supp} h'$ and $\operatorname{supp}(\lambda h) \subseteq \operatorname{supp} h$ for $\lambda \in \mathbb{R}$, and each $\operatorname{supp} h$ is a closed subset of M;

(c) the wave operator P maps $\mathcal{F}$ into $\mathcal{F}$ and $\mathcal{D}$ into $\mathcal{D}$, and $P(\iota f) = \iota(Pf)$ for $f \in \mathcal{D}$ (so $P_{\mathcal{D}}$ is P on test functions);

(d) the pairing $\langle f, h\rangle = \int_M fh\,dp$ of $f \in \mathcal{D}$ and $h \in \mathcal{F}$ vanishes when $\operatorname{supp} f \cap \operatorname{supp} h = \emptyset$, and it is nondegenerate in h: if $\langle f, h\rangle = 0$ for all $f \in \mathcal{D}$, then $h = 0$;

(e) (integration by parts) for $h, h' \in \mathcal{F}$ whose supports meet in a compact subset of M, $\int_M (Ph)h'\,dp = \int_M h\,(Ph')\,dp$; in particular $\langle Pf, h\rangle = \langle f, Ph\rangle$ for $f \in \mathcal{D}$ and $h \in \mathcal{F}$.

Proof. (a) Reflexivity is $0 \le 0$; transitivity is the triangle inequality; if $p \le q \le p$ then $q^0 = p^0$ and $|\vec{q} - \vec{p}| \le 0$.

(b) $\{h + h' \neq 0\} \subseteq \{h \neq 0\} \cup \{h' \neq 0\}$; take closures.

(c) Derivatives of smooth functions are smooth and do not enlarge supports.

(d) Locality: the integrand vanishes everywhere. Nondegeneracy: if $\int fh = 0$ for all $f \in \mathcal{D}$, then $h = 0$ almost everywhere by the fundamental lemma of the calculus of variations, hence everywhere by continuity.

(e) Integrating by parts twice along each coordinate direction gives the identity when one of h, h' has compact support. In general both integrands vanish outside the compact set $\operatorname{supp} h \cap \operatorname{supp} h'$; multiplying h' by a test function equal to 1 near that set changes neither integrand and makes h' compactly supported.

□

8.2 The retarded and advanced Green operators

Definition 8.4 (Cone measures and Green operators)**.** The *future cone measure* μ^+ on M is the image of the measure $dy/(4\pi|y|)$ on $\mathbb{R}^3$ under $y \mapsto (|y|, y)$, and the *past cone measure* μ^- is the image of μ^+ under $w \mapsto -w$. For $f \in \mathcal{D}$,

$$(G^{\pm} f)(p) = \int_M f(p - w)\, \mu^{\pm}(dw).$$

Proposition 8.5 (The retarded potential)**.** Lean ✓ *For $f, g \in \mathcal{D} = C_c^\infty(M)$, the Green operators $G^{\pm}$ of Definition 8.4 satisfy:*

(a) $(G^{\pm} f)(t, x) = \frac{1}{4\pi} \int_{\mathbb{R}^3} f(t \mp |y|, x - y)\, |y|^{-1}\, dy$*;*

(b) $G^{\pm} f$ *is smooth,* $G^{\pm} : \mathcal{D} \to \mathcal{F}$ *is linear, and* $\partial_v G^{\pm} f = G^{\pm} \partial_v f$ *for every direction v; hence* $P G^{\pm} f = G^{\pm} P f$*;*

(c) $\operatorname{supp} G^{\pm} f \subseteq J^{\pm}(\operatorname{supp} f)$*;*

(d) $\langle f, G^+ g \rangle = \langle g, G^- f \rangle$*.*

Proof. (a) Integration against an image measure; for G^- it gives $f(t + |y|, x + y)$, and the substitution $y \mapsto -y$ gives the formula.

(b) $\mu^{\pm}$ are finite on bounded sets, since $\int_{|y|<r} dy/(4\pi|y|) = r^2/2$. So $G^{\pm} f$ is the convolution of the smooth, compactly supported f with a locally finite measure, which is smooth. The derivative is taken under the integral, dominated by a constant on a compact set.

(c) If $(G^+ f)(t, x) \neq 0$, the integrand of (a) is not identically zero, so $(t - |y|, x - y) \in \operatorname{supp} f$ for some y, a point below (t, x) on its light cone. So $\{G^+ f \neq 0\} \subseteq J^+(\operatorname{supp} f)$, which is closed by Lemma 8.10(b); likewise for G^-.

(d) $(p, w) \mapsto f(p) g(p - w)$ is continuous and vanishes unless $p \in \operatorname{supp} f$ and $w \in p - \operatorname{supp} g$, a compact set on which $dp\, \mu^+(dw)$ is finite. So Fubini applies, and with translation invariance of Lebesgue measure both sides equal $\iint f(p) g(p - w)\, dp\, \mu^+(dw)$.

□

G^+ samples f in the causal past of the point and is supported in the causal future of $\operatorname{supp} f$: it is the retarded operator in the naming of Remark 7.3.

8.3 The fundamental solution

Theorem 8.6 (The fundamental solution). Lean ✓ *For every test function $f \in C_c^\infty(M)$ and the wave operator $P = \partial_t^2 - \nabla^2$,*

$$\int_{\mathbb{R}^3} \frac{(Pf)(|y|, y)}{|y|}\, dy = 4\pi\, f(0).$$

Proof. Spherical means. Let $(\mathsf{A}f)(s, y)$ be the mean of $f(s, \cdot)$ over the sphere of radius $|y|$ about 0. The weight $1/|y|$ and Lebesgue measure are rotation invariant, so the left side does not change when Pf is replaced by $\mathsf{A}Pf$; and $\mathsf{A}Pf = P\mathsf{A}f$, because rotations commute with ∇^2. Put $\bar f(s, \rho) = (\mathsf{A}f)(s, \rho e_1)$ for $\rho \in \mathbb{R}$, with e_1 a unit vector of $\mathbb{R}^3$. It is smooth and compactly supported on $\mathbb{R}^2$, $(\mathsf{A}f)(s, y) = \bar f(s, |y|)$, and $\bar f(0, 0) = f(0)$.

Radial calculus. For $y \neq 0$, $P[\bar f(s, |y|)] = (\bar f_{ss} - \bar f_{\rho\rho} - \frac{2}{\rho}\bar f_\rho)(s, |y|)$ with $\rho = |y|$, and $\int_{\mathbb{R}^3} \Phi(|y|)\, dy = 4\pi \int_0^\infty \rho^2 \Phi(\rho)\, d\rho$ for integrable $\Phi(|y|)$. The left side becomes $4\pi \int_0^\infty \rho\,(\bar f_{ss} - \bar f_{\rho\rho} - \frac{2}{\rho}\bar f_\rho)(\rho, \rho)\, d\rho$.

A null ray. $\rho(\bar f_{ss} - \bar f_{\rho\rho} - \frac{2}{\rho}\bar f_\rho) = (\partial_s + \partial_\rho)(\partial_s - \partial_\rho)(\rho \bar f)$, and $d/d\rho$ along the diagonal $s = \rho$ is $\partial_s + \partial_\rho$. So the integrand is the derivative along the diagonal of $(\partial_s - \partial_\rho)(\rho \bar f)$, which vanishes for large ρ and equals $-\bar f(0, 0)$ at $\rho = 0$. The integral is $\bar f(0, 0) = f(0)$. □

Remark 8.7 (The method of spherical means). The theorem says that $\delta(t - |y|)/(4\pi|y|)$ is a fundamental solution of P. The proof is the classical method of spherical means [18, §2.4.1], which reduces the wave equation in $3 + 1$ dimensions to the one-dimensional wave equation for $\rho \bar f$.

The formalization writes the spherical mean as an average over the group $O(3)$ of orthogonal operators on $\mathbb{R}^3$ with its Haar probability measure, $(\mathsf{A}f)(s, y) = \int_{O(3)} f(s, Q^{-1}y)\, dQ$. Mathlib has Haar measure on compact groups; for its measure on spheres it lacks the rotation invariance, and it lacks the divergence theorem on balls, on which the textbook proof rests. formalization $O(3)$ acts transitively on each sphere: if $|y| = |y'|$ and $y \neq y'$, the reflection $1 - 2zz^\top/|z|^2$ with $z = y' - y$ maps y' to y.

Corollary 8.8 (The Green identities). Lean ✓ *For $f \in \mathcal{D}$, the operators $G^\pm$ of Definition 8.4 and the wave operator P satisfy $G^+Pf = f$, $G^-Pf = f$, $PG^+f = f$ and $PG^-f = f$.*

Proof. P commutes with translations and with the reflections of time and of space. Fix (t, x) and let $f_{t,x}(s, z) = f(t - s, x - z)$; then $(Pf_{t,x})(|y|, y) = (Pf)(t - |y|, x - y)$. So by Proposition 8.5(a) and Theorem 8.6, $(G^+Pf)(t, x) = f_{t,x}(0) = f(t, x)$. For G^- use $(s, z) \mapsto f(t + s, x - z)$. Finally, $PG^\pm f = G^\pm Pf$ by Proposition 8.5(b). □

8.4 Cutoffs and the Green data

What the cutoffs do. The observable $W([f])$ depends on the test function f only through the solution Ef it generates, since $[f]$ is the class of f modulo $\ker E$. The time-slice axiom asks that every such observable already belong to the algebra of a thin slab of time S. To show this, one moves the smearing of f into S without changing Ef.

Take χ equal to 0 before S and to 1 after it, and replace f by $f' = -P(\chi Ef)$. Since $PEf = 0$, the function $P(\chi Ef)$ can be nonzero only where χ changes, that is inside S, and only where Ef lives, inside the causal shadow of $\operatorname{supp} f$: this is condition (c1). And $f - f' = P((1 - \chi)G^+f + \chi G^-f)$ is P of a test function, which E annihilates, so $Ef' = Ef$: this is condition (c2). Physically, f' is f transported along the equation of motion into the slab.

The slab must be allowed to be thin and to sit anywhere, and the family of slabs must be carried to itself by the symmetries. Here the slabs are the time intervals $t_1 \le \tau \le t_2$ of all inertial observers.

Definition 8.9 (Inertial slabs)**.** A vector n is *future unit*, short for future-directed unit timelike, if $n^0 > 0$ and $\eta(n,n) = -1$; its time function is $\tau_n(p) = -\eta(n,p) = n^0 p^0 - \vec{n}\cdot\vec{p}$, the time of the inertial observer with four-velocity n.

A *cutoff* is (n, t_1, t_2) with n future unit and $t_1 < t_2$. Its *slab* is $S = \{t_1 \le \tau_n \le t_2\}$, and χ is multiplication by $\chi(p) = \vartheta((\tau_n(p) - t_1)/(t_2 - t_1))$, with ϑ a smooth function equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$ (Mathlib's smooth transition function formalization).

Lemma 8.10 (Time functions and compactness)**.** Lean ✓ *Let n be future unit, τ_n its time function (Definition 8.9), and $J^{\pm}$ the causal future and past for the order $\le$ of Definition 8.1.*

(a) $n^0 > |\vec{n}|$, and $\tau_n(v) \ge (n^0 - |\vec{n}|)v^0 \ge 0$ for v future causal.

(b) For $Y \subseteq M$ compact, $J^{\pm}(Y)$ are closed, and $J^+(Y) \cap \{\tau_n \le t_2\}$ and $J^-(Y) \cap \{\tau_n \ge t_1\}$ are compact for all $t_1, t_2 \in \mathbb{R}$.

Proof. (a) $(n^0)^2 = 1 + |\vec{n}|^2$, and $\tau_n(v) \ge n^0 v^0 - |\vec{n}||\vec{v}| \ge (n^0 - |\vec{n}|)v^0$ by Cauchy–Schwarz.

(b) $J^+(Y)$ is the sum of Y and the closed cone of future causal vectors, which is closed. For $Y \neq \emptyset$ and $q = y + v$ with $y \in Y$ and $\tau_n(q) \le t_2$, (a) bounds $v^0 \ge |\vec{v}|$ by $(t_2 - \min_Y \tau_n)/(n^0 - |\vec{n}|)$, so the set is bounded; for $Y = \emptyset$ it is empty. The past case follows by $p \mapsto -p$. □

Proposition 8.11 (Conditions (c1) and (c2))**.** Lean ✓ *For every cutoff (n, t_1, t_2) of Definition 8.9, with slab S and multiplication χ, and every $f \in \mathcal{D}$, with $E = G^- - G^+$ for the operators of Definition 8.4:*

(c1) $\operatorname{supp} P(\chi E f) \subseteq S \cap J(\operatorname{supp} f)$*;*

(c2) $(1 - \chi)G^+ f + \chi G^- f \in \mathcal{D}$*.*

Proof. (c1). P is local: functions equal near p have equal P at p. Below the slab $\chi E f = 0$ near p; above it $\chi E f = E f$ near p, and $PEf = f - f = 0$ (Corollary 8.8). Outside $J(\operatorname{supp} f)$, the open cone $p + I^-$ misses $J^+(\operatorname{supp} f)$, since $p' \le q$ and $q - p \in I^-$ would give $p' \le p$ by transitivity. So $\chi G^+ f = 0$ there and $P(\chi G^+ f) = 0$ on $p + I^-$. The points $p - s e_0$, $s > 0$, tend to p, so $P(\chi G^+ f)(p) = 0$ by continuity. Likewise $P(\chi G^- f)(p) = 0$ with the open cone $p + I^+$. So $P(\chi E f)$ vanishes outside $S \cap J(\operatorname{supp} f)$, which is closed ($\tau_n$ is continuous; Lemma 8.10(b)) and therefore contains the support.

(c2). The function vanishes outside $\big(J^+(\operatorname{supp} f) \cap \{\tau_n \le t_2\}\big) \cup \big(J^-(\operatorname{supp} f) \cap \{\tau_n \ge t_1\}\big)$, which is compact by Lemma 8.10. □

Theorem 8.12 (The free massless field)**.** Lean ✓ *The objects of Definition 8.1, the operators $G^{\pm}$ of Definition 8.4 and the cutoffs of Definition 8.9 form Green data (Definition 7.1). If the CCR algebra of the derived phase space has an irreducible representation, the CCR net of the free massless field is generating, causal and has the time-slice property over the derived causal index, and its algebra is primitive.*

Proof. (G1)–(G5) are Proposition 8.3; (G6) is Proposition 8.5(b)–(d) and Corollary 8.8; (G7) is Proposition 8.11. Then Theorem 7.11. □

As after Theorem 7.11, generation, causality and the time-slice property hold without the hypothesis; only primitivity uses it. The data are not trivial: for $f \ge 0$ nonzero, $G^+ f$ vanishes before $\operatorname{supp} f$ while $G^- f$ does not, so $Ef \neq 0$, $V \neq 0$, and $\sigma \neq 0$ by nondegeneracy (Proposition 7.10). Regions are all subsets of M. One whose interior is empty contains the support of no nonzero test function, so its algebra is $\mathbb{C}1$.

Remark 8.13 (Which time slices). The cutoffs are slabs between parallel spacelike hyperplanes, one family for each inertial frame n. Remark 7.2 uses all Cauchy temporal functions, of which each τ_n is one. Fewer cutoffs give fewer pairs $O \triangleleft O'$, so the derived dependence here is contained in the one derived from all Cauchy temporal functions (both with closed supports).

- *Flat Cauchy surfaces.* For $\Sigma = \{\tau_n = c\}$, the first argument of Remark 7.6 runs with $\mathcal{T} = \tau_n - c$ and Lemma 8.10(b). So, for open regions with compact closure, the dependence of Example 1.4 with such a Σ implies the one here, and the time-slice axiom in that form holds.

- *Curved Cauchy surfaces.* Here the implication can fail.

  Dependence in the sense of Example 1.4. Let ϕ be smooth on $\mathbb{R}^3$ with $\phi(\vec{x}) = |\vec{x}|^2$ for $|\vec{x}| \le 2$ and $|\nabla\phi| \le 5$, so that $\Sigma = \{t = \phi(\vec{x})/10\}$ is a smooth spacelike Cauchy surface. Such ϕ exist: take $\phi(\vec{x}) = g(|\vec{x}|^2)$ with g smooth, $g(s) = s$ for $s \le 4$, and g' decreasing from 1 to 0 on $[4, 5]$ and zero beyond; then $|\nabla\phi| = 2|\vec{x}|\, g'(|\vec{x}|^2) \le 2\sqrt{5}$. Let $D = \{|t| + |\vec{x}| < 1\}$ and $O' = \{|t - \phi(\vec{x})/10| < 1/40,\ |\vec{x}| < 2\}$, both open with compact closure.

  A causal curve from D meets Σ where $|\vec{x}| < 1 + |\phi(\vec{x})|/10$. For $|\vec{x}| \ge 2$, $|\phi(\vec{x})| \le 4 + 5(|\vec{x}| - 2)$ would give $|\vec{x}| < 4/5$, which is impossible; so $|\vec{x}| < 2$, $\phi(\vec{x}) = |\vec{x}|^2$, and $|\vec{x}| < 1.13$. So every endless causal curve through D meets Σ in O', and $D \triangleleft O'$ in the sense of Example 1.4.

  No inertial slab serves. Now take $f \in \mathcal{D}(D)$ with $\{0\} \times \{|\vec{x}| \le 3/4\} \subseteq \operatorname{supp} f$. Then $J(\operatorname{supp} f)$ contains every line $\{(t, \vec{x}) : t \in \mathbb{R}\}$ with $|\vec{x}| \le 3/4$, and a slab meets such a line in a segment whose lowest point has t an affine function of $\vec{x}$. For $S \cap J(\operatorname{supp} f) \subseteq O'$ that affine function would have to stay within $1/40$ of $|\vec{x}|^2/10$ on the ball $|\vec{x}| \le 3/4$. But every affine function differs from $|\vec{x}|^2/10$ by at least $(3/4)^2/20 = 9/320 > 1/40$ somewhere on it (compare $\vec{x} = 0$ with $\vec{x} = \pm\frac{3}{4}e_1$). So $D \triangleleft O'$ fails for the inertial slabs.

  The inclusion $\mathcal{A}(D) \subseteq \mathcal{A}(O')$ itself still holds, on paper. The local algebras do not depend on the cutoffs. With all Cauchy temporal functions as cutoffs, Remark 7.6 gives $D \triangleleft O'$ for the derived index, Proposition 7.9 gives phase-space data over it, and the time-slice property of Theorem 4.7 gives the inclusion. It is not a consequence of Theorem 8.12.

Cutoffs from curved Cauchy temporal functions are not treated in these notes. The family of inertial slabs is carried to itself by Poincaré maps (Proposition 8.17), which covariance needs. Neither bullet is formalized.

8.5 Uniqueness and Poincaré covariance

Theorem 8.14 (Uniqueness of the Green operators)**.** Lean ✓ *If a map $G' : \mathcal{D} \to \mathcal{F}$ satisfies $PG'f = f$ and $\operatorname{supp} G'f \subseteq J^+(\operatorname{supp} f)$ for every $f \in \mathcal{D}$, then G' is the operator G^+ of Definition 8.4; likewise with J^- and G^-.*

Proof. Compact overlap. Let $f, g \in \mathcal{D}$. The supports of G^-g and $G'f$ lie in the closed sets $J^-(\operatorname{supp} g)$ and $J^+(\operatorname{supp} f)$. If $q \le q'$ then $q^0 \le q'^0$, so $J^-(\operatorname{supp} g) \subseteq \{\tau_{e_0} \le t_2\}$ with t_2 an upper bound of the times in $\operatorname{supp} g$. The intersection is therefore a closed subset of $J^+(\operatorname{supp} f) \cap \{\tau_{e_0} \le t_2\}$, which is compact by Lemma 8.10(b).

Pairing. By Proposition 8.3(e), Corollary 8.8 and Proposition 8.5(d),

$$\langle g, G'f\rangle = \int (PG^-g)G'f = \int G^-g\, PG'f = \langle f, G^-g\rangle = \langle g, G^+f\rangle,$$

and nondegeneracy gives $G'f = G^+f$. □

Definition 8.15 (Orthochronous Poincaré group)**.** An *orthochronous Lorentz map* is a linear $\Lambda : \mathbb{R}^4 \to \mathbb{R}^4$ with $\eta(\Lambda u, \Lambda w) = \eta(u, w)$ and $(\Lambda e_0)^0 > 0$. The *orthochronous Poincaré group* $\mathcal{P}^\uparrow$ consists of the maps $\gamma(p) = \Lambda p + a$, $a \in \mathbb{R}^4$, under composition; it contains the space reflections.

Lemma 8.16 (Orthochronous Lorentz maps)**.** Lean ✓ *Let $\gamma(p) = \Lambda p + a$ be in $\mathcal{P}^{\uparrow}$.*

(a) Λ is invertible, Λ^{-1} and composites of orthochronous Lorentz maps are orthochronous Lorentz maps, and $\mathcal{P}^{\uparrow}$ is a group.

(b) $\gamma p \le \gamma q$ iff $p \le q$.

(c) If n is future unit, so is Λn, and $\tau_n(\Lambda^{-1}p) = \tau_{\Lambda n}(p)$.

(d) $|\det \Lambda| = 1$, so γ preserves Lebesgue measure.

(e) $P(h \circ \gamma) = (Ph) \circ \gamma$ for smooth h.

Proof. (a) η is nondegenerate, so $\Lambda u = 0$ implies $u = 0$; Λ^{-1} and composites preserve η. Orthochrony of Λ^{-1} and of composites is shown in (b) and (c).

(b) Let $n' = \Lambda^{-1}e_0$. Then $\eta(n', n') = \eta(e_0, e_0) = -1$ and $n'^0 = -\eta(\Lambda^{-1}e_0, e_0) = -\eta(e_0, \Lambda e_0) = (\Lambda e_0)^0 > 0$, so n' is future unit and Λ^{-1} is orthochronous. For v future causal, $(\Lambda v)^0 = -\eta(\Lambda v, e_0) = -\eta(v, n') = \tau_{n'}(v) \ge (n'^0 - |\vec{n}'|)v^0 \ge 0$ by Lemma 8.10(a), and $\eta(\Lambda v, \Lambda v) = \eta(v, v) \le 0$; so Λv is future causal. The converse is the same for Λ^{-1}.

(c) $\eta(\Lambda n, \Lambda n) = -1$, and $(\Lambda n)^0 \ge (n'^0 - |\vec{n}'|)n^0 > 0$ by the inequality in (b), since $n^0 > |\vec{n}|$. With $n = \Lambda' e_0$ this gives orthochrony of $\Lambda\Lambda'$. Also $-\eta(n, \Lambda^{-1}p) = -\eta(\Lambda n, p)$.

(d) In the standard basis $\Lambda^{\top}\eta\Lambda = \eta$ and $\det \eta = -1$, so $(\det \Lambda)^2 = 1$.

(e) $Ph = -\operatorname{tr}(\eta^{-1}\partial^2 h)$ with $\partial^2 h$ the matrix of second derivatives, which transforms as $\Lambda^{\top}\partial^2 h\,\Lambda$. And $\Lambda\eta^{-1}\Lambda^{\top} = \eta^{-1}$ follows from $\Lambda^{\top}\eta\Lambda = \eta$: here $\eta^{-1} = \eta$, $\Lambda^{-1} = \eta\Lambda^{\top}\eta$, so $\Lambda\eta\Lambda^{\top}\eta = 1$. Translations commute with P.

□

Proposition 8.17 (Poincaré maps are symmetries)**.** Lean ✓ *Every $\gamma = (\Lambda, a) \in \mathcal{P}^{\uparrow}$ gives a symmetry of the Green data (Definition 7.12) with point map γ, $\gamma_{\mathcal{D}} f = f \circ \gamma^{-1}$, $\gamma_{\mathcal{F}} h = h \circ \gamma^{-1}$, and on cutoffs $(n, t_1, t_2) \mapsto (\Lambda n,\, t_1 + \tau_{\Lambda n}(a),\, t_2 + \tau_{\Lambda n}(a))$.*

Proof. Order, supports and pairing. The order is preserved by Lemma 8.16(b); γ is a homeomorphism, so it maps supports to supports; and the pairing is invariant by (d).

Slabs. By (c), $\gamma(S) = \{q : t_1 \le \tau_n(\Lambda^{-1}(q - a)) \le t_2\}$ is the slab of the image cutoff. The cutoff map is a bijection, with inverse the map of γ^{-1}, because the shifts add under composition: $\tau_{\Lambda\Lambda' n}(\Lambda a' + a) = \tau_{\Lambda' n}(a') + \tau_{\Lambda\Lambda' n}(a)$ by (c).

Green operators. $G' = \gamma_{\mathcal{F}} G^{+} \gamma_{\mathcal{D}}^{-1}$ satisfies $PG'f = f$ by (e) and $\operatorname{supp} G'f \subseteq J^{+}(\operatorname{supp} f)$ by (b), so $G' = G^{+}$ by Theorem 8.14; likewise for G^{-}. □

Theorem 8.18 (Poincaré covariance)**.** Lean ✓ *The CCR net of Theorem 8.12 is covariant under $\mathcal{P}^{\uparrow}$: with $\kappa_\gamma(O) = \gamma(O)$ and α_γ the automorphism of the CCR algebra induced by the symmetry of Proposition 8.17, $\gamma \mapsto \kappa_\gamma$ and $\gamma \mapsto \alpha_\gamma$ are group homomorphisms and every $(\kappa_\gamma, \alpha_\gamma)$ is a covariance pair (Definition 1.14).*

Proof. Proposition 7.13 gives, for each γ, an isomorphism of phase-space data with $L_\gamma[f] = [\gamma_{\mathcal{D}} f]$. Since $\gamma_{\mathcal{D}}$ composes as γ does, $L_{\gamma\gamma'} = L_\gamma L_{\gamma'}$ and $L_{\mathrm{id}} = \mathrm{id}$; likewise $\kappa_{\gamma\gamma'} = \kappa_\gamma \kappa_{\gamma'}$. Theorem 4.11 turns these isomorphisms into covariance pairs with $\alpha_{\mathrm{id}} = \mathrm{id}$ and $\alpha_{\gamma\gamma'} = \alpha_\gamma \alpha_{\gamma'}$. □

Every isometry of Minkowski space is affine, so $\mathcal{P}^{\uparrow}$ is the group of isometries that preserve the time orientation. This is covariance in the sense of Definition 1.14 under that group, Axiom 5 of [1] for this spacetime, restricted to the isometries that preserve the time orientation (Remark 1.8).

Two other maps are left out. Time reversal exchanges G^{+} and G^{-}, so it is not a symmetry of Green data in the sense of Definition 7.12. Dilations $p \mapsto \lambda p$, $\lambda > 0$, also preserve the order and the slabs, and, with $f \mapsto \lambda^{-3} f(\cdot/\lambda)$ on test functions, the form σ. Definition 7.12 does not express rescalings, and they are not used.

8.6 Green operators of a local operator

The proofs of this chapter on cutoffs, uniqueness and covariance use only these properties of the wave operator and its Green operators: the operator is local and can be integrated by parts; its Green operators satisfy the Green identities, are formally adjoint and are supported in the causal future and past; and, for covariance, the operator is Poincaré invariant.

This section states the construction on Minkowski space for every operator with these properties (Theorem 8.21), so that for the Klein–Gordon operator only the Green operators remain to be constructed. In this section P is any such operator and $G^{\pm}$ are its Green operators; in Chapter 9, from Definition 9.3 on, P is the wave operator and $G^{\pm}$ are the extended Green operators of Definition 9.3. In the formalization the results of this chapter on cutoffs, uniqueness and covariance are derived as the instance of Theorem 8.21 in Proposition 8.22, with their statements unchanged. formalization

Definition 8.19 (Local operators and their Green operators)**.** Let M, $\leq$, $\mathcal{D}$, $\mathcal{F}$, ι, supp and the pairing be as in Definition 8.1. A linear map $P : \mathcal{F} \to \mathcal{F}$ is *local* if, for all $h, h' \in \mathcal{F}$ and $p \in M$, $h = h'$ on a neighbourhood of p implies $(Ph)(p) = (Ph')(p)$. It is *Poincaré invariant* if $P(h \circ \gamma^{-1}) = (Ph) \circ \gamma^{-1}$ for all $h \in \mathcal{F}$ and all $\gamma \in \mathcal{P}^{\uparrow}$ (Definition 8.15).

Let P be local, and let $P_{\mathcal{D}} : \mathcal{D} \to \mathcal{D}$ be linear with $\iota P_{\mathcal{D}} = P\iota$ and $\langle P_{\mathcal{D}} f, h\rangle = \langle f, Ph\rangle$ for $f \in \mathcal{D}$ and $h \in \mathcal{F}$ (formal self-adjointness). Since ι is injective, $P_{\mathcal{D}}$ is determined by P. *Green operators for* P are linear maps $G^{\pm} : \mathcal{D} \to \mathcal{F}$ such that, for all $f, g \in \mathcal{D}$:

(a) $PG^{\pm} f = f$ and $G^{\pm} P_{\mathcal{D}} f = f$;

(b) $\operatorname{supp} G^{+} f \subseteq J^{+}(\operatorname{supp} f)$ and $\operatorname{supp} G^{-} f \subseteq J^{-}(\operatorname{supp} f)$;

(c) $\langle f, G^{+} g\rangle = \langle g, G^{-} f\rangle$.

Condition (c) and $G^{\pm} P_{\mathcal{D}} f = f$ follow from $PG^{\pm} f = f$, (b), locality and formal self-adjointness, on paper: $\langle f, G^{+} g\rangle = \int (PG^{-} f)\, G^{+} g = \int G^{-} f\, PG^{+} g = \langle g, G^{-} f\rangle$ by Lemma 8.20, and then $\langle g, G^{+} P_{\mathcal{D}} f\rangle = \langle P_{\mathcal{D}} f, G^{-} g\rangle = \langle f, PG^{-} g\rangle = \langle g, f\rangle$ for all g, so $G^{+} P_{\mathcal{D}} f = f$ by nondegeneracy (likewise for G^{-}). They are listed because Definition 7.1 lists them, and for the Klein–Gordon operator they are proved directly.

Lemma 8.20 (Locality and integration by parts)**.** Lean ✓ *Let $P : \mathcal{F} \to \mathcal{F}$ be local (Definition 8.19). Then $\operatorname{supp} Ph \subseteq \operatorname{supp} h$ for every $h \in \mathcal{F}$. For $h_1, h_2 \in \mathcal{F}$ whose supports meet in a compact set, $(Ph_1)h_2$ and $h_1\, Ph_2$ are integrable on M; if moreover $P_{\mathcal{D}}$ is as in Definition 8.19, then $\int_M (Ph_1)h_2\, dp = \int_M h_1\, (Ph_2)\, dp$.*

Proof. Supports. If $p \notin \operatorname{supp} h$, then $h = 0$ near p, so $(Ph)(p) = (P0)(p) = 0$ by locality. The complement of $\operatorname{supp} h$ is open, so Ph vanishes on it, and $\operatorname{supp} Ph \subseteq \operatorname{supp} h$.

Integrability. Let $K = \operatorname{supp} h_1 \cap \operatorname{supp} h_2$. By the first part both products are continuous and vanish outside K, which is compact.

Integration by parts. Take $\psi \in \mathcal{D}$ equal to 1 on a neighbourhood U of K. On U, $\psi h_1 = h_1$, so $P(\psi h_1) = Ph_1$ there by locality; off K both $(Ph_1)h_2$ and $(P(\psi h_1))h_2$ vanish. Hence $(Ph_1)h_2 = (P(\psi h_1))h_2$, and likewise $h_1\, Ph_2 = (\psi h_1)\, Ph_2$. With $\iota P_{\mathcal{D}} = P\iota$ and formal self-adjointness, $\int (Ph_1)h_2 = \langle P_{\mathcal{D}}(\psi h_1), h_2\rangle = \langle \psi h_1, Ph_2\rangle = \int h_1\, Ph_2$. □

Theorem 8.21 (Green data from Green operators)**.** Lean ✓ *Let $G^{\pm}$ be Green operators for a local operator P (Definition 8.19).*

(a) The data of Definition 8.1 with P, $P_{\mathcal{D}}$, $G^{\pm}$ and the inertial slabs of Definition 8.9 are Green data (Definition 7.1). If the CCR algebra of the derived phase space has an irreducible representation, the CCR net is generating, causal and has the time-slice property over the derived causal index, and its algebra is primitive; as after Theorem 7.11, only primitivity uses the hypothesis.

(b) If a map $G' : \mathcal{D} \to \mathcal{F}$ satisfies $PG'f = f$ and $\operatorname{supp} G'f \subseteq J^+(\operatorname{supp} f)$ for every $f \in \mathcal{D}$, then $G' = G^+$; likewise with J^- and G^-.

(c) If P is Poincaré invariant, every $\gamma \in \mathcal{P}^\uparrow$ gives a symmetry of these Green data with the maps of Proposition 8.17, and the CCR net is covariant under $\mathcal{P}^\uparrow$ in the sense of Theorem 8.18.

Proof. (a) (G1)–(G5) are Proposition 8.3(a), (b), (d) and $\iota P_{\mathcal{D}} = P\iota$, and (G6) is Definition 8.19(a)–(c). The proof of Proposition 8.11 uses of P only locality, continuity of $P(\chi G^\pm f)$ and $PEf = f - f = 0$, together with the supports of $G^\pm$ (Definition 8.19(b)) and the closedness of $J(\operatorname{supp} f)$, so it gives (c1); (c2) uses no property of P. The rest is Theorem 7.11.

(b) The proof of Theorem 8.14, with Lemma 8.20 for the integration by parts, Definition 8.19(a)–(c) for $G^\pm$, and Proposition 8.3(d) for nondegeneracy.

(c) In the proof of Proposition 8.17 only the Green operators involve P. For them, the map $G'f = G^+(f \circ \gamma) \circ \gamma^{-1}$ of that proof satisfies $PG'f = f$ by invariance and $\operatorname{supp} G'f \subseteq J^+(\operatorname{supp} f)$ by Lemma 8.16(b), so $G' = G^+$ by (b); likewise for G^-. The proof of Theorem 8.18 does not involve P.

□

Proposition 8.22 (The wave operator)**.** Lean ✓ *The wave operator P of Definition 8.1 is local and Poincaré invariant, and the operators $G^\pm$ of Definition 8.4 are Green operators for it (Definition 8.19). Their Green data (Theorem 8.21) are the Green data of Theorem 8.12.*

Proof. Locality and invariance. $(Ph)(p)$ is a combination of second derivatives at p, so P is local; it is Poincaré invariant by Lemma 8.16(e).

Green operators. $\iota P = P\iota$ and formal self-adjointness are Proposition 8.3(c), (e), and (a)–(c) of Definition 8.19 are Corollary 8.8 and Proposition 8.5(c), (d). The two Green data coincide component by component. □

9 The Klein–Gordon field on Minkowski space

Chapter 8 constructs Green data for the wave operator P on $3+1$ Minkowski space. This chapter adds a mass term. For every real m^2 it constructs the retarded and advanced Green operators of the Klein–Gordon operator $P + m^2$, which gives the free field with that operator: its CCR net is generating, causal and has the time-slice property for the inertial slabs, its algebra is primitive given an irreducible representation, its Green operators are unique, and it is covariant under the orthochronous Poincaré group (Theorem 9.18). At $m^2 = 0$ the Green data are those of Chapter 8 (Corollary 9.19).

The Green operators are the series $G^{\pm}_{m^2} = \sum_{j\geq 0}(-m^2)^j (G^{\pm})^{j+1}$ in the Green operators $G^{\pm}$ of the wave operator (Definition 8.4). Formally $(P + m^2)^{-1} = P^{-1}(1 + m^2 P^{-1})^{-1}$, and the series is the geometric series of the second factor. It is a sum of smooth functions $(G^{\pm})^{j+1} f$, not of distribution kernels (Remark 9.28).

Already its second term applies G^+ to $G^+ f$, which is not compactly supported. The chapter therefore first extends $G^{\pm}$ to functions supported in the causal future or past of a compact set and bounds the iterates by $(t - t_0)^{2j}/(2j)!$; these factorials make the series converge for every real m^2 (Proposition 9.12). It ends by comparing propagation: for $m^2 = 0$ the Green operators propagate only along the light cone, and for every $m^2 \neq 0$ they also propagate inside it (Theorem 9.27).

9.1 The Klein–Gordon operator

Definition 9.1 (Klein–Gordon operator)**.** For a real number m^2, the Klein–Gordon operator $P_{m^2} = P + m^2$ acts on $\mathcal{F}$ and on $\mathcal{D}$ by $P_{m^2} h = Ph + m^2 h$, with P the wave operator of Definition 8.1. The parameter m^2 is written as a square, as in Remark 2.1, whose operator P is P_{m^2} here, but may be negative: $m^2 > 0$ is the free field of mass m, and $m^2 < 0$ is the tachyonic case.

Lemma 9.2 (Properties of the Klein–Gordon operator)**.** Lean ✓ *For every $m^2 \in \mathbb{R}$ the operator P_{m^2} of Definition 9.1 satisfies:*

(a) $P_0 = P$, on $\mathcal{F}$ and on $\mathcal{D}$;

(b) $\iota P_{m^2} f = P_{m^2} \iota f$ for $f \in \mathcal{D}$;

(c) $\langle P_{m^2} f, h\rangle = \langle f, P_{m^2} h\rangle$ for $f \in \mathcal{D}$ and $h \in \mathcal{F}$;

(d) P_{m^2} is local, and it is Poincaré invariant (Definition 8.19).

Proof. Each property holds for P (Proposition 8.22) and for multiplication by the constant m^2, and is preserved by sums. □

9.2 The Green operators beyond test functions

By Theorem 8.21, the axioms, uniqueness and covariance for the field with P_{m^2} follow once Green operators for P_{m^2} exist; they are the series of Definition 9.10, built on the extension of $G^{\pm}$ in Proposition 9.6. The value $G^+ h(p)$ samples h only on the causal past of p. Cutting h off far from that past turns it into a test function without changing $G^+ h$ near p, so the results of Chapter 8 for test functions carry over.

Definition 9.3 (Extended Green operators; functions supported in a causal cone)**.** A function $h : M \to \mathbb{R}$ is *supported in the future of a compact set* if $\operatorname{supp} h \subseteq J^+(K)$ for some compact $K \subseteq M$, and *supported in the past of a compact set* if $\operatorname{supp} h \subseteq J^-(K)$ for some compact K. $\mathcal{F}^+$ and $\mathcal{F}^-$ are the smooth such functions; test functions lie in both.

The *extended Green operators* are the Green operators $G^{\pm}$ of Definition 8.4, extended to all functions: for $h : M \to \mathbb{R}$ and $p \in M$, $G^{\pm}h(p) = \int_M h(p-w)\,\mu^{\pm}(dw)$, taken as 0 when it does not converge absolutely formalization. For continuous h this is

$$G^{\pm}h(p) = \frac{1}{4\pi}\int_{\mathbb{R}^3} h(p^0 \mp |y|, \vec{p}-y)\,\frac{dy}{|y|}$$

by the proof of Proposition 8.5(a). On test functions they are the operators of Definition 8.4, by Proposition 8.5(a). Their iterates are $(G^{\pm})^0 h = h$ and $(G^{\pm})^{j+1}h = G^{\pm}((G^{\pm})^j h)$.

Lemma 9.4 (Causality of $G^{\pm}$). Lean ✓ *Let $h, h' : M \to \mathbb{R}$ be continuous and $p \in M$. If $h = h'$ on $J^-(\{p\})$, then the extended Green operator G^+ of Definition 9.3 satisfies $G^+h(p) = G^+h'(p)$. If $h = h'$ on $J^+(\{p\})$, then $G^-h(p) = G^-h'(p)$.*

Proof. The sample point $q = (p^0 - |y|, \vec{p}-y)$ of $G^+h(p)$ has $p - q = (|y|, y)$ future causal, so $q \le p$, and the two integrands agree for every y. The sample points of $G^-h(p)$ are $(p^0+|y|, \vec{p}-y) \ge p$. □

Lemma 9.5 (Localization). Lean ✓ *Let M, $\le$, $J^{\pm}$ and $\mathcal{D}$ be as in Definition 8.1, and* dist *the distance of the norm* $\max(|q^0|, |\vec{q}|)$ *on M.*

(a) For compact $K, K' \subseteq M$, the set $J^+(K) \cap J^-(K')$ is compact.

(b) For compact $A \subseteq M$ there is $\psi \in \mathcal{D}$ with $\psi = 1$ on $\{q : \mathrm{dist}(q, A) < 1\}$.

(c) Let $h, \psi : M \to \mathbb{R}$ and $K, U \subseteq M$ with $\operatorname{supp} h \subseteq J^+(K)$ *and $\psi = 1$ on $\{q : \mathrm{dist}(q, A) < 1\}$, where $A = J^+(K) \cap J^-(\overline{U})$. Then for every $p \in U$, $\psi h = h$ on a neighbourhood of every point of $J^-(\{p\})$.*

Proof. (a) The set is closed (Lemma 8.10(b)). It lies in $J^+(K) \cap \{q^0 \le t_2\}$, with t_2 an upper bound of the times in K', which is compact by Lemma 8.10(b) for the future unit vector e_0.

(b) A smooth bump function equal to 1 on the closed 1-neighbourhood of A and supported in its 2-neighbourhood, which is compact.

(c) Let $q \le p$. If $q \in \operatorname{supp} h$, then $q \in A$ and $\psi = 1$ near q. Otherwise $h = 0$ near q, and so is ψh.

□

Proposition 9.6 ($G^{\pm}$ on $\mathcal{F}^{\pm}$). Lean ✓ *For the extended Green operators $G^{\pm}$ and the spaces $\mathcal{F}^{\pm}$ of Definition 9.3:*

(a) if h is continuous with $\operatorname{supp} h \subseteq J^+(K)$*, K compact, then* $\operatorname{supp} G^+h \subseteq J^+(K)$*;*

(b) for $h \in \mathcal{F}^+$, G^+h is smooth, and $\partial_v G^+h = G^+\partial_v h$ for every direction v;

(c) for $h \in \mathcal{F}^+$, $G^+h \in \mathcal{F}^+$;

(d) for $h \in \mathcal{F}^+$, $PG^+h = h$ and $G^+Ph = h$;

(e) with $r(t, \vec{x}) = (-t, \vec{x})$, $G^-h(p) = G^+(h \circ r)(rp)$ for continuous h, and (a)–(d) hold for G^- on $\mathcal{F}^-$ with J^- in place of J^+.

Proof. Supports. If $G^+h(p) \ne 0$, some sample point $q \le p$ has $h(q) \ne 0$, so $p \in J^+(K)$ by transitivity; $J^+(K)$ is closed. This gives (a), and with (b) below, (c).

Localizing. Fix p', let U be the open unit ball about it in the norm of Lemma 9.5, and take ψ from Lemma 9.5(b), (a) with $A = J^+(K) \cap J^-(\overline{U})$. For $p \in U$, $h = \psi h$ on $J^-(\{p\})$ by Lemma 9.5(c), so $G^+h = G^+(\psi h)$ on U by Lemma 9.4. Since $\psi h \in \mathcal{D}$, Proposition 8.5(b) gives

smoothness on U. For the derivative, $\partial_v h = \partial_v(\psi h)$ near every point of $J^-(\{p\})$ by Lemma 9.5(c), so $\partial_v G^+ h = G^+ \partial_v(\psi h) = G^+ \partial_v h$ on U. This is (b).

Green identities. On U, $PG^+h = PG^+(\psi h) = \psi h$ by locality of P and Corollary 8.8, and $\psi h(p) = h(p)$. Since $Ph = P(\psi h)$ near every point of $J^-(\{p\})$, also $G^+Ph(p) = G^+P(\psi h)(p) = h(p)$. This is (d).

The advanced case. μ^- is the image of μ^+ under r, since $dy/|y|$ is invariant under $y \mapsto -y$; this gives the formula for G^-. The involution r reverses the order, so it exchanges $\mathcal{F}^+$ and $\mathcal{F}^-$; it commutes with P, and $\partial_v(h' \circ r) = (\partial_{rv} h') \circ r$ for smooth h'. Applying the retarded statements to $h \circ r$ gives the advanced ones, (e). □

Lemma 9.7 (Power bound). Lean ✓ *For every $s \geq 0$ and every integer $n \geq 0$,*

$$\int_0^s \rho(s-\rho)^n \, d\rho = \frac{s^{n+2}}{(n+1)(n+2)}.$$

Let $h : M \to \mathbb{R}$ be continuous, $n \geq 0$ an integer, and $t_0, T, C \in \mathbb{R}$, with $h(q) = 0$ for $q^0 < t_0$ and $|h(q)| \leq C(q^0 - t_0)^n/n!$ for $t_0 \leq q^0 \leq T$. Then the extended Green operator G^+ of Definition 9.3 satisfies

$$|G^+h(p)| \leq C \, \frac{(p^0 - t_0)^{n+2}}{(n+2)!} \qquad \text{for } t_0 \leq p^0 \leq T.$$

Proof. The integral. Substitute $\rho \mapsto s - \rho$ and integrate $(s-\rho)\rho^n$ term by term.

The bound. Put $s = p^0 - t_0$. The sample point $(p^0 - |y|, \vec{p} - y)$ has time at most T, and h vanishes there unless $|y| \leq s$, when its time exceeds t_0 by $s - |y|$. In polar coordinates,

$$|G^+h(p)| \leq \frac{C}{4\pi} \int_{|y| \leq s} \frac{(s-|y|)^n}{n!\,|y|} \, dy = \frac{C}{n!} \int_0^s \rho(s-\rho)^n \, d\rho = C \, \frac{s^{n+2}}{(n+2)!}.$$

□

Corollary 9.8 (Iterates). Lean ✓ *Let h be smooth with $\operatorname{supp} h \subseteq J^+(K)$, K compact, let $j \geq 0$, and let $(G^+)^j$ be the iterate of Definition 9.3. Then:*

(a) $(G^+)^j h$ is smooth with support in $J^+(K)$, and for every $t_0 \in \mathbb{R}$ it vanishes on $\{q^0 < t_0\}$ when h does;

(b) if $t_0, T, C \in \mathbb{R}$, h vanishes on $\{q^0 < t_0\}$ and $|h| \leq C$ on $\{q^0 \leq T\}$, then $|(G^+)^j h(p)| \leq C(p^0 - t_0)^{2j}/(2j)!$ for $t_0 \leq p^0 \leq T$;

(c) $\partial_v (G^+)^j h = (G^+)^j \partial_v h$ for every direction v; so if $t_0, T, C' \in \mathbb{R}$, h vanishes on $\{q^0 < t_0\}$ and $|\partial_v h| \leq C'$ on $\{q^0 \leq T\}$, then $|\partial_v (G^+)^j h(p)| \leq C'(p^0 - t_0)^{2j}/(2j)!$ for $t_0 \leq p^0 \leq T$.

Proof. Induction on j.

(a) Proposition 9.6(b), (c); the vanishing before t_0 follows from Lemma 9.4, since all sample points of $G^+h(p)$ lie at times at most p^0.

(b) Lemma 9.7 with $n = 2j$, since $2j + 2 = 2(j+1)$.

(c) Proposition 9.6(b), applied j times, and (b) for $\partial_v h$, which vanishes on the open set $\{q^0 < t_0\}$.

□

Proposition 9.9 (Formal adjointness of $G^\pm$ on $\mathcal{F}^\pm$). Lean ✓ *For $h_1 \in \mathcal{F}^-$ and $h_2 \in \mathcal{F}^+$ (Definition 9.3), the functions $h_1 \, G^+ h_2$ and $(G^- h_1) \, h_2$ are integrable, and $\int_M h_1 \, G^+ h_2 \, dp = \int_M (G^- h_1) \, h_2 \, dp$.*

Proof. Integrability. Let $\operatorname{supp} h_1 \subseteq J^-(K')$ and $\operatorname{supp} h_2 \subseteq J^+(K)$. By Proposition 9.6 both products vanish outside $A = J^+(K) \cap J^-(K')$, which is compact (Lemma 9.5(a)); they are continuous, hence integrable.

Cutting off. Take $\psi \in \mathcal{D}$ equal to 1 near A. For $p \in A$ and $q \le p$, either $h_2(q) = 0$ or $q \in J^+(K) \cap J^-(K') = A$; so $\psi h_2 = h_2$ on $J^-(\{p\})$, and likewise $\psi h_1 = h_1$ on $J^+(\{p\})$. Hence $G^+h_2(p) = G^+(\psi h_2)(p)$ and $G^-h_1(p) = G^-(\psi h_1)(p)$ by Lemma 9.4, and the localized products also vanish outside A. So Proposition 8.5(d) for the test functions ψh_1, ψh_2 gives $\int h_1\, G^+h_2 = \int \psi h_1\, G^+(\psi h_2) = \int G^-(\psi h_1)\, \psi h_2 = \int (G^-h_1)\, h_2$. □

9.3 The Green operators as a series

Definition 9.10 (The series)**.** For $m^2 \in \mathbb{R}$ and $f : M \to \mathbb{R}$, with the iterates $(G^\pm)^{j+1}$ of the extended Green operators (Definition 9.3),

$$G^\pm_{m^2} f(p) = \sum_{j \ge 0} (-m^2)^j (G^\pm)^{j+1} f(p),$$

with $(-m^2)^0 = 1$, the sum taken pointwise (and as 0 where the series does not converge absolutely formalization). For test functions the series converges absolutely (Proposition 9.12), so the convention is not used for them.

Lemma 9.11 (Linearity of the iterates)**.** Lean ✓ *For the extended Green operators $G^\pm$ and their iterates (Definition 9.3): for continuous h supported in the future of a compact set, the integrand $y \mapsto h(p^0 - |y|, \vec{p} - y)/|y|$ of $G^+h(p)$ is integrable for every p, and G^+ is additive on such functions. For smooth h_1, h_2 supported in the future of compact sets and every j, $(G^+)^j(h_1 + h_2) = (G^+)^j h_1 + (G^+)^j h_2$. For every $h : M \to \mathbb{R}$ and $\lambda \in \mathbb{R}$, $(G^\pm)^j(\lambda h) = \lambda\, (G^\pm)^j h$ and $(G^\pm)^{j+1} h = (G^\pm)^j (G^\pm h)$.*

Proof. Integrability and additivity. Let $\operatorname{supp} h \subseteq J^+(K)$; if K is empty, $h = 0$. Otherwise let $t_K = \min_K q^0$. A sample point in $J^+(K)$ has time at least t_K, so the integrand vanishes unless $|y| \le p^0 - t_K$. There it is bounded by $\max_B |h|/|y|$, with B the compact set of the sample points $(p^0 - |y|, \vec{p} - y)$, $|y| \le p^0 - t_K$, and $1/|y|$ is integrable on balls of $\mathbb{R}^3$. Additivity of G^+ follows.

Iterates. Additivity of $(G^+)^j$ follows by induction with Proposition 9.6(b), (c). Homogeneity holds for every integral, and the last identity is associativity of iteration. □

Proposition 9.12 (Convergence)**.** Lean ✓ *Let $m^2, t_0 \in \mathbb{R}$, $C \ge 0$, and let $f \in \mathcal{D}$ vanish on $\{q^0 < t_0\}$ with $|f| \le C$. For the iterates of Definition 9.3 and the series of Definition 9.10:*

(a) for every $j \ge 0$, $p \in M$ and $T \ge t_0$ with $p^0 \le T$,

$$\left|(-m^2)^j (G^+)^{j+1} f(p)\right| \le C\, (T - t_0)^2\, \frac{\left(\sqrt{|m^2|}\, (T - t_0)\right)^{2j}}{(2j)!}.$$

(b) the series $G^+_{m^2} f(p)$ converges absolutely at every p, and on each set $\{q^0 \le T\}$ its terms are bounded by a summable sequence that does not depend on p; since every test function vanishes before some time and is bounded, this holds for every test function;

(c) $G^+_{m^2}(f + g) = G^+_{m^2} f + G^+_{m^2} g$ for $f, g \in \mathcal{D}$, and $G^+_{m^2}(\lambda f) = \lambda\, G^+_{m^2} f$ for every $f : M \to \mathbb{R}$ and $\lambda \in \mathbb{R}$.

Proof. The bound. Corollary 9.8(b) with $j + 1$ in place of j gives $C(p^0 - t_0)^{2j+2}/(2j+2)!$, which is at most the right side, since $(2j+2)! \ge (2j)!$ and $p^0 - t_0 \le T - t_0$; for $p^0 < t_0$ the term vanishes (Corollary 9.8(a)).

Convergence. On $\{q^0 \le T\}$ the bound is the j-th term of $C(T - t_0)^2 \cosh \xi$ with $\xi = \sqrt{|m^2|}(T - t_0)$, a convergent series, since $\sum_j \xi^{2j}/(2j)! = \cosh \xi$ for every real ξ.

Linearity. Each term is additive on $\mathcal{D}$ (Lemma 9.11), and absolutely convergent series add termwise. Each term is homogeneous. □

Proposition 9.13 (Derivatives and smoothness). Lean ✓ *Let $e_0, \dots, e_3$ be the standard basis of $\mathbb{R}^4$ and $e^0, \dots, e^3$ its dual basis. For every $f \in \mathcal{D}$ and $m^2 \in \mathbb{R}$, the series $G^+_{m^2} f$ of Definition 9.10 satisfies:*

(a) $G^+_{m^2} f$ is differentiable at every $p \in M$, with derivative $\sum_{i=0}^{3} G^+_{m^2}(\partial_{e_i} f)(p)\, e^i$;

(b) $\partial_v G^+_{m^2} f = G^+_{m^2} \partial_v f$ for every $v \in \mathbb{R}^4$;

(c) $G^+_{m^2} f$ is smooth.

Proof. (a) Fix p and the open unit ball U about it. Each term $H_j = (-m^2)^j (G^+)^{j+1} f$ is smooth with $\partial_{e_i} H_j = (-m^2)^j (G^+)^{j+1} \partial_{e_i} f$ (Corollary 9.8). By Proposition 9.12 for the test functions $\partial_{e_i} f$, with t_0 below $\operatorname{supp} f$ and $T = \max(t_0, p^0 + 1)$, the norms of the derivatives of H_j are bounded on U by a summable sequence in j.

A series of differentiable functions that converges at one point of a convex open set, with derivatives bounded there by a summable sequence, is differentiable on the set with the summed derivative. Summing gives the formula.

(b) Write $v = \sum_i v^i e_i$ and use (a), additivity and homogeneity (Proposition 9.12).

(c) By induction on n, $G^+_{m^2} f$ is n times continuously differentiable for every test function f: by (a) its derivative is $p \mapsto \sum_i G^+_{m^2}(\partial_{e_i} f)(p)\, e^i$, and each $\partial_{e_i} f$ is a test function. □

Theorem 9.14 (Green identities). Lean ✓ *For every $f \in \mathcal{D}$ and $m^2 \in \mathbb{R}$, the series $G^+_{m^2}$ of Definition 9.10 and the operator P_{m^2} of Definition 9.1 satisfy $P_{m^2} G^+_{m^2} f = f$ and $G^+_{m^2} P_{m^2} f = f$.*

Proof. Commuting with P. P is a combination of second directional derivatives, so Proposition 9.13(b) twice, additivity and homogeneity give $P G^+_{m^2} f = G^+_{m^2} P f$.

Telescoping. By Lemma 9.11 and Corollary 8.8, $(G^+)^{j+1} P f = (G^+)^j (G^+ P f) = (G^+)^j f$. Hence, shifting the index of an absolutely convergent series,

$$G^+_{m^2} P f = \sum_{j \geq 0} (-m^2)^j (G^+)^j f = f + \sum_{j \geq 0} (-m^2)^{j+1} (G^+)^{j+1} f = f - m^2 G^+_{m^2} f.$$

So $P_{m^2} G^+_{m^2} f = G^+_{m^2} P f + m^2 G^+_{m^2} f = f$, and $G^+_{m^2} P_{m^2} f = f$ by additivity and homogeneity. □

Proposition 9.15 (Supports). Lean ✓ *For every $f \in \mathcal{D}$ and $m^2 \in \mathbb{R}$, the series of Definition 9.10 satisfies* $\operatorname{supp} G^+_{m^2} f \subseteq J^+(\operatorname{supp} f)$.

Proof. Every term vanishes outside $J^+(\operatorname{supp} f)$ (Corollary 9.8(a)), so the sum does, and $J^+(\operatorname{supp} f)$ is closed. □

Proposition 9.16 (The advanced series). Lean ✓ *For every $f \in \mathcal{D}$, $m^2 \in \mathbb{R}$ and $j \geq 0$, with $r(t, \vec{x}) = (-t, \vec{x})$ and the iterates and series of Definitions 9.3 and 9.10, $(G^-)^j f = ((G^+)^j (f \circ r)) \circ r$ and $G^-_{m^2} f = (G^+_{m^2}(f \circ r)) \circ r$. Consequently:*

(a) the series $G^-_{m^2} f(p)$ converges absolutely at every p;

(b) $G^-_{m^2}$ is additive on $\mathcal{D}$ and homogeneous;

(c) $\partial_v G^-_{m^2} f = G^-_{m^2} \partial_v f$;

(d) $G^-_{m^2} f$ is smooth;

(e) $P_{m^2} G^-_{m^2} f = f$ and $G^-_{m^2} P_{m^2} f = f$;

(f) $\operatorname{supp} G^-_{m^2} f \subseteq J^-(\operatorname{supp} f)$.

Proof. The identities. The one-step identity is Proposition 9.6. By induction, applying it to $h = ((G^+)^j(f \circ r)) \circ r$ gives $(G^-)^{j+1} f = ((G^+)^{j+1}(f \circ r)) \circ r$, since $r \circ r$ is the identity. Summing gives the series identity.

Consequences. The reflection commutes with P and exchanges J^+ and J^-, so the consequences follow from Propositions 9.12, 9.13 and 9.15 and Theorem 9.14 for $f \circ r$. □

Proposition 9.17 (Formal adjointness of the series)**.** Lean ✓ *For $f, g \in \mathcal{D}$ and $m^2 \in \mathbb{R}$, the series of Definition 9.10 satisfy $\int_M f\, G^+_{m^2} g\, dp = \int_M (G^-_{m^2} f)\, g\, dp$.*

Proof. Exchanging sum and integral. On the compact set $\operatorname{supp} f$, the terms of $G^+_{m^2} g$ are bounded by a summable sequence: Proposition 9.12 for g, with t_0 below $\operatorname{supp} g$ and $T = \max(t_0, T')$ with T' an upper bound of the times in $\operatorname{supp} f$. So $\int f\, G^+_{m^2} g = \sum_j (-m^2)^j \int f\, (G^+)^{j+1} g$ by dominated convergence. Likewise on the other side, where the terms of $G^-_{m^2} f$ are bounded on $\operatorname{supp} g$ through $(G^-)^{j+1} f = ((G^+)^{j+1}(f \circ r)) \circ r$ (Proposition 9.16) and Proposition 9.12 for $f \circ r$.

Moving one G at a time. For $0 \le i \le j$, Proposition 9.9 with $h_1 = (G^-)^i f \in \mathcal{F}^-$ and $h_2 = (G^+)^{j-i} g \in \mathcal{F}^+$ gives $\int (G^-)^i f\, (G^+)^{j+1-i} g = \int (G^-)^{i+1} f\, (G^+)^{j-i} g$. Here h_1 is supported in the past of $\operatorname{supp} f$ by the advanced case of Proposition 9.6, applied i times, and h_2 in the future of $\operatorname{supp} g$ by Corollary 9.8(a). The $j+1$ steps give $\int f\, (G^+)^{j+1} g = \int (G^-)^{j+1} f\, g$; summing gives the claim. □

9.4 The Klein–Gordon field

Theorem 9.18 (The Klein–Gordon field)**.** Lean ✓ *For every $m^2 \in \mathbb{R}$, the series $G^\pm_{m^2}$ of Definition 9.10, restricted to test functions, are Green operators (Definition 8.19) for the Klein–Gordon operator P_{m^2} of Definition 9.1. Consequently:*

(a) they give Green data with the inertial slabs, whose CCR net is generating, causal and has the time-slice property over the derived causal index; if the CCR algebra has an irreducible representation, its algebra is moreover primitive;

(b) every map $G' : \mathcal{D} \to \mathcal{F}$ with $P_{m^2} G' f = f$ and $\operatorname{supp} G' f \subseteq J^+(\operatorname{supp} f)$ for all $f \in \mathcal{D}$ equals $G^+_{m^2}$, and likewise $G^-_{m^2}$ with J^-;

(c) the CCR net is covariant under the orthochronous Poincaré group $\mathcal{P}^\uparrow$, with $\kappa_\gamma(O) = \gamma(O)$ and α_γ induced by the symmetry of Theorem 8.21(c).

Proof. $G^\pm_{m^2}$ map $\mathcal{D}$ linearly into $\mathcal{F}$ (Propositions 9.12, 9.13 and 9.16). Condition (a) of Definition 8.19 is Theorem 9.14 and Proposition 9.16, (b) is Propositions 9.15 and 9.16, and (c) is Proposition 9.17; P_{m^2} has the required properties by Lemma 9.2. Then Theorem 8.21; generation, causality and time slice hold without the hypothesis, as after Theorem 7.11. □

The cutoffs, and with them the derived causal index, do not depend on m^2. So Remark 8.13 applies unchanged: the time-slice property implies the time-slice axiom in the form of Example 1.4 for flat Cauchy surfaces, on paper; for a curved Cauchy surface the implication can fail, and the axiom then rests on all Cauchy temporal functions as cutoffs (Remark 7.6), which are not formalized.

Corollary 9.19 (The massless field is the case $m^2 = 0$)**.** Lean ✓ *The series of Definition 9.10 satisfy $G^\pm_0 f = G^\pm f$ for every $f : M \to \mathbb{R}$, and the Green data of Theorem 9.18 with $m^2 = 0$ are equal to the Green data of the free massless field (Theorem 8.12). So every statement of Chapter 8 about the massless field is a statement about P_0.*

Proof. All terms with $j \ge 1$ carry the factor $0^j = 0$. By Lemma 9.2(a), the Green data of Theorem 9.18 at $m^2 = 0$ are those of Theorem 8.21 for P and the extended Green operators, which are the Green data of Theorem 8.12 by Proposition 8.22. Green data are determined by their components, since (G1)–(G7) are properties, so equal components give equal Green data. □

9.5 Huygens' principle

For $m^2 = 0$ the operator G^+ samples f only on the past light-cone surface of the point, so $G^+ f$ vanishes at every point strictly inside the future cone of all of $\operatorname{supp} f$: signals travel exactly at the speed of light (Proposition 9.21). For $m^2 \neq 0$ the second term $-m^2 (G^+)^2 f$ already reaches inside the cone, and Theorem 9.26 shows that, for suitable f and p, the rest of the series does not cancel it.

Definition 9.20 (Light-cone surface, chronological order, Huygens' principle)**.** $C^+ = \{v \in M : v^0 \geq |\vec{v}|\}$ is the closed future cone, and $\partial C^+ = \{v : v^0 = |\vec{v}|\}$ is its boundary, the future light-cone surface. For $p, q \in M$, $p \ll q$ means $q - p \in I^+$, that is $|\vec{q} - \vec{p}| < q^0 - p^0$ (I^+ as in Definition 8.1). For $A, B \subseteq M$, $A + B = \{a + b : a \in A,\ b \in B\}$ and $-B = \{-b : b \in B\}$; so $\operatorname{supp} f + \partial C^+$ is the union of the light-cone surfaces $q + \partial C^+$, $q \in \operatorname{supp} f$. A linear map $G : \mathcal{D} \to \mathcal{F}$ with the supports of G^+ in Definition 8.19(b) satisfies *Huygens' principle* if $Gf(p) = 0$ for every $f \in \mathcal{D}$ and every $p \in M$ with $q \ll p$ for all $q \in \operatorname{supp} f$; one with the supports of G^- satisfies it if $Gf(p) = 0$ whenever $p \ll q$ for all $q \in \operatorname{supp} f$.

Proposition 9.21 (Huygens' principle for the wave operator)**.** Lean ✓ *For every $f \in \mathcal{D}$, the extended Green operators of Definition 9.3 and the sets of Definition 9.20 satisfy $\operatorname{supp} G^+ f \subseteq \operatorname{supp} f + \partial C^+$ and $\operatorname{supp} G^- f \subseteq \operatorname{supp} f + (-\partial C^+)$. In particular $G^+ f(p) = 0$ whenever $q \ll p$ for all $q \in \operatorname{supp} f$, and $G^- f(p) = 0$ whenever $p \ll q$ for all $q \in \operatorname{supp} f$.*

Proof. Supports. Every sample point of $G^+ f(p)$ is $p - (|y|, y)$ with $(|y|, y) \in \partial C^+$. If $p \notin \operatorname{supp} f + \partial C^+$, no sample point lies in $\operatorname{supp} f$, and $G^+ f(p) = 0$. The set $\operatorname{supp} f + \partial C^+$ is closed, as the sum of a compact and a closed set, so it contains $\operatorname{supp} G^+ f$. For G^- the sample points are $p - (-|y|, y)$ with $(-|y|, y) \in -\partial C^+$.

Points inside the cone. If $q \ll p$ for all $q \in \operatorname{supp} f$, then $p - q \in I^+$, which misses ∂C^+, so $p \notin \operatorname{supp} f + \partial C^+$. The advanced case is the same with $-I^+$. □

Lemma 9.22 (Positivity of G^+)**.** Lean ✓ *Let $h : M \to \mathbb{R}$ be continuous and nonnegative, $p \in M$, and let $y \mapsto h(p^0 - |y|, \vec{p} - y)/|y|$ be integrable on $\mathbb{R}^3$. If $h(p^0 - |y_1|, \vec{p} - y_1) > 0$ for some $y_1 \neq 0$, then the extended Green operator of Definition 9.3 satisfies $G^+ h(p) > 0$.*

Proof. The integrand is nonnegative, and continuous and positive at $y_1 \neq 0$, hence positive on a ball about y_1, which has positive measure. □

Proposition 9.23 (Positivity of $(G^+)^2$)**.** Lean ✓ *Let $f \in \mathcal{D}$ be nonnegative with $f(q_0) > 0$, and let $q_0 \ll p$ (Definition 9.20). Then the iterate of Definition 9.3 satisfies $(G^+)^2 f(p) > 0$.*

Proof. A point on both cones. Put $s = p^0 - q_0^0$ and $d = \vec{p} - \vec{q}_0$, so $|d| < s$. Choose a unit vector $\hat{d}$ with $d = |d|\hat{d}$ (any unit vector if $d = 0$) and let $y_1 = \frac{1}{2}(s + |d|)\hat{d} \neq 0$. The sample point $q = (p^0 - |y_1|, \vec{p} - y_1)$ has $q^0 - q_0^0 = \frac{1}{2}(s - |d|) > 0$ and $\vec{q} - \vec{q}_0 = \frac{1}{2}(|d| - s)\hat{d}$. So $z_1 = \vec{q} - \vec{q}_0$ satisfies $|z_1| = q^0 - q_0^0 > 0$, and $(q^0 - |z_1|, \vec{q} - z_1) = q_0$.

Two applications of Lemma 9.22. For $h = f$ at q with z_1, whose integrand is integrable by Lemma 9.11, it gives $G^+ f(q) > 0$. The function $G^+ f$ is continuous, nonnegative (its integrand is) and supported in the future of a compact set (Proposition 9.6), so its integrand is integrable (Lemma 9.11). For $h = G^+ f$ at p with y_1 the lemma gives $(G^+)^2 f(p) > 0$. □

Lemma 9.24 (The tail)**.** Lean ✓ *Let $m^2, t_0, C \in \mathbb{R}$, and let f be smooth with compact support, $f(q) = 0$ for $q^0 < t_0$, and $|f| \leq C$. For $p^0 \geq t_0$ and $s = p^0 - t_0$, the series of Definition 9.10 satisfies*

$$|G^+_{m^2} f(p) - G^+ f(p) + m^2 (G^+)^2 f(p)| \leq C\,(m^2)^2\, s^6 \cosh\!\big(\sqrt{|m^2|}\, s\big).$$

Proof. The left side is $|\sum_{j \geq 2} (-m^2)^j (G^+)^{j+1} f(p)|$. With $\xi = \sqrt{|m^2|}\, s$, Proposition 9.12 with $T = p^0$ bounds it by $C s^2 \sum_{j \geq 2} \xi^{2j}/(2j)!$; here $C \geq 0$, since $|f| \leq C$. And $\sum_{j \geq 2} \xi^{2j}/(2j)! = \xi^4 \sum_{k \geq 0} \xi^{2k}/(2k+4)! \leq \xi^4 \cosh \xi$, since $(2k+4)! \geq (2k)!$, and $C s^2 \xi^4 = C (m^2)^2 s^6$. □

Lemma 9.25 (Dilation). Lean ✓ *Let $\lambda > 0$, $f : M \to \mathbb{R}$ and $f_\lambda(q) = f(q/\lambda)$, with the extended Green operators, iterates and series of Definitions 9.3 and 9.10.*

(a) For continuous f and every p, $G^+ f_\lambda(\lambda p) = \lambda^2\, G^+ f(p)$.

(b) For smooth f with compact support and every $m^2 \in \mathbb{R}$, $(G^+)^j f_\lambda(q) = \lambda^{2j} (G^+)^j f(q/\lambda)$ for all j and q, and $G^+_{m^2} f_\lambda(\lambda p) = \lambda^2\, G^+_{\lambda^2 m^2} f(p)$.

Proof. (a) Substitute $y = \lambda z$: $dy = \lambda^3 dz$ and $|y| = \lambda |z|$, which leaves λ^2. One integral converges absolutely exactly when the other does.

(b) Induction on j, using (a) for the continuous function $(G^+)^j f$ (Corollary 9.8) and homogeneity. Then $\sum_j (-m^2)^j \lambda^{2j+2} (G^+)^{j+1} f(p) = \lambda^2 \sum_j (-\lambda^2 m^2)^j (G^+)^{j+1} f(p)$. □

Theorem 9.26 (Failure of Huygens' principle for $m^2 \neq 0$). Lean ✓ *For every $m^2 \neq 0$, with the series of Definition 9.10 and $\ll$ as in Definition 9.20, there are a test function f and a point p with $q \ll p$ for all $q \in \operatorname{supp} f$ and $G^+_{m^2} f(p) \neq 0$. Likewise there are $f \in \mathcal{D}$ and p with $p \ll q$ for all $q \in \operatorname{supp} f$ and $G^-_{m^2} f(p) \neq 0$.*

Proof. A fixed configuration. Let f_0 be a smooth bump function with $0 \le f_0 \le 1$, $f_0(0) = 1$ and support in $\{\max(|q^0|, |\vec{q}|) \le \frac{1}{4}\}$, and let $p_0 = (1, \vec{0})$. Every $q \in \operatorname{supp} f_0$ has $|\vec{p}_0 - \vec{q}| \le \frac{1}{4} < \frac{3}{4} \le p_0^0 - q^0$, so $q \ll p_0$. Hence $G^+ f_0(p_0) = 0$ (Proposition 9.21) and $c = (G^+)^2 f_0(p_0) > 0$ (Proposition 9.23 with $q_0 = 0$).

Small parameters. Lemma 9.24 with $C = 1$, $t_0 = -\frac{1}{4}$ and $s = \frac{5}{4}$ gives $|G^+_{m_0^2} f_0(p_0) + m_0^2 c| \le (m_0^2)^2 s^6 \cosh s$ for $|m_0^2| \le 1$, since cosh is increasing on $[0, \infty)$. If moreover $m_0^2 \neq 0$ and $|m_0^2|\, s^6 \cosh s < c$, the right side is less than $|m_0^2|\, c$, so $G^+_{m_0^2} f_0(p_0) \neq 0$.

Every m^2. Choose $\lambda > 0$ so small that $m_0^2 = \lambda^2 m^2$ satisfies both conditions, and put $f = (f_0)_\lambda$, $p = \lambda p_0$. Then $\operatorname{supp} f = \lambda \operatorname{supp} f_0$, and $q \ll p$ if and only if $q/\lambda \ll p_0$. By Lemma 9.25, $G^+_{m^2} f(p) = \lambda^2 G^+_{m_0^2} f_0(p_0) \neq 0$.

The advanced case. Take $g = f \circ r$ and the point rp. Then $G^-_{m^2} g(rp) = G^+_{m^2} f(p) \neq 0$ by Proposition 9.16, and $q \ll p$ for all $q \in \operatorname{supp} f$ becomes $rp \ll q'$ for all $q' \in \operatorname{supp} g$. □

Theorem 9.27 (Huygens' principle holds exactly at $m^2 = 0$). Lean ✓ *For $m^2 \in \mathbb{R}$, the series $G^+_{m^2}$ of Definition 9.10 and $\ll$ as in Definition 9.20, the following are equivalent:*

(i) $G^+_{m^2} f(p) = 0$ for every $f \in \mathcal{D}$ and every $p \in M$ with $q \ll p$ for all $q \in \operatorname{supp} f$;

(ii) $m^2 = 0$.

The same holds for $G^-_{m^2}$, with $p \ll q$ in (i).

Proof. (ii)⇒(i) is Corollary 9.19 with Proposition 9.21; (i)⇒(ii) is Theorem 9.26. □

Condition (i) is not vacuous: the configuration of Theorem 9.26 has $\operatorname{supp} f \neq \emptyset$ and p in the open future cone of each of its points.

For $m^2 \neq 0$ the Green data are not trivial: for the test function f and the point p of the second sentence of Theorem 9.26 (the advanced case), $G^+_{m^2} f(p) = 0$ because $p \notin J^+(\operatorname{supp} f)$, so $Ef(p) = G^-_{m^2} f(p) \neq 0$, and $\sigma \neq 0$ by Proposition 7.10.

By Theorem 9.18(b), Huygens' principle is a property of P_{m^2}, not of the series. For operators with the supports of Definition 8.19(b), it is equivalent to the support form of Proposition 9.21, on paper: if $p \notin \operatorname{supp} f + \partial C^+$, then $\operatorname{supp} f$ lies in the union of the disjoint open sets $p - I^+$ and $M \setminus (p - C^+)$, and splitting f accordingly gives one part covered by (i) and one part whose support misses $J^-(\{p\})$.

Remark 9.28 (Kernels). Condition (i) of Theorem 9.27 says that the convolution kernel of $G^+_{m^2}$, a distribution in $(t,x) \in M$ (its Schwartz kernel is $k(p-q)$), vanishes on the open future cone I^+ (on paper; kernels are not formalized). For $m^2 = 0$ the kernel is $\delta(t-|x|)/(4\pi|x|)$ (Remark 8.7), and the kernel of $(G^+)^2$ is $1/(8\pi)$ times the indicator function of C^+, so that $c = \frac{1}{8\pi}\int_{p_0 - C^+} f_0$ in Theorem 9.26.

For $m^2 > 0$, with $m = \sqrt{m^2}$, the kernel has, on I^+, the additional term

$$-\frac{m}{4\pi}\,\frac{J_1(m\sqrt{t^2-|x|^2})}{\sqrt{t^2-|x|^2}},$$

with J_1 the Bessel function [20] (pinpoint to be checked). For $m^2 < 0$ the additional term is

$$\frac{\sqrt{-m^2}}{4\pi}\,\frac{I_1(\sqrt{-m^2}\sqrt{t^2-|x|^2})}{\sqrt{t^2-|x|^2}},$$

with the modified Bessel function I_1. Summing the kernels of the powers $(G^+)^{j+1}$, which are locally integrable functions for $j \geq 1$, gives these terms.

The proof above uses neither formula, only the first two terms of the series and a bound on the rest. For $m^2 < 0$ and $f \geq 0$ every term $(-m^2)^j (G^+)^{j+1} f$ is nonnegative, so Proposition 9.23 alone gives $G^+_{m^2} f(p) > 0$ at every p in the open future cone of a point where $f > 0$, without the tail bound or dilation. For $m^2 > 0$, $f \geq 0$ and p in the open future cone of every point of $\operatorname{supp} f$, Lemma 9.24 gives $G^+_{m^2} f(p) < 0$ for the dilated configurations of Theorem 9.26 with small λ. The additional kernel term changes sign at the zeros of $J_1(m\sqrt{t^2-|x|^2})$, and so does $G^+_{m^2} f(p)$ for f concentrated near a point (on paper). Bessel functions are not formalized.

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Lean declarations

The linked Lean ✓ tag marks a result whose Lean proof has been checked and whose correspondence with the statement here has been reviewed. Definitions and the unmarked Examples 1.7 and 3.8 are listed for navigation; only marked statements were reviewed. Files marked (in BosonicLean) belong to the library BosonicLean, on which this one depends, and the declarations whose names begin with `BosonicLean` are defined there.

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Chapter 8. The free massless field on Minkowski space

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